

Human Capital and Industrial Reallocation

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Abstract

The skill composition of the local workforce is pivotal to the location and expansion of industries. Yet, its role in reshaping industrial structure remains less understood, since workforce skills and local production patterns evolve jointly. This paper studies how exogenous shifts in county human capital distribution affect employment and establishment shares of industries with different workforce education profiles. I construct education-specific instruments using a two-step ancestry-based strategy that predicts county ancestry stocks from historical origin-by-destination immigration patterns over 1860-2010 and maps them into working-age populations by educational attainment for 1970-2010. Relative to low-skill population share, one-percentage-point increases in medium- and high-skill shares raise employment share of high-skill industries by 0.18 and 0.54 percentage points and reduce its low-skill counterpart by 0.39 and 0.32 percentage points, respectively. Establishment shares exhibit similar reallocation. Positive responses arise in nontradable sectors, whereas contractions occur primarily in tradable industries. A decomposition of changes in county mean schooling attributes 26 percent of the adjustment to between-industry reallocation and 74 percent to within-industry upgrading. A CES model with imperfect substitution across skill groups and sectoral differences in skill intensity provides a benchmark for the between-industry response. The findings establish industrial composition as an important margin through which local labor markets adjust to changes in relative skill supply.

Keywords: Labor Supply, Skills, Human Capital, Industrial Composition, Local Labor Markets.

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1 Introduction

How local labor markets respond to changes in the relative supply of skills is a central question in labor economics. Existing work documents adjustment through wages, technology, task assignment, worker mobility, local demand, amenities, and structural change.¹ An additional channel may arise when variation in human capital across local labor markets changes the industrial composition of economic activity. In response to a rise in local human capital, industries that use skilled labor more intensively may enter and expand, whereas less skill-intensive industries may contract. At the same time, skilled workers may sort towards locations where skill-intensive industries are already concentrated, making the causal effect of human capital difficult to identify. Despite the strong correlation between workforce skills and industry skill intensity, causal evidence on whether changes in relative skill supply reshape industrial composition remains limited.²

This paper studies how exogenous shifts in the human capital of local populations affect the industrial skill composition of employment and establishments across US counties. Examining both outcomes reveals whether shifts in local human capital induce both a reallocation of workers and corresponding changes in establishment composition. Thus, the fundamental question is whether increases in local population shares with varying human capital levels shift the distribution of employment and establishments towards industries that employ more highly educated workers across counties.

This question has direct implications for workforce development and place-based policies. Investments in education and training are often motivated by the expectation that a more skilled workforce will attract or expand skill-intensive economic activity and create higher-paying employment opportunities.³ Such policies can generate a lasting impact in local industrial structure only if the composition of employment and establishments responds to the augmented skill supply. Otherwise, the gains may dissipate if firms fail to enter, or if skilled workers relocate elsewhere. Understanding how shifts in the relative supply of skills shape industrial composition is, therefore, pivotal to evaluating policies intended to raise earnings and expand skill-intensive employment.

The empirical challenge pertains to isolating variation in local population human capital that is not generated by contemporaneous labor demand conditions. I address this challenge by

¹See [Katz and Murphy \(1992\)](#), [Acemoglu \(2011\)](#), [Autor and Dorn \(2013\)](#), [Hollanders and Ter Weel \(2002\)](#), [Moretti \(2010\)](#), [Diamond \(2016\)](#), and [Boddin and Kroeger \(2025\)](#).

²See [Bartik \(1991\)](#), [Notowidigdo \(2020\)](#), and [Giannone \(2022\)](#) for related evidence on local labor supply and industrial adjustment.

³For relevant discussions of local human capital, productivity, and wage premia, see [Moretti \(2004b\)](#) and [Greenstone et al. \(2010\)](#).

exploiting three documented features of historical immigration: (i) origin-specific selection in immigrant human capital, (ii) the inter-generational transmission of human capital, and (iii) the persistence of settlement patterns through ethnic enclaves and social networks. Together, these channels provide a source of long-run variation in county human capital that is plausibly exogenous to contemporaneous worker sorting and firm location decisions.

First, immigrants from different origin countries have arrived with systematically distinct skill levels and occupational specializations. This selection process (Borjas, 1987; Abramitzky et al., 2012; Clemens and Mendola, 2024) has generated cross-location differences in the composition of human capital by shifting the distribution of skills across industries (Goldin, 1994; Abramitzky et al., 2014; Boberg-Fazlić and Sharp, 2024). Second, historical settlement patterns of immigrants has produced heterogeneity in educational attainment across localities through the inter-generational transmission of human capital (Abramitzky and Braggion, 2006; Sequeira et al., 2020). Finally, immigrants have historically clustered in areas with pre-existing ethnic enclaves, demonstrating the non-random nature of immigrant settlement (Altonji and Card, 1991; Card, 2001). Social networks have further amplified these patterns by attracting subsequent cohorts of immigrants with similar skill endowments (Munshi, 2003). Collectively, these mechanisms contribute to the persistence and path dependence of the spatial variation in contemporary human capital.⁴

Motivated by these empirical regularities, I proceed with my identification strategy in two steps. First, using detailed data on ancestry groups (Fulford et al., 2019), I predict ancestry stocks for each decennial census year from 1970 to 2010. Following the panel implementation of Terry et al. (2026), which builds on the origin-by-destination “push-pull leave-out” method within a shift-share instrumental variable (SSIV) approach developed in Burchardi et al. (2019), I exploit historical immigration patterns from 1860 to 2010 to isolate variation in county ancestry stocks.⁵ This construction uses leave-out variation in both the push and pull components to reduce endogeneity arising from origin-destination-specific migration forces and local economic conditions.⁶

⁴See Allen and Donaldson (2020) and Voth (2021) for the persistence and path dependence of historical shocks.

⁵Note that Burchardi et al. (2019) predict the number of individuals with reported ancestries at the county level, reflecting the changing nature of ethnic identity as a social construct. By contrast, I use a different measure of ancestry in the spirit of Fulford et al. (2019) that highlights genealogical construction by taking internal migration, generational transmission, and population growth of ancestries into account.

⁶Although the design differs from conventional Card-type instruments, persistence in immigration and settlement patterns remains a potential threat to identification rather than one mechanically eliminated by the construction (Jaeger et al., 2018). Goldsmith-Pinkham et al. (2020) and Borusyak et al. (2022) develop alternative identification frameworks for canonical shift-share designs based on exposure-share exogeneity and shock orthogonality, respectively. See Borusyak et al. (2025) for a comprehensive treatment. Although Adao et al. (2019) study a canonical shift-share setting, their analysis highlights the risk of over-rejection when locations with similar exposure patterns have correlated regression residuals.

Second, I use the predicted ancestry stocks to generate plausibly exogenous variation in working-age populations by educational attainment at the county level. I predict the total number of individuals within each education group by utilizing the full vector of predicted ancestry stocks over the 1970-2010 period. These predicted populations isolate the ancestry-driven component of local human capital, allowing me to estimate its effect on the industrial composition of employment and establishments.

I make two main contributions to the relevant literature. My first contribution is to provide the causal evidence on how exogenous shifts in human-capital-specific local working-age population impact the industrial skill composition of employment and establishment shares at the county level. Existing studies primarily examine how local labor markets absorb changes in relative skill supply through wages, employment, technology adoption, and within-firm adjustments ([Acemoglu, 1998](#); [Moretti, 2004b](#); [Lewis, 2011](#); [Dustmann and Glitz, 2015](#); [Modestino et al., 2016](#)). I identify a distinct compositional margin by investigating whether an increase in local population human capital shifts workers and establishments towards industries that rely more intensively on highly educated labor. Considering employment and establishments together allows me to assess whether the reallocation of workers is accompanied by corresponding adjustment in the composition of local production units. I further distinguish between tradable and nontradable industries to estimate the degree to which this adjustment differs between activities exposed to global competition and those more closely tied to local demand.

My second contribution is methodological. I construct skill-specific instruments for county working-age populations that isolate plausibly exogenous variation in local human capital composition from contemporaneous labor demand conditions. Since workers sort towards locations offering higher returns to their skills, and firms may locate where the labor they demand is relatively abundant, local human capital is jointly determined with industrial composition. I address this endogeneity through a two-step ancestry-based identification strategy. First, I use historical origin-by-destination immigration patterns to predict county ancestry stocks. Second, I map these predicted stocks into local working-age populations by educational attainment. While existing approaches generally instrument a single summary measure, such as average schooling or the local college-graduate share ([Moretti, 2004a](#); [Sand, 2013](#)), my approach constructs separate, time-varying instruments for education-specific population groups. This allows me to identify shifts in the relative composition of local human capital rather than changes only in its average level or college-educated share. The instruments draw on persistent variation in historical immigrant settlement, the inter-

generational transmission of human capital, and the reinforcement of settlement patterns through ethnic enclaves and social networks. The resulting variation enables me to estimate how county-level industrial composition responds to shifts in local population human capital.

To guide the empirical analysis, I develop a constant elasticity of substitution (CES) model with two representative firms, one white-collar and one blue-collar (Autor et al., 1998). The white-collar firm is relatively intensive in high-skill labor, whereas the blue-collar firm is relatively intensive in low-skill labor. Each firm produces a sector-specific output in a perfectly competitive market, taking output prices and common skill-specific wages as given. Within each firm, high- and low-skill workers are imperfect substitutes (Katz and Murphy, 1992; Card and Lemieux, 2001). In equilibrium, an increase in the relative supply of high-skill labor raises the employment share of the high-skill-intensive sector and lowers that of the low-skill-intensive sector. Therefore, the model isolates the between-sector employment reallocation margin and provides the main empirical predictions for the two polar skill groups.

I find that, relative to low-skill population share, increases in medium- and high-skill population shares reallocate county employment and establishment shares towards high-skill industries. A one-percentage-point rise in medium- or high-skill population shares raises employment share of high-skill industries by approximately 0.2–0.5 percentage points and reduces low-skill industry share by 0.3–0.4 percentage points, while medium-skill industries exhibit smaller and mixed responses. Establishment composition adjusts similarly: high-skill industry share rises by roughly 0.3–1.0 percentage point, low-skill share falls by 0.1–0.7 percentage points, and medium-skill share declines modestly following an increase in high-skill population share. These findings identify a labor-supply-driven compositional margin that complements evidence linking local human capital to industrial reallocation and specialization (Simon, 2004; Davis and Dingel, 2020), plant productivity and technology adoption (Moretti, 2004b; Beaudry et al., 2010), and local employment growth and employer skill requirements (Blien et al., 2006; Modestino et al., 2016). Unlike studies centered on productivity, aggregate employment growth, or within-firm skill demand, I show that exogenous shifts in local human capital reshape the county-level distribution of both employment and establishments across industries with different workforce education profiles.

To examine heterogeneity by industry tradability, I estimate the baseline specification separately for tradable and nontradable industries. The positive responses are concentrated in the nontradable sector, whereas the negative responses arise primarily among tradable industries. This pattern is consistent with a local demand channel, through which increases in population human

capital support the expansion of nontradable goods and services (Moretti, 2010; Mian and Sufi, 2014). The contraction of tradable industries, by contrast, is consistent with a loss of local cost competitiveness as low-skill labor becomes relatively scarce. The decomposition by establishment size further demonstrates that the positive responses are concentrated among medium-sized establishments in the nontradable sector, while the negative responses fall mainly on small establishments in both sectors. Taken together, these findings indicate that tradability and establishment size shape the local adjustment of industrial reallocation: nontradable industries provide the principal expansion margin, whereas contractions are mainly borne by tradable industries and smaller establishments.

To shed light on the adjustment mechanisms, I conduct two complementary exercises. First, I quantify the relative importance of between- and within-industry adjustment using a Foster–Haltiwanger–Krizan decomposition (Foster et al., 2001) of the change in county mean years of schooling associated with an increase in high-skill labor supply. After normalizing the within- and between-industry components to sum to one, between-industry reallocation, the margin emphasized by the CES model, accounts for approximately 27 percent of the total, while within-industry upgrading accounts for the remaining 73 percent. Thus, shifts in local human capital are absorbed through both changes in employment across industries and changes in workforce education within industries. Second, I examine whether the magnitude of industrial reallocation varies with counties’ initial industry composition. The estimates show that counties with initially smaller high-skill industry shares exhibit larger reallocation responses to increases in high-skill population share, suggesting that initial industrial structure shapes the scope for subsequent adjustment.

Finally, I conduct an extensive set of robustness checks covering alternative instrument constructions and samples, alternative industry tradability and skill classifications, the inclusion of controls, tests probing the exclusion restriction, spatial correlation in the regression errors, commuting zones as an alternative geographic unit, and permutation-based inference. The main findings remain largely robust across these specifications.

This paper connects to four strands of research. First, it relates to the labor literature on how local economies absorb changes in the relative supply of skills. Existing work documents adjustment through wages, technology adoption, factor intensity, employer skill requirements, and changes in the scale and composition of production units (Acemoglu, 1998; Lewis, 2011; Dustmann and Glitz, 2015; Modestino et al., 2016). Related studies link local human capital to productivity, employment growth, industrial specialization, and employment structure (Moretti, 2004b; Simon,

2004; Hollanders and Ter Weel, 2002; Blien et al., 2006; Davis and Dingel, 2020; Boddin and Kroeger, 2025). The closest connections are to Dustmann and Glitz (2015), who examine within- and across-firm responses to changes in local skill supply, and to Simon (2004) and Davis and Dingel (2020), who study the relationship between local human capital and industrial specialization. I bridge these strands by showing that exogenous increases in local population human capital shift county-level employment and establishment shares towards skill-intensive industries.

Second, the paper speaks to research on the persistence of historical events in shaping present-day economic outcomes (Nunn, 2014; Allen and Donaldson, 2020; Voth, 2021; Abramitzky and Braggion, 2006; Bleakley and Lin, 2012; Sequeira et al., 2020; Fulford et al., 2019). Relative to these studies, I highlight a new labor market implication of historical persistence and path dependence. Historical immigrant settlement patterns, together with origin-specific differences in human capital and its intergenerational transmission, provide long-run variation in the educational composition of county populations. I use the ancestry-predicted component of this variation to identify the response of local industrial composition to relative skill supply. Historical immigration is, therefore, the source of identifying variation rather than the treatment of interest.

Third, the empirical strategy draws on the shift-share and ancestry-based instrumental-variables literature. Earlier studies combine historical settlement patterns with subsequent origin-specific inflows to construct immigration-induced labor-supply shocks (Altonji and Card, 1991; Card, 2001; Ottaviano and Peri, 2012; Dustmann et al., 2017). I build on the origin-by-destination “push-pull leave-out” framework of Burchardi et al. (2019) and its panel application in Terry et al. (2026) to predict county ancestry stocks while accounting for origin- and destination-specific confounding shocks. I then map these stocks into separate, time-varying instruments for working-age populations by educational attainment. Whereas existing approaches generally instrument a single measure of local human capital, including mean schooling or the college-graduate share (Moretti, 2004a; Sand, 2013), this construction identifies shifts in its relative educational composition.

Finally, the theoretical framework builds on CES models in which workers with different skills are imperfect substitutes (Katz and Murphy, 1992; Card and Lemieux, 2001), and firms differ in the intensity with which they use each skill group (Autor et al., 1998). While this literature commonly emphasizes relative wages and within-firm skill demand, I derive a prediction for the sectoral composition of employment. An increase in the relative supply of a skill group expands the employment share of the sector that uses that group more intensively and contracts the share of the other sector. Thus, the model links population skill composition to the between-industry

reallocation estimated empirically, while establishment composition and within-industry upgrading represent complementary margins outside the benchmark framework.

The remainder of the paper is organized as follows. Section 2 presents the theoretical framework, introducing the model and deriving the comparative statics. Section 3 outlines the empirical strategy, addressing endogeneity concerns in human capital and describing the identification strategy. Section 4 details the data sources, explains the skill and tradability classifications, and reports summary statistics. Section 5 presents the main results and discussion. Section 6 tests the adjustment mechanisms. Section 7 examines the robustness of the findings. Section 8 concludes.

2 Theoretical Framework

In this section, I outline a theoretical framework that guides the subsequent empirical analysis. The model features two firm types, each producing a sector-specific output and employing one of the two skill groups more intensively, following Autor et al. (1998). Each firm combines labor of those groups using a constant elasticity of substitution (CES) technology that allows for imperfect substitution between them (Katz and Murphy, 1992; Card and Lemieux, 2001). Unlike the standard approach, which emphasizes wage adjustment, the framework focuses on how changes in relative labor supplies shift the equilibrium allocation of production across firm types, thereby shaping local industrial composition and patterns of regional development.

2.1 Setup

There exist two representative firm types, blue and white collar, indexed by $i \in \{b, w\}$. Each firm type specializes in producing a sector-specific output, Q_i , which it sells at price p_i in a perfectly competitive market. Production combines high-skill labor ($L_{i,h}$) and low-skill labor ($L_{i,l}$) according to a CES technology:

$$Q_i = \left(\beta_{i,h} L_{i,h}^{\frac{\sigma-1}{\sigma}} + \beta_{i,l} L_{i,l}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \quad (1)$$

Each firm type takes the economy-wide skill wages, ω_h and ω_l , and its output price, p_i , as given. Because both firm types face the same market wage for a given skill, each equates the value of that skill's marginal product to the same wage in equilibrium.

Since the production technology exhibits constant returns to scale, free entry and perfect

competition imply zero profits for each active firm type, so that its unit cost equals its output price. For tractability, I assume that the two sectoral outputs have the same price, $p_b/p_w = 1$. Because only relative prices matter, I choose this common output price as the numeraire and set $p_b = p_w = 1$. The two zero-profit conditions then imply equality of unit costs across firm types, yielding the equilibrium condition that determines the relative skill wage.

The equilibrium labor allocations must also satisfy the economy-wide labor-market-clearing conditions

$$L_{b,h} + L_{w,h} = N_h, \quad (2)$$

$$L_{b,l} + L_{w,l} = N_l. \quad (3)$$

These clearing conditions are imposed across both firm types rather than within either firm's individual optimization problem. The firms's first-order conditions, the zero-profit conditions, and the two market-clearing conditions jointly characterize the common skill wages and the equilibrium labor allocations.

2.2 Comparative Statics

The differences in relative productivity parameters and the assumption of imperfect substitutability between high- and low-skill labor imply that $B_b > 1 > B_w$, $B_b - B_w > 0$, and $Z > 0$. By construction, the labor supply shares satisfy $n_h + n_l = 1$, while the sectoral employment shares satisfy $s_w + s_b = 1$. Consequently, there is only one independent labor supply share and one independent sectoral employment share. Writing $N_h + N_l = N$, $N_h = n_h N$, and $N_l = (1 - n_h)N$, the equilibrium allocations are homogeneous of degree one in (N_h, N_l) . The corresponding employment shares are, therefore, homogeneous of degree zero, allowing total labor supply to be normalized to $N = 1$ without loss of generality. Substituting $n_l = 1 - n_h$, and differentiating $s_h(n_h)$ and $s_b(n_h) = 1 - s_w(n_h)$ directly yields

$$\frac{ds_w}{dn_h} = \frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} > 0, \quad (4)$$

$$\frac{ds_b}{dn_h} = -\frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} < 0, \quad (5)$$

$$\frac{ds_b}{dn_l} = \frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} > 0, \quad (6)$$

$$\frac{ds_w}{dn_l} = -\frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} < 0, \quad (7)$$

where $B_b = (\beta_{b,l}/\beta_{b,h})^\sigma$, $B_w = (\beta_{w,l}/\beta_{w,h})^\sigma$, and $Z = [(\beta_{b,l}^\sigma - \beta_{w,l}^\sigma)/(\beta_{w,h}^\sigma - \beta_{b,h}^\sigma)]^{\frac{\sigma}{\sigma-1}}$. These derivatives satisfy the accounting identities $s_w + s_b = 1$ and $n_h + n_l = 1$, so $ds_w/dn_h + ds_b/dn_h = 0$ and $ds_w/dn_l = -ds_b/dn_l$. The signs of these derivatives follow directly from $B_b > 1 > B_w$, $B_b - B_w > 0$, and $Z > 0$. Because the equilibrium shares (s_w, s_b) are linear in the labor supply shares (n_h, n_l) , the reallocation slope $(B_b + Z)(B_w + Z)/[Z(B_b - B_w)]$ is constant over the range of skill supplies supporting an interior equilibrium. Under the maintained assumption that technologies and relative output prices are common across counties, this slope is also common across counties.⁷

An increase in the relative supply of high-skill workers raises the sectoral employment share of white-collar firms, while a rise in low-skill labor supply reduces it, as shown in equations (4) and (7). Conversely, the sectoral employment share of blue-collar firms increases with greater low-skill labor supply and declines when high-skill labor becomes more abundant, as shown in equations (6) and (5). These results follow from differences in relative productivity across labor types. White-collar firms have a comparative advantage in employing high-skill labor ($\beta_{w,h} > \beta_{b,h}$), whereas blue-collar firms are relatively more productive with low-skill labor ($\beta_{b,l} > \beta_{w,l}$) more intensively. Thus, a higher supply share of high-skill labor reallocates production towards the high-skill-intensive sector, raising its employment share and lowering that of the low-skill-intensive sector. Likewise, a larger supply share of low-skill labor expands the relative size of blue-collar firms. These responses reflect firm specialization combined with imperfect substitutability between labor types.⁸

The magnitude of these comparative statics is governed by the underlying production parameters, including σ , which determines the degree of substitutability between high- and low-skill labor within each firm type. The benchmark model isolates the between-sector adjustment margin: a change in relative skill supply reallocates employment towards the sector that uses the increasingly abundant skill group more intensively.⁹

⁷The full derivations of the model are presented in Appendix A.

⁸The two-skill formulation isolates the model's central mechanism by contrasting the two polar skill groups. The empirical analysis additionally distinguishes medium-skill workers, whom I interpret as occupying an intermediate position between low- and high-skill labor. Introducing a third skill group would add an additional first-order condition and substitution margin, but the qualitative reallocation mechanism carries over under a monotone ordering of factor intensities: low-skill-intensive firms employ low-skill labor relatively more intensively, high-skill-intensive firms employ high-skill labor relatively more intensively, and medium-skill labor lies between the two. Thus, the two-skill model provides the clearest predictions for the movement of employment away from low-skill-intensive activity and towards more skill-intensive activity.

⁹The analysis focuses on the case in which high- and low-skill labor are imperfect but finite substitutes, $1 < \sigma < \infty$. Appendix A derives the corresponding skill-specific sectoral allocation shares, showing how changes in labor supply composition affect the fractions of high- and low-skill workers employed in each firm type.

The empirical analysis extends beyond this benchmark in three respects. First, I examine whether the reallocation of employment is accompanied by corresponding adjustment in establishment composition. Second, I assess within-industry changes in workforce education alongside the between-industry margin emphasized by the model. Finally, I test whether the magnitude of industrial reallocation varies with counties' initial industry composition and other local characteristics. These exercises capture adjustment margins and sources of heterogeneity that are not explicitly modeled in the benchmark framework.

3 Empirical Strategy

3.1 Estimating Equation and Potential Threats to Identification

Guided by the theoretical framework, the empirical analysis tests whether changes in the relative supply of different skill groups translate into shifts in the industrial composition of local labor markets. Specifically, I examine how the shares of medium- and high-skill population, measured relative to low-skill population, affect employment and establishment shares of industries of varying skill intensity across US counties over time. Considering both employment and establishment outcomes allows me to capture adjustment along two related margins of local industry structure, reflecting changes in the allocation of workers and establishments within local economies. I estimate the following equation

$$y_{e,d,t} = \alpha_{s(d)} + \alpha_t + \mathbf{n}_{d,t}\beta_e + \varepsilon_{e,d,t}, \quad (8)$$

where $\mathbf{n}_{d,t}$ is a vector of medium- and high-skill population shares in county d at time t , measured by human capital levels. For each decennial census year, I define this variable as the number of working-age individuals with certain educational attainment by county relative to the entire working-age county population. I classify individuals with a high school degree and less as low skill, those with some college education as medium skill, and those with college degree and above as high skill. $y_{e,d,t}$ is the outcome of interest, which is employment and establishment shares of industry skill type e in county d at time t in separate specifications. The classification of industries into skill cells closely follows the human capital classification. In particular, I classify industries with average years of education corresponding to a high-skill degree and less as low skill, those with mean education equivalent to some college as medium skill, and those with average education comparable to college degree and above as high skill.

I define the employment share variable for each skill type as the number of workers in industries of each skill type by county divided by the entire county workforce in each census year. I construct the establishment share variable of each skill type as the number of establishments in industries of each skill type by county relative to the total number of establishments by county in each census year. $\alpha_{s(d)}$ and α_t represent state and time (decennial census year) fixed effects controlling for any time-invariant state-wide factors and aggregate shocks, as well as time-specific trends. The identifying variation, therefore, arises from changes across counties within the same state over time. The error term $\varepsilon_{e,d,t}$ captures all omitted factors. Standard errors are clustered at the county level to account for arbitrary serial correlation within counties, which is the level of treatment variation.¹⁰

The vector β_e , where $e \in \{medium, high\}$, contains the coefficients of interest identifying the effects of medium- and high-skill population shares on the outcome. The OLS estimates of β_e , however, are likely biased because unobserved confounding factors may affect the industrial skill composition of both employment and establishments in counties, despite the inclusion of state and time fixed effects. This problem may lead to reverse causality and omitted variable bias, which may emerge due to the local labor market and firm creation opportunities being likely correlated with local productivity shocks. Consequently, these opportunities may attract individuals with certain educational attainment to a specific locality. For instance, the major concentration of the oil and gas industry in Houston (Harris county) may draw in high-skill petroleum engineers. Furthermore, as this sector experiences a boom, it may result in an expansion of oil refineries (*i.e.*, establishments). Therefore, these local confounding shocks would generate spurious correlations between population human capital shares and the outcomes of interest. In the following subsection, I underline the mechanisms as motivating facts to account for this endogeneity concern.

3.2 What Explains Today’s Varying Human Capital Levels across US Counties?

The first mechanism that contributes to the variation in the current economic conditions in US localities is the heterogeneity in skills and occupational specialization of immigrants in both historical and modern times. To the extent that retrospective immigrant selection process (Borjas, 1987; Abramitzky et al., 2012) across different origins is instrumental in affecting future economic outcomes in destinations, one can illustrate shifts in the skill composition of industries through

¹⁰Clustering at the state level would be excessively conservative given the relatively small number of states and the sub-state nature of the analysis.

variation in immigrant human capital.¹¹ For instance, some immigrants historically provided unskilled labor (Bergquist, 2007; Goldin, 1994) and crucial skills for industries (Malone, 1935), while others provided knowledge and innovation in various industries (Wittke, 1939; Boberg-Fazlić and Sharp, 2024).

Findings in Abramitzky et al. (2014) reveal substantial heterogeneity in the 1900-level occupational distribution of immigrants. The similar patterns on the task specialization, innovation, comparative advantage, and skills can still be observed among the relatively recent immigrants (Peri and Sparber, 2009; Hunt and Gauthier-Loiselle, 2010; Chiquiar and Hanson, 2005; Chiswick and Taengnoi, 2007; Hanson and Liu, 2023; Clemens and Mendola, 2024).

Building on the channel identified above, the second mechanism pertains to persistence and path dependence, the long-lasting impact of historical events (Nunn, 2014; Allen and Donaldson, 2020; Voth, 2021). From the human capital perspective, one likely channel through which the spatial differences in human capital levels exist at present can be attributed to historical immigration. In particular, settlements of earlier immigrants have induced varying degrees of educational attainment among local population via the transmission of human capital. For example, Sequeira et al. (2020) find that counties with a higher exposure to historical immigration have a larger level of educational attainment today.¹² Abramitzky and Braggion (2006) explain the comparative economic performance in the subsequent periods through the historical positive and negative selections among servants to mainland American colonies and those to West Indies, respectively.¹³

The final mechanism connects to historical immigration leading to the emergence of ethnic enclaves (Altonji and Card, 1991; Card, 2001). The location choice of prospective immigrants is heavily dictated by pre-existing immigrant settlements in a certain location. This can be corroborated by the diverse historical ancestry compositions across regions of the US such as the prevalence of Mexican ancestry in the Southwest, Italian ancestry in the Northeast, and German ancestry in the Midwest (Abramitzky and Boustan, 2017). These social networks are another contributing factor in attracting immigrants to specific areas. The existence of these networks is pivotal to immigrants arriving in the succeeding periods, since they might receive job referrals or acquire information on local economic conditions from their fellow countrymen (Munshi, 2003).

¹¹The immigrant selection process is built upon the selection of workers into occupations pioneered in Roy (1951).

¹²For the contributions of immigrants to the US educational system, see Faust (1916).

¹³Valencia Caicedo (2019) documents the role of eighteenth century Jesuit presence in places across South America. Rocha et al. (2017) demonstrate the importance of historical immigrant settlement policies in Brazilian municipalities in underpinning current higher educational attainment. Droller (2018) finds that counties with higher shares of historical European immigrants have higher GDP per capita today due to better educational attainment among immigrants in Argentina.

3.3 The Construction of Valid Instruments

Motivated by the mechanisms highlighted above, I generate the quasi-experimental variation in ancestry relying on historical settlement patterns at the granular level within a version of the shift-share instrumental variable (SSIV) approach. Using this predicted ancestry, I isolate plausibly exogenous working-age population with varying levels of educational attainment. These instruments correct for the previously identified endogeneity problem in human capital.

3.3.1 Predicting Ancestry Stocks

First, I predict the total number of individuals with ancestry o (*i.e.*, country), who resides in d (*i.e.*, county) at time t (*i.e.*, census year), $\mathcal{A}_{o,d,t}$, by following the econometric procedure developed in [Burchardi et al. \(2019\)](#) but using the geneological ancestry measure of [Fulford et al. \(2019\)](#).¹⁴ This method predicts ancestry stocks via the interactions of the historical “push” and “pull” factors along with a leave-out strategy. Thus, I identify granular variation in $\mathcal{A}_{o,d,t}$ that is plausibly exogenous to both county- and country-county-specific factors, by estimating

$$\mathcal{A}_{o,d,t} = \alpha_{o,r(d)} + \alpha_{c(o),d} + X'_{o,d}\phi + \sum_{\tau=1860}^t \gamma_{r(d),\tau} \left(I_{o,-r(d),\tau} \frac{I_{-c(o),d,\tau}}{I_{-c(o),\tau}} \right) + u_{o,d,t}, \quad (9)$$

where $\mathcal{A}_{o,d,t}$ is the total number of individuals with ancestry o settled in county d at time t . $I_{o,-r(d),\tau}$ is the “push” factor that is the total number of immigrants from o at time τ settled in counties *outside* of the census division where d is located. $\frac{I_{-c(o),d,\tau}}{I_{-c(o),\tau}}$ is the “pull” factor: that is, the denominator shows the total number of immigrants in the United States at time τ coming from *outside* of the continent where o is located, whereas the numerator displays the total number of immigrants in location d at time τ , again from *outside* of the continent where o is located.¹⁵ $\alpha_{o,r(d)}$ and $\alpha_{c(o),d}$ represent a series of country $\times$ destination census division and continent of origin $\times$ destination county fixed effects. $X'_{o,d}$ is a vector of the time-invariant distance and difference in latitude controls between o and d . The identification strategy in this step allows me to estimate equation (9) separately for each decennial census year between 1970 and 2010 utilizing all origin

¹⁴[Burchardi et al. \(2019\)](#) predict county-level ancestries based on reported identities that evolve as social constructs, while I employ an ancestry measure following [Fulford et al. \(2019\)](#) that reflects geneological lineage by incorporating internal migration, inter-generational transmission, and ancestry-specific population growth.

¹⁵See this [website](#) for the allocation of states to census divisions.

countries and all destination counties in my sample.¹⁶ Finally, I obtain predicted ancestry stocks

$$\hat{\mathcal{A}}_{o,d,t} = \sum_{\tau=1860}^t \hat{\gamma}_{\tau} \left(I_{o,-r(d),\tau} \frac{I_{-c(o),d,\tau}}{I_{-c(o),\tau}} \right), \quad (10)$$

where $\hat{\gamma}_{\tau}$ are the coefficients that I estimate from equation (9). Note that I take the residuals of the interactions of push and pull factors with respect to all fixed effects and controls included in equation (9) similar to [Terry et al. \(2026\)](#). This procedure is necessary to conclude that $\hat{\mathcal{A}}_{o,d,t}$ hinges solely on local variations in relative predicted ancestry stocks by eliminating any remaining confounding factors between large and small counties as well as countries. This addresses the first threat, reverse causality, highlighted above.

The following example provides some intuition. Using this approach, I predict a large stock of ancestry from Mexico relative to other ancestry groups from the Americas to Los Angeles county in the Pacific census division relative to other census divisions in the West in 1970 if the following occur. First, a considerably large number of Mexicans migrate to the United States in 1910, into census divisions excluding the Pacific census division in the West. Second, Los Angeles county is an appealing destination to foreign immigrants from *any* origins excluding the Americas in 1910. This method ensures that Mexican migrants with particular skill sets are left out to mitigate endogenous sorting into destinations.

On the other hand, had I used observed (realized) ancestry stocks in “push” and “pull” factors without the “leave-out” element (as in the canonical shift-share approach), this would induce a spurious correlation between the local ancestry composition and the outcomes owing to the persistence in local productivity or amenity shocks. Such shocks attract immigrants with different human-capital mixes, thereby inducing non-random destination choices.

3.3.2 Isolating Exogenous Population with Varying Levels of Educational Attainment

After predicting ancestry stocks, I proceed to the second step to isolate human-capital-specific working-age population by county over the 1970-2010 period, which is a key contribution in my identification strategy. To this end, I predict the total number of individuals with varying levels of educational attainment, corresponding to low-, medium-, and high-skill types, in US counties by utilizing the quasi-experimental variation in ancestry estimated in the previous step. Therefore, the

¹⁶The sample comprises 146 origin countries and 3,141 destination counties harmonized using the 1990-level country and county definitions and borders.

estimating equation becomes

$$N_{e,d,t} = \alpha_{s(d)} + \alpha_t + \sum_o \mu_{e,o} \cdot \hat{\mathcal{A}}_{o,d,t} + \nu_{e,d,t}, \quad (11)$$

where $N_{e,d,t}$ is the total number of individuals with educational attainment e residing in county d at time t . $\alpha_{s(d)}$ and α_t represent state and time fixed effects, respectively. $\hat{\mathcal{A}}_{o,d,t}$ is the predicted ancestry stock obtained in equation (10). I estimate equation (11) for individuals with each educational attainment separately. Finally, I obtain the main instruments for working-age population share with human capital level e in county d at time t , $n_{e,d,t}$, in equation (8) as follows:

$$\hat{N}_{e,d,t} = \sum_o \hat{\mu}_{e,o} \cdot \hat{\mathcal{A}}_{o,d,t}. \quad (12)$$

These instruments address endogeneity in the skill-specific local working-age population by exploiting exogenous variation from previously identified channels that shape local human capital. The channels are (i) the heterogeneity in immigrants' skills, comparative advantages, along with occupational specialization, (ii) the transmission of human capital from historical immigration to local population via persistence, and (iii) the formation of ethnic enclaves and social networks induced by immigration.¹⁷

A potential concern is that the instruments might capture the persistence of ethnic or cultural traits, or simply reflect geographic patterns of historical immigrant settlement, rather than differences in human capital. In this context, however, the interpretation is distinct: immigrant groups arrived at different times and settled in different locations with markedly heterogeneous levels of education, literacy, and occupational skills. For instance, [Borjas and Katz \(2007\)](#) document that Mexican immigrants historically lagged in educational attainment relative to other groups, contributing to persistent wage gaps. Similarly, 45% of Southern European and 33% of Eastern European arrivals aged 14 and older were illiterate in 1899–1910 ([Goldin, 1994](#)). Linked census records further show systematic, country-specific differences in initial occupation-based earnings, with Southern and Eastern Europeans starting lower and Germans and Nordics closer to natives, consistent with differential skill endowments rather than cultural assimilation ([Abramitzky et al., 2014](#)). These historical patterns suggest that predicted ancestry captures the long-run transmis-

¹⁷Appendix Figures [D.7](#), [D.8](#), and [D.9](#) illustrate the bivariate maps of endogenous human-capital-specific working-age population shares, and Appendix Figures [D.10](#), [D.11](#), and [D.12](#) display the bivariate maps of exogenous skill-specific working-age population across US counties in 2010.

sion of human capital embedded in early settler populations, as opposed to cultural persistence or spatial clustering.¹⁸

Identifying Assumption. The ancestry prediction follows the leave-out push-pull design developed in [Burchardi et al. \(2019\)](#). The push component uses immigration from origin country o to census divisions outside the division containing county d , while the pull component captures the share of immigrants from origins outside the continent containing o who settle in county d . These leave-out restrictions are designed to reduce contamination from origin-destination-specific shocks, endogenous immigrant settlement, and the persistence of local immigration patterns that complicates conventional Card-type instruments. The construction does not, however, eliminate the possibility that the underlying push and pull factors are correlated with persistent local conditions that independently affect subsequent industrial composition.

For notational simplicity, let $\varepsilon_{d,t}$ denote the structural error in a given industrial composition equation, suppressing its outcome skill index. Define the historical push-pull interaction as

$$\mathcal{Z}_{o,d,\tau} = I_{o,-r(d),\tau} \left(\frac{I_{-c(o),d,\tau}}{I_{-c(o),\tau}} \right). \quad (13)$$

A sufficient primitive condition for instrument validity is

$$\mathbb{E} [\mathcal{Z}_{o,d,\tau} \varepsilon_{d,t} \mid \alpha_{s(d)}, \alpha_t] = 0 \quad \forall o, \tau \in \mathcal{T}_t, \quad (14)$$

where $\mathcal{T}_t$ denotes the set of historical periods used to predict ancestry in census year t . Put differently, conditional on state and year fixed effects, immigration from a given origin to other census divisions, interacted with the county-specific leave-out pull factor, must be orthogonal to unobserved determinants of industrial skill composition in county d at time t . This condition rules out historical or persistent local factors that both affect the push-pull interactions and independently shape subsequent industrial composition.

Predicted ancestry is an intermediate input into the final instruments. In the second step, I map the predicted ancestry stocks into working-age populations by educational attainment. Accordingly, validity of the main specification requires

$$\mathbb{E} [\hat{N}_{e,d,t} \varepsilon_{d,t} \mid \alpha_{s(d)}, \alpha_t] = 0 \quad \forall e. \quad (15)$$

¹⁸While cultural persistence may matter in other settings ([Alesina and Tabellini, 2024](#); [Gagliarducci and Tabellini, 2025](#)), the channel here operates through inherited human capital differences. To address the concern that culture, rather than human capital, drives the results, I present corroborating evidence in Section 7.

This restriction requires the education-specific population instruments to affect county industrial composition only through the corresponding shifts in local human-capital composition.¹⁹

The following example illustrates the identifying restriction. Suppose that agriculture, a low-skill industry, experiences a positive productivity shock in Kern County at time t . The shock may increase the local employment share of low-skill industries and attract Mexican immigrants to the county. Since the push component excludes Mexican immigration to the census division containing Kern County, the local productivity shock does not directly enter that component. The instrument could, nevertheless, remain endogenous if the same shock affects the pull component by attracting immigrants from other continents to Kern County, or if persistent agricultural conditions jointly shape historical immigrant settlement and subsequent industrial composition. The identifying assumption requires these channels to be absent, conditional on state and year fixed effects. Under this restriction, the predicted ancestry stocks generate valid education-specific population instruments for estimating the effect of local human capital composition on industrial outcomes.

3.4 Instrument Validation Exercises and Instrument Performance

To assess whether the instruments are correlated with pre-existing local economic trends, I estimate a series of cross-sectional balance regressions. For each census year $t \in \{1970, 1980, 1990, 2000, 2010\}$, I estimate

$$\Delta y_{e,d,t'}^{\text{std}} = \alpha_{s(d)} + \hat{\mathbf{N}}_{d,t}^{\text{std}} \boldsymbol{\psi}_{e,t} + \epsilon_{e,d,t}, \quad (16)$$

where $\Delta y_{e,d,t'}^{\text{std}}$ is a pre-trend outcome for each industry-skill type e in county d at time t' (*e.g.*, a standardized measure of the change in employment share of each industry-skill type between 1940 and 1960 or a standardized measure of the change in establishment share of each industry-skill type between 1960 and 1965). The vector $\hat{\mathbf{N}}_{d,t}^{\text{std}}$ contains the standardized exogenous medium- and high-skill population measures for census year t , and $\alpha_{s(d)}$ denotes state fixed effects.

The purpose of equation (16) is to test whether the skill-specific exogenous population measures are systematically related to economic trends that predate the estimation period.²⁰ Each

¹⁹The component-level restriction in equation (14) is sufficient but stronger than necessary. Instrument validity ultimately requires orthogonality of the weighted combinations that form $\hat{N}_{e,d,t}$. In canonical shift-share settings, Goldsmith-Pinkham et al. (2020) emphasize identification based on the exogeneity of exposure shares, whereas Borusyak et al. (2022) develop identification based on the quasi-random assignment of shocks. Since the present design relies on leave-out push-pull interactions rather than a canonical shift-share construction, I state the relevant moment conditions directly.

²⁰Since the research design exploits persistence in the geographic distribution of ancestry and human capital, pre-period levels may be mechanically correlated with the instruments. Therefore, I emphasize changes in outcomes prior to the estimation period as a more informative diagnostic of differential pre-trends.

element of $\psi_{e,t}$ measures the conditional association between a one-standard-deviation increase in the corresponding instrument and the pre-period change in the outcome. Under the identifying assumption, the instruments should not systematically predict these earlier changes, implying $\psi_{e,t} = \mathbf{0}$. A consistent pattern of nonzero coefficients would instead raise concern that the instruments capture persistent local trends that also affect subsequent industrial composition.

Figures 1 and 2 display the resulting estimates. The coefficients are generally small, and none is statistically distinguishable from zero, providing no evidence that the predicted medium- or high-skill population measures are systematically associated with pre-period changes in employment or establishment composition, conditional on state fixed effects.²¹ Although balance tests cannot definitively establish the exclusion restriction, these findings reduce the concern that the ancestry-based instruments proxy for persistent pre-existing local economic trends, a concern emphasized in Jaeger et al. (2018).²²

Figure 3 displays binned scatter plots of residualized exogenous working-age population of each human capital level plotted against residualized endogenous working-age population of each human capital level. Conditional on state and time fixed effects, binned scatter plots demonstrate that these variables are well-aligned.²³

3.5 Relationship to Shift-Share Instruments and Rotemberg Weights

The Rotemberg-weight decomposition developed in Goldsmith-Pinkham et al. (2020) is designed for instruments with the canonical shift-share form, $\mathcal{G}_d = \sum_o r_{o,d} g_o$, where $r_{o,d}$ is a predetermined measure of local exposure to shock g_o . In that setting, the IV estimand can be represented as a weighted average of shock-specific just-identified estimands, and the Rotemberg weights reveal which shocks contribute most strongly to identification.

The instruments used here do not have this structure. Identification proceeds in two stages. First, county-level ancestry stocks are predicted using the leave-out push-pull specification in equation (9). The push variable captures immigration from origin country o to destinations outside the census division containing county d , while the pull variable captures the historical attractiveness of county d to immigrants from origins outside the continent containing o . Predicted ancestry, therefore, arises from interactions between origin-specific migration pressure and destination-specific

²¹ Appendix Figures D.1 and D.2 present additional balance tests.

²² Section 7 reports additional exercises that probe the plausibility of the exclusion restriction.

²³ Appendix Table C.1 further provides the first-stage regression estimates. Formally, I estimate the following first-stage equation: $n_{e,d,t} = \alpha_{s(d)} + \alpha_t + \hat{N}_{d,t} \kappa_e + \xi_{e,d,t}$.

attractiveness, both constructed using geographic leave-out elements. The right-hand side does not contain a predetermined local exposure share of the type used in a canonical Bartik instrument.

Second, the predicted ancestry stocks are mapped into skill-specific population through equation (11), and the final instruments are the fitted values defined in equation (12). The coefficients μ in this mapping are estimated regression parameters rather than pre-period exposure shares. Although the resulting fitted values can be written algebraically as linear combinations of the underlying predictors, the associated coefficients do not have the exposure-share interpretation required for the conventional Rotemberg decomposition. Thus, the final instruments are generated predictors obtained from a sequence of leave-out regressions, rather than shift-share instruments in the sense considered in Goldsmith-Pinkham et al. (2020).

For this reason, conventional Rotemberg weights would not provide an economically meaningful diagnostic in this setting. Instead, I assess the identifying variation through the leave-out construction, pre-period balance tests, first-stage diagnostics, alternative geographic units and instrument constructions, and robustness exercises that directly probe the exclusion restriction.

4 Data

4.1 Sources

Ancestry. I use county-level data on ancestry stocks for 1970–2010 from Fulford et al. (2019). I briefly summarize how they construct expected ancestry for each person and aggregate it into population estimates to map the geographic distribution of ancestry. They build ancestry distributions at both national and county levels using census micro-samples for each decade from 1850 to 2010. The expected ancestry for first-generation immigrants is based on their countries of birth. For their US-born children (*i.e.*, second-generation immigrants), ancestry depends on the parents’ birthplaces. Additionally, ancestry is traced through parents’ birth states or respondents’ own counties of residence, capturing internal migration and inter-generational transmission. For later generations, expected ancestry shares are constructed from past ancestry shares, reflecting how they evolve as internal migrants relocate and pass their ancestries to their children. In my sample, I harmonize all counties and county groups to 1990 boundaries for consistency across decennial census years.²⁴

This approach emphasizes genealogical reconstruction by accounting for internal migration,

²⁴The crosswalk files come from Burchardi et al. (2019) and Eckert et al. (2020). See Appendix B for additional details.

generational transmission, and differential group growth. In contrast, studies using self-reported ancestry in the census since 1980 capture the changing nature of ethnic identity as a social construct.

Immigration. I utilize data on immigration obtained from the individual decadal census files of the Integrated Public Use of Microdata Series (IPUMS USA and NHGIS: [Ruggles et al., 2022](#); [Manson et al., 2022](#)) samples from 1850-2000 as well as the 2006-2010 five-year sample of the American Community Survey (ACS). I weigh observations by the person weights in these sources. My dataset on immigration covers 3,141 US counties, 146 foreign countries, and 15 census waves. I provide additional details on the construction of immigration stock data in subsection B.1 of Data Appendix.

Working-Age Population, Employment, and Establishments. I acquire data on working-age population and employment at the county level using the individual census rounds from 1970-2000, and the 2006-2010 five-year waves. I define working-age population as individuals aged 16 to 64. As before, I weigh observations by the person weights. I obtain data on the number of establishments at the county level from the 1970-2010 extracts of the County Business Patterns (CBP) data published by the Census Bureau. I supplement these data with those compiled in [Eckert et al. \(2020, 2022\)](#). The Census Bureau defines an establishment as a single physical location where services and production activities are undertaken.²⁵ I provide additional details on the constructions of working-age population, employment, and establishment data in subsections B.2, B.3, and B.4 of Data Appendix, respectively.

4.2 Skills, Human Capital, and Tradability Classification

A central component of my analysis is to allocate industries into skill cells. For this purpose, I construct the industry-based continuous skill measure in the following steps. First, I create an education variable and assign values to it based on the educational attainment (years of education) of each individual. Second, I aggregate the 1990-level census industry codes to a balanced panel of industries utilizing an industry concordance generated in [Autor et al. \(2019\)](#). Finally, I calculate mean education for workers employed by each industry in the balanced panel in each census year. I define industries with average years of education corresponding to a high school degree and less as low skill, those with average education corresponding to some college degree as medium skill, and those with mean education comparable to college degree and more as high skill. Appendix Tables C.2, C.3, and C.4 present the industry-skill classifications for low-, medium-, and high-skill,

²⁵An establishment is not necessarily identical to a firm or enterprise, which may own one or more establishments.

respectively.

The classification of working-age individuals is identical to the procedure described above. Motivated by the task-based framework of [Autor et al. \(2003\)](#), I distinguish between low-, medium-, and high-skill workers based on educational attainment. High-skill workers are more likely to perform abstract, nonroutine tasks that complement capital, whereas medium-skill workers are disproportionately employed in routine occupations that are more susceptible to technological substitution ([Goos and Manning, 2007](#)). Combining medium- and high-skill workers would, therefore, obscure economically meaningful differences in their task content and exposure to technological change.

Since tradability shapes how local shocks transmit, with nontradables absorbing local demand and tradables facing global competition, I also group industries into tradable and nontradable sectors. I classify agriculture, mining, and manufacturing industries as tradable, whereas I categorize all service industries as nontradable similar to the classification scheme in [Burstein et al. \(2020\)](#). Appendix Tables [C.5](#) and [C.6](#) present tradable and nontradable sectors, while Appendix Tables [C.7](#), [C.8](#), and [C.9](#) show the skill classification across these sectors.

4.3 Summary Statistics

Table [1](#) reports summary statistics. Panel A shows substantial variation in the skill distribution of working-age population shares and in mean education levels across counties. Most working-age individuals hold a high school degree or less, a quarter have some college education, and the remainder possess a college degree or higher. Panel B presents employment and establishment shares, as well as the mean education of workers across industries, used in the baseline and tradability analyses.

The distribution of employment and establishment shares in medium-skill industries is similar; however, a relatively larger share of workers are employed in low-skill industries, while establishments are relatively more concentrated in high-skill industries. Employment and establishment shares in low-skill industries are comparable across tradable and nontradable sectors, but most workers and establishments are concentrated in medium- and high-skill nontradable industries. Although the mean education of medium-skill workers is similar across both sectors, workers in low- and high-skill nontradable industries tend to have slightly more schooling than those in corresponding tradable industries.

5 Results

In this section, I present my empirical findings by exploiting exogenous working-age population of each human capital level to explore the causal impacts of population shares on employment and establishment shares of industries across counties in the baseline and tradability analyses. Furthermore, I conduct a heterogeneity analysis across major industries along with establishment size domains and probe the robustness of my baseline estimates on a series of sensitivity checks. All regressions include state and time fixed effects to absorb time-invariant differences across states and common aggregate shocks, allowing identification to come from within-state, cross-county variation over time. Standard errors are clustered at the county level to control for serial correlation within counties across years in all specifications, which is the level at which the treatment varies.²⁶

5.1 Baseline Analysis: Estimates for Employment and Establishment Shares

Prior to the causal results, I report OLS estimates in Table 2 presenting the associations between population shares and employment shares in Panel A and establishment shares in Panel B based on estimating equation (8). The OLS estimates of medium- and high-skill population shares on employment and establishment shares of all industry-skill types are statistically significant. These results, however, reflect only correlations between the endogenous distribution of human capital and the industrial skill composition of counties rather than causal effects.

Table 3 reports IV estimates for the impacts of population shares on employment shares in Panel A and on establishment shares in Panel B. Endogenous medium- and high-skill population shares are instrumented with exogenous population of corresponding human capital levels as described in Section 3. All Anderson-Rubin (AR) Wald F -test p -values for the statistically significant point estimates are around 0%, suggesting that endogenous population shares are jointly significant, and that my instruments, exogenous population of given human capital levels, are valid. The table also presents Sanderson-Windmeijer (SW) first-stage F -statistics for each endogenous variable, which are sufficiently greater than the conventional threshold.²⁷

In particular, I find that, relative to low-skill population share, a one-percentage-point exogenous increase in medium- and high-skill population shares leads to a decrease in employment

²⁶Clustering at a higher level (*e.g.*, by state or census division) would be overly conservative given the limited number of states and the sub-state variation exploited in the analysis.

²⁷The Sanderson–Windmeijer F -statistics provide a measure of instrument relevance in models with multiple endogenous regressors, testing the joint significance of the excluded instruments for each endogenous variable conditional on the others (Sanderson and Windmeijer, 2016).

share of low-skill industries by 0.39 and 0.32 percentage points, respectively. By contrast, this increase results in a rise in employment share of high-skill industries by corresponding 0.18 and 0.54 percentage points. Additionally, a one-percentage-point rise in medium-skill population share boosts employment share of medium-skill industries by 0.21 percentage points, while a rise in high-skill population by the same magnitude reduces employment share of those industries by 0.23 percentage points. One can observe similar patterns across the estimates for establishment shares. Specifically, establishment share of low-skill industries drops by 0.75 and 0.11 percentage points as medium- and high-skill population shares grow by a one percentage point. Conversely, establishment share of high-skill industries increases by 0.96 and 0.33 percentage points in response to a one-percentage-point increase in medium- and high-skill population shares. A one-percentage-point increase in high-skill population share leads to a 0.18 percentage points decline in establishment share of medium-skill industries.

These findings are consistent with the qualitative reallocation mechanism derived in Section 2. Increases in the medium- and high-skill population shares raise the employment and establishment shares of high-skill-intensive industries, while reducing the corresponding shares of low-skill-intensive industries. In the two-sector benchmark, an increase in the relative supply of high-skill labor expands the white-collar sector, which uses high-skill labor relatively intensively, and contracts the blue-collar sector, which is relatively intensive in low-skill labor. The employment-share estimates, therefore, map directly into the sectoral reallocation predicted by the model. The establishment-share estimates reveal a parallel adjustment in the composition of local establishments, with activity shifting towards industries that use the increasingly abundant skill groups more intensively. The medium-skill results provide an empirical analogue to this mechanism, treating medium skill as an intermediate category rather than as a separate factor explicitly modeled in the two-skill benchmark. Taken together, the estimates support the predicted direction of reallocation across industrial skill types, although their relative magnitudes do not, by themselves, identify the elasticity of substitution between skill groups.

The employment share results are related to, but differ from, existing evidence linking local human capital to industrial specialization. [Simon \(2004\)](#) finds that employment growth in skill-intensive industries is particularly responsive to the local stock of human capital, while [Davis and Dingel \(2020\)](#) show that larger, skill-abundant cities specialize in activities that use skilled labor more intensively. These studies primarily characterize differential industry growth or cross-sectional patterns of spatial sorting and specialization. By contrast, I isolate exogenous changes in the skill

composition of the local population and estimate how they alter industry employment shares within local economies over time. The analysis, therefore, identifies a labor-supply-driven reallocation margin underlying the specialization patterns documented in the earlier literature. The establishment share findings point to a parallel adjustment in the local organization of production. [Moretti \(2004b\)](#) demonstrates that increases in the local supply of college-educated workers raise plant productivity, while [Beaudry et al. \(2010\)](#) show that skill-abundant cities adopt skill-complementary technologies more intensively. [Doms et al. \(2010\)](#) further document a positive relationship between local educational attainment and business activity. Consistent with these findings, my estimates show that increases in county human capital alter not only the allocation of workers but also the industrial composition of establishments. Relative to studies emphasizing productivity, technology adoption, business formation, or aggregate employment growth, my results reveal a distinct compositional margin, whereby local production units shift towards industries that use more highly skilled labor intensively.

5.2 Baseline Analysis: Economic Magnitudes

To facilitate comparisons across outcomes with different sample means, I convert the baseline estimates into elasticities implied by the linear specification. For outcome share y and population share x , the elasticity evaluated at the estimation-sample means is $\hat{\mathcal{E}}_{y,x} = \hat{\beta}^{IV} \left(\frac{\bar{x}}{\bar{y}} \right)$. This quantity gives the percentage change in the outcome share associated with a one-percent increase in the population share, evaluated at the sample means. Therefore, it is a local elasticity implied by the linear specification rather than a constant elasticity.

Appendix Table [C.10](#) reports the complete set of implied elasticities and standardized estimates. Both measures are mechanical transformations of the baseline coefficients that provide alternative representations of economic magnitude rather than new estimates or hypothesis tests. The elasticity calculations reveal different patterns for employment and establishment composition. For employment shares, the largest elasticity is the response of the high-skill industry share to the high-skill population share: a one-percent increase in the latter raises the former by approximately 0.36 percent. For establishment shares, the largest elasticities are associated with the medium-skill population share, which reduces the low-skill establishment share and increases the high-skill establishment share by approximately 0.80 and 0.77 percent, respectively. By contrast, the medium-skill outcome shares are comparatively less responsive. Thus, employment composition is most responsive to high-skill labor supply along the high-skill industry margin, whereas estab-

lishment composition responds more strongly to medium-skill labor supply through shifts between low- and high-skill industries.

The elasticity calculations are complemented by instrument-scaled standardized estimates, $\hat{S}_{y,x}^{SD} = \hat{\beta}_{y,x}^{IV} \left[\frac{SD(\hat{x}^\perp)}{SD(y^\perp)} \right]$, where $\hat{x}^\perp$ is the first-stage fitted component of the endogenous population share after removing state and year fixed effects, and $y^\perp$ is the corresponding residualized outcome. The standardized estimate expresses the outcome response to a one-standard-deviation increase in the instrument-predicted component of each endogenous population share. In the employment-share specifications, the response of high-skill industry employment to the high-skill population share is the largest standardized effect. The standardized establishment-share effects are smaller overall, although the strongest establishment response remains the increase in the high-skill establishment share associated with the medium-skill population share. These estimates facilitate comparisons across outcomes with different residual variation, while the elasticities provide the more direct measure of economic magnitude.

5.3 Tradability Analysis: Employment and Establishment Shares

The theory and empirical evidence in the existing literature suggest that local labor supply shocks generate heterogeneous adjustments across firms, occupations, and industries (Acemoglu, 2011; Diamond, 2016; Burstein et al., 2020). Motivated by this evidence, I distinguish between tradable and nontradable industries to capture heterogeneous adjustment margins. Nontradable industries are inherently local, with employment responding to shifts in local demand and supply, while tradable industries can reallocate across regions or absorb shocks through domestic and global markets. Failing to separate the two obscures these mechanisms.²⁸ In this subsection, I present my results pertinent to the effects of working-age population shares of all human capital levels, instrumented similarly as before, on the outcomes of interest in the tradable and nontradable sectors. I include state and time fixed effects and cluster standard errors by county in all specifications.

Table 4 reports the regression results of how population shares affect employment shares in the tradable and nontradable sectors in Panels A and B, respectively. The positive baseline effects originate in the nontradable sector, whereas the tradable sector absorbs the adverse impacts. Higher medium- and high-skill population shares lead to substantial declines in employment share of low-skill tradable industries, but sizable increases in employment share of nontradable industries of

²⁸This distinction is standard in the literature on local labor markets and trade (Autor and Dorn, 2013; Mian and Sufi, 2014).

the corresponding skill types. In addition, a larger high-skill population share reduces employment share of medium-skill tradable industries, while modestly raising high-skill tradable employment share. Finally, an increase in high-skill population share has a small negative effect on low-skill nontradable employment share.

Table 5 presents the effects of population shares on establishment shares in the tradable and nontradable sectors in Panel A and Panel B, respectively. Relative to the baseline, the negative effects are concentrated in the tradable sector, while the positive effects arise in the nontradable sector. Higher medium- and high-skill population shares substantially reduce establishment share of low-skill tradable industries, and moderately lower share of high-skill tradable industries. By contrast, increases in medium- and high-skill population shares produce large gains in establishment share of high-skill nontradable industries. An increase in the high-skill population share slightly reduces the shares of high-skill tradable and low-skill nontradable industries. The positive effect of a larger medium-skill population share on low-skill nontradable establishments and the negative effect of a larger high-skill population share on medium-skill nontradable establishments are similar in magnitude, while the adverse effect on low-skill nontradable establishments is modest.

The estimates reveal a clear contrast between tradable and nontradable industries. Increases in the local shares of medium- and high-skill workers lead to an expansion of nontradable industries, consistent with the fact that these industries serve local markets and can respond to increases in local income and demand (Moretti, 2010; Mian and Sufi, 2014). Tradable industries, on the other hand, contract in response to the same changes in labor supply. Since tradable producers compete in national and international markets, their activity is less closely tied to local demand and may be more sensitive to changes in local production costs and relative factor availability (Autor and Dorn, 2013). Overall, the results suggest that nontradable industries provide the principal margin through which local economies expand following an increase in human capital, whereas the contraction of low-skill-intensive activity is concentrated in the tradable sector. This pattern is consistent with the model’s reallocation mechanism, in which changes in relative skill supply shift activity across industries with different factor intensities. Tradability determines how that reallocation is expressed locally: nontradable industries can expand with local demand, while tradable industries face stronger external competition and have less scope to pass higher relative production costs into local prices.

5.4 Employment Heterogeneity by Nativity

I use plausibly exogenous variation in county ancestry composition to isolate ancestry-driven shifts in local human capital. The identification strategy builds on the intergenerational persistence of human capital and knowledge discussed in Section 3, allowing historical differences across ancestry groups to generate long-run variation in local human-capital endowments. The estimates capture the effects of the component of contemporary population human capital predicted by historical ancestry patterns.

Since the instruments are constructed at the county level, they do not separately generate native- and immigrant-specific human-capital shocks. Nevertheless, I examine whether the resulting employment reallocation is present among both groups. Appendix Table C.11 reports estimates for native workers in Panel A and immigrant workers in Panel B. The qualitative pattern appears in both groups despite the estimated responses being generally more pronounced among immigrants. This heterogeneity is consistent with immigrants adjusting more readily across industries due to greater mobility, stronger employment networks, or faster occupational transitions, although the analysis does not separately identify these channels. More broadly, the results indicate that ancestry-induced changes in local human capital affect the employment composition of both native and immigrant workers rather than operating exclusively through one nativity group.

5.5 Employment and Establishment Heterogeneity by Industry

The existing literature shows that historical shocks can produce persistent changes in sectoral composition, industrialization, and local labor markets (Peters, 2022; Droller, 2018; Rocha et al., 2017). Motivated by this evidence, I examine how exogenous shifts in local human capital reallocate employment and establishments across broad industry groups. Unlike the baseline analysis, which classifies industries by skill intensity, this subsection reports results for major industry categories.²⁹ All specifications include state and year fixed effects, and standard errors are clustered by county.

Appendix Figure D.3 plots the coefficients for employment in Panel A and establishment shares in Panel B. Increases in the medium- and high-skill population shares raise the service-sector share of both employment and establishments, while reducing the shares of mining and manufacturing. The employment results also show a decline in the agricultural share. These findings provide a sector-level counterpart to the baseline results, with economic activity shifting away from

²⁹Since establishment coverage in the County Business Patterns data is limited for agriculture over the study period, I exclude agriculture from the establishment share analysis.

agriculture and goods-producing industries and towards services.

The pattern is also consistent with the tradability results reported above. The expansion is concentrated in services, which are largely oriented towards local markets, while the contraction occurs primarily in sectors more exposed to national and international competition. This interpretation aligns with evidence that changes in local human capital and demand can reallocate activity towards services and other nontradable industries and away from manufacturing (Moretti, 2010; Autor and Dorn, 2013; Notowidigdo, 2020; Mian and Sufi, 2014). The results differ from those of Peters (2022), who find that a refugee-induced labor-supply shock in Germany increased manufacturing and reduced agriculture. In the present setting, however, the adjustment is primarily service-intensive, reinforcing the broader finding that increases in local human capital expand nontradable activity.

5.6 Establishment Heterogeneity by Size

In this subsection, I examine heterogeneity by establishment size, both for all industries and separately for the tradable and nontradable sectors. The purpose of this exercise is to identify which types of establishments account for the compositional response to ancestry-driven shifts in local human capital, and whether the reallocation across industry-skill types differs systematically by establishment size. Following the literature, I classify establishments with fewer than 20 employees as small, those with 20 to 500 employees as medium-sized, and those with more than 500 employees as large.³⁰ As in the baseline specification, I instrument the endogenous medium- and high-skill population shares with their corresponding exogenous population measures. All regressions include state and year fixed effects, and standard errors are clustered at the county level.

Establishments in All Industries. Appendix Table C.12 reports the results for small, medium-sized, and large establishments in Panels A, B, and C, respectively. The negative responses of establishment shares in low- and medium-skill industries are concentrated primarily among small establishments, followed by medium-sized establishments in low-skill industries. By contrast, the positive responses in medium- and high-skill industries are captured by medium-sized establishments, with stronger effects in medium-skill industries. The coefficients for large establishments are estimated less precisely and provide limited evidence of a systematic response at that margin.

Taken together, the estimates indicate that the adjustment in establishment composition is heterogeneous across both size classes and industry-skill types. Most of the statistically detectable

³⁰For a similar classification, see Olney (2013).

reallocation occurs among small- and medium-sized establishments, while large establishments exhibit comparatively little precisely estimated adjustment. This pattern is consistent with firm-dynamics models, in which adjustment costs, entry and exit, and responsiveness to local market conditions vary with establishment size (Hopenhayn, 1992; Haltiwanger et al., 2013). Although the present analysis does not separately identify entry, exit, and changes in the size of continuing establishments, the estimates suggest that small and medium-sized establishments account for the majority of compositional responses to shifts in local human capital.

Establishments in the Tradable and Nontradable Sectors. Appendix Tables C.13 and C.14 present the corresponding estimates for tradable and nontradable industries. Within each table, Panels A, B, and C present the results for small, medium-sized, and large establishments, respectively. In the tradable sector, the decline in establishment share of low-skill industries documented in the baseline findings is concentrated almost entirely among small establishments. The corresponding estimates for medium-sized and large establishments are generally less precise, providing little evidence of a comparable response among larger production units.

A similar pattern arises in the nontradable sector. The negative effects of the high-skill population share on establishments in low- and medium-skill industries are concentrated among small establishments. The positive responses in medium-skill industries, by contrast, are accounted for primarily by medium-sized establishments. These findings suggest that establishment size is an important dimension of the adjustment process in both sectors. Small establishments account for much of the contraction in less skill-intensive industries, whereas medium-sized establishments explain a larger share of the expansion in more skill-intensive activities. Therefore, the results are consistent with a reallocation process, in which smaller establishments constitute a particularly responsive margin, while the distinction between tradable and nontradable activity shapes where the adjustment occurs.

6 Mechanisms of Industrial Adjustment

6.1 Within- and Between-Industry Decomposition

The CES framework in Section 2 predicts that skill-supply shocks raise county-level mean education through between-industry reallocation: employment shifts towards sectors that already employ more educated workers. A distinct margin is within-industry upgrading, whereby incumbent industries raise the average schooling of their own workforces. My baseline estimates identify the between-

industry margin by construction because the dependent variables are industry employment shares by skill group and, therefore, abstract from changes in workforce composition within industries. To assess both channels jointly, I apply the Foster-Haltiwanger-Krizan (FHK; Foster et al., 2001) decomposition of the change in county-level mean years of schooling. Let j index industry, $E_{j,d,t}$ employment in industry j of county d at time t , $E_{d,t} = \sum_j E_{j,d,t}$ total county employment, $\eta_{j,d,t} \equiv E_{j,d,t}/E_{d,t}$ the industry employment share, and $e_{j,d,t}$ mean years of schooling in industry j at (d,t) . County-level mean schooling is $\bar{e}_{d,t} = \sum_j \eta_{j,d,t} e_{j,d,t}$, and the FHK decomposition of $\Delta \bar{e}_{d,t}$ becomes:

$$\Delta \bar{e}_{d,t} = \mathcal{W}_{d,t} + \mathcal{B}_{d,t} + \mathcal{C}_{d,t}, \quad (17)$$

$$\mathcal{W}_{d,t} = \sum_j \eta_{j,d,t-1} \Delta e_{j,d,t}, \quad (18)$$

$$\mathcal{B}_{d,t} = \sum_j \Delta \eta_{j,d,t} e_{j,d,t-1}, \quad (19)$$

$$\mathcal{C}_{d,t} = \sum_j \Delta \eta_{j,d,t} \Delta e_{j,d,t}. \quad (20)$$

$\mathcal{W}_{d,t}$ captures within-industry skill upgrading (*i.e.*, changes in worker education occurring within industries, holding industry employment shares fixed at their initial values), $\mathcal{B}_{d,t}$ captures between-industry reallocation (*i.e.*, reallocation towards industries that had initially highly educated workers), and $\mathcal{C}_{d,t}$ is the interaction (or covariance) term that captures industries that simultaneously expand and upgrade (or contract and downgrade).

As discussed in Section 2, the benchmark model isolates the reallocation of employment across sectors with different skill intensities. In the data, however, changes in county-level mean schooling may also arise through changes in workforce composition within industries. The FHK decomposition allows me to quantify both margins without imposing a structural mapping from the elasticity of substitution, σ , to their relative magnitudes.

For observations with $\Delta \bar{e}_{d,t} \neq 0$, I summarize the contribution of between-industry reallocation using the between-to-total ratio³¹

$$\lambda_{d,t} \equiv \frac{\mathcal{B}_{d,t}}{\mathcal{B}_{d,t} + \mathcal{W}_{d,t} + \mathcal{C}_{d,t}} = \frac{\mathcal{B}_{d,t}}{\Delta \bar{e}_{d,t}}. \quad (21)$$

³¹Since the decomposition components may have different signs, $\lambda_{d,t}$ need not lie between zero and one and can become large when the change in mean schooling is close to zero. I therefore winsorize this measure at the 1st and 99th percentiles and treat it as a supplementary summary statistic, with the component-level estimates providing the primary evidence.

To estimate the response of each adjustment margin, I instrument $\mathbf{n}_{d,t}$, the vector containing the medium- and high-skill population shares, with their exogenous counterparts as before and estimate

$$g_{d,t} = \alpha_{s(d)} + \alpha_t + \mathbf{n}_{d,t} \boldsymbol{\beta}_e^g + \vartheta_{d,t}^g \quad (22)$$

separately for each $g \in \{\Delta\bar{e}, \mathcal{W}, \mathcal{B}, \mathcal{C}, \lambda\}$, where $\alpha_{s(d)}$ and α_t denote state and year fixed effects. The benchmark model predicts that an increase in the high-skill population share raises the between-industry component, so that $\beta_h^{\mathcal{B}} > 0$. The coefficients on $\mathcal{W}$ and $\mathcal{C}$ measure additional adjustment margins not explicitly generated by the benchmark model. Therefore, I interpret the relative magnitudes of the component estimates as evidence on the empirical importance of the two channels, rather than as a structural estimate of σ .

Table 6 reports the IV estimates for each component of the decomposition. For the high-skill population share, the estimated coefficients are 0.42 for $\mathcal{B}_{d,t}$, 1.18 for $\mathcal{W}_{d,t}$, and -0.04 for $\mathcal{C}_{d,t}$. The ratio of the between- and within-industry coefficients is, therefore, $\hat{\beta}_h^{\mathcal{B}}/\hat{\beta}_h^{\mathcal{W}} = 0.36$. Normalizing the two principal adjustment components to sum to one, within-industry upgrading accounts for approximately 74 percent of the combined within- and between-industry response, while between-industry reallocation accounts for approximately 26 percent. When the cross term is included, the corresponding contributions to the total estimated change in mean schooling are about 76 percent, 27 percent, and -3 percent for the within, between, and cross components, respectively. Both the within- and between-industry coefficients are statistically significant. The positive between-industry response is consistent with the reallocation mechanism emphasized by the benchmark model, whereas the positive within-industry response captures an additional adjustment margin beyond the model’s primary channel.

The coefficients from the regression using $\lambda_{d,t}$ are negative and statistically significant for both the medium- and high-skill population shares. This result indicates that the ratio declines, on average, as the supply shares of medium- and high-skill workers increase. Since $\lambda_{d,t}$ is a ratio that may be negative, exceed one, or become large when $\Delta\bar{e}_{d,t}$ is close to zero, I treat these estimates as supplementary.³² In particular, the negative coefficients should not be interpreted as conventional percentage-point changes in the share of adjustment occurring between industries. The component-level estimates provide the more direct evidence on the relative importance of within- and between-industry responses.

³²Because raw $\lambda_{d,t}$ is highly sensitive to county-year observations for which $\Delta\bar{e}_{d,t}$ is close to zero, I winsorize the ratio at its 1st and 99th percentiles for estimation.

For the high-skill population share, the estimated cross-term coefficient is -0.04 , small relative to the within- and between-industry coefficients and statistically insignificant. This result indicates little systematic interaction between changes in industry employment shares and changes in industry-level mean schooling along the high-skill margin. The cross-term coefficient for the medium-skill population share is, however, -0.44 , statistically significant, and larger in absolute value than the corresponding between-industry coefficient. It accounts for nearly 21 percent of the total medium-skill response in absolute terms.

The sizable negative cross term does not reflect a failure of the FHK decomposition, but rather, it indicates that changes in industry employment shares systematically coincide with changes in within-industry mean schooling in the opposite direction. For instance, industries that expand following a medium-skill supply shock may experience smaller increases, or declines, in their own mean schooling, while contracting industries may experience within-industry upgrading. Since this interaction component is quantitatively important, the medium-skill within- and between-industry coefficients do not, by themselves, provide a complete characterization of the adjustment. Hence, I place greater emphasis on the high-skill results, for which the interaction component is small, while interpreting the medium-skill components more cautiously.

6.2 Initial Industry Composition and the Magnitude of Reallocation

The comparative statics in equations (4)–(7) predict that an increase in the high-skill labor-supply share raises the employment share of white-collar firms and lowers that of blue-collar firms. Under the maintained assumption that technologies and relative output prices are common across counties, every term on the right-hand side of equation (4) is a structural constant. The equilibrium relationship is, therefore, linear in labor-supply shares, and the reallocation slope is common across counties. Since the model is static, initial industry composition does not enter the equilibrium relationship, and the comparative-static prediction is invariant to counties’ initial conditions.

Nevertheless, I examine whether the observed response varies with initial industry composition. This exercise is motivated by the possibility that adjustment over a finite horizon is state dependent. In particular, counties with initially smaller white-collar industries may have greater scope for expansion following a high-skill supply shock, whereas counties with initially larger white-collar industries may have less remaining room for further reallocation. Under this auxiliary “room-to-grow” interpretation, the marginal effect of the high-skill labor-supply share may decline with the county’s initial white-collar employment share.

Thus, I treat initial industry composition as a source of heterogeneity in the observed response rather than as an additional comparative-static prediction of the benchmark model. A negative interaction between high-skill population share and white-collar industries would indicate that the magnitude of reallocation depends on the county's initial industrial structure, contrary to the uniform-slope benchmark. Such a pattern would be consistent with state-dependent adjustment or limits on short-run sectoral expansion, although it would not uniquely identify a particular adjustment mechanism.

To examine whether the reallocation response varies with initial industry composition, I estimate

$$s_{w,d,t} = \alpha_{s(d)} + \alpha_t + \mathbf{n}_{d,t}\boldsymbol{\beta} + \boldsymbol{\iota}_{d,t}\boldsymbol{\gamma} + \delta s_{w,d}^0 + v_{d,t}, \quad (23)$$

where $s_{w,d,t}$ denotes employment share of white-collar or high-skill-intensive industry w in county d at time t , $n_{h,d,t}$ is high-skill population share, $\mathbf{n}_{d,t} \equiv (n_{m,d,t}, n_{h,d,t})$ is the row vector containing medium- and high-skill population shares, $\boldsymbol{\iota}_{d,t} \equiv (n_{m,d,t} \times s_{w,d}^0, n_{h,d,t} \times s_{w,d}^0)$ is the corresponding row vector of interactions, and $s_{w,d}^0$ is the corresponding county industry employment share in 1970, which I treat as predetermined. $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ represent the associated column vectors of coefficients. I instrument medium- and high-skill population shares contained in $\mathbf{n}_{d,t}$ with their respective exogenous counterparts, while I instrument the two interaction terms contained in $\boldsymbol{\iota}_{d,t}$ using the corresponding exogenous population interacted with $s_{w,d}^0$. The specification contains four endogenous regressors and four excluded instruments and is, therefore, exactly identified.

Marginal effects are given by

$$\frac{\partial s_{w,d,t}}{\partial n_{h,d,t}} = \beta_h + \gamma_h s_{w,d}^0, \quad \frac{\partial s_{w,d,t}}{\partial n_{m,d,t}} = \beta_m + \gamma_m s_{w,d}^0. \quad (24)$$

Under the auxiliary state-dependent adjustment hypothesis, $\gamma_h < 0$: the marginal effect of the high-skill population share declines with the county's initial share of employment in high-skill-intensive industries. Thus, counties with initially smaller high-skill-intensive industries experience larger reallocation responses. By contrast, the uniform-slope benchmark implies $\gamma_h = 0$.³³

Table 7 presents the estimates. The interaction between the high-skill population share and the initial white-collar employment share is approximately -2.7 and is statistically significant.³⁴ The

³³The coefficient γ_m captures analogous heterogeneity in the response to the medium-skill population share, although it is not a direct prediction of the two-skill benchmark model.

³⁴The estimated marginal effect of the high-skill population share is $\frac{\partial s_{w,d,t}}{\partial n_{h,d,t}} = 0.844 - 2.687s_{w,d}^0$. Since both variables are measured as shares, a 10-percentage-point increase in $s_{w,d}^0$ reduces the effect of a 1-percentage-point

negative interaction is consistent with the auxiliary prediction $\gamma_h < 0$, indicating that counties with initially smaller high-skill-intensive industries experience larger reallocation responses. On the other hand, the interaction between the medium-skill population share and initial industry composition is positive but statistically insignificant, providing no evidence of systematic heterogeneity along the medium-skill margin.

The tercile estimates provide a less parametric assessment of the heterogeneity summarized by the interaction term. I divide counties into terciles of the initial white-collar employment share, $s_{w,d}^0$, and estimate the baseline specification in equation (8) separately within each group. Appendix Table C.15 reports these estimates. For the employment-share outcome, the estimated high-skill coefficient declines monotonically from approximately 0.62 in the bottom tercile to 0.58 in the middle tercile and 0.42 in the top tercile. This ordering is consistent with the auxiliary state-dependent adjustment hypothesis, in which counties with initially smaller high-skill-intensive industries exhibit larger reallocation responses.

As a placebo-style test of the composition gradient, I repeat the exercise using the natural log of aggregate county employment and present the estimates in Appendix Table C.16. An exogenous increase in high-skill labor supply raises aggregate employment, but the tercile estimates display no monotonic relationship with initial industry composition. The coefficients are 0.07, 0.15, and 0.10 in the bottom, middle, and top terciles, respectively. The absence of a corresponding gradient for aggregate employment suggests that the declining response in the employment-share regressions reflects heterogeneity in industrial reallocation rather than a general relationship between initial industry composition and the scale of local employment growth.

7 Robustness

I implement a battery of robustness checks and show that the results are robust to (i) alternative instrument constructions, (ii) alternative sample selections, (iii) an alternative classification of industry tradability, (iv) alternative skill classifications, (v) the inclusion of controls, (vi) additional exercises probing the exclusion restriction, (vii) spatial correlation in the regression errors, (viii) the use of commuting zones as an alternative geographic unit, and (ix) permutation tests.

Alternative Instruments/Exogenous Population of Different Human Capital Levels.

To assess the persistence and robustness of the main results, I examine whether ancestry-induced

increase in the high-skill population share by approximately 0.27 percentage points. The coefficient 0.844 is the estimated effect when $s_{w,d}^0 = 0$.

exogenous shifts in local human capital have had lasting effects on the industrial skill composition across US counties. This exercise sheds light on how historical shocks continue to shape local labor market structures. I predict ancestry stocks from 1970 to 2010 while restricting the push and pull factors to the 1860–1900 and 1860–1950 periods in separate specifications.³⁵ Using these predicted ancestry stocks, I isolate human-capital-specific exogenous local population following the baseline approach. Appendix Tables C.17 and C.18 present the corresponding regression results for employment and establishment shares in Panels A and B, respectively. The coefficient estimates across both periods closely resemble those in the baseline analysis, suggesting that the results are not sensitive to the historical window used to construct the instruments.

Overall, these findings point to two conclusions. First, historical immigrant settlement patterns have left lasting imprints on local labor markets through long-term human capital transmission. Second, the main results remain robust to alternative definitions of the immigration period. The consistency of the coefficients across these windows shows that the relationship between past immigrant settlement and today’s industrial structure reflects persistent, deep-rooted channels of human capital transmission rather than artifacts of a particular historical period.

Various Sample Selections. Another concern is that the baseline estimates could be driven by the largest ancestry groups or counties that historically received the most immigrants. To ensure robustness to their exclusion, I conduct two exercises. First, I omit the five largest ancestry groups when isolating the human-capital-specific exogenous population and re-estimate the baseline regressions. Second, I exclude the five largest immigrant-receiving counties when constructing the instruments and repeat the analysis.³⁶ Appendix Tables C.19 and C.20 present the corresponding regression results, showing that the estimates remain highly consistent with the baseline results.

Alternative Tradability Classification. Next, I employ an alternative measure of tradability by re-classifying industries into tradable and nontradable sectors. The goal is to assess the sensitivity of the baseline estimates to the choice of classification. I adopt the industry tradability measure proposed in Mian and Sufi (2014), which is based on geographical Herfindahl–Hirschman Indices (HHIs); industries that are more geographically concentrated are considered more tradable.³⁷ Appendix Tables C.21 and C.22 report the effects on employment and establishment shares using this alternative measure. On the whole, the estimated coefficients remain broadly consistent with the

³⁵These periods roughly correspond to the Age of Mass Migration.

³⁶The five largest ancestry groups are English, German, Irish, Italian, and Mexican, while the five largest destinations are Los Angeles, Cook, Harris, Miami-Dade, and Kings counties.

³⁷See Appendix Table I in the online appendix of their paper for the full classification scheme.

baseline results, indicating that the findings are not sensitive to the tradability classification.

Alternative Skill Classifications. The baseline classification ranks industries by the mean educational attainment of their workers, recalculated in each census year. Since educational attainment increased broadly across industries between 1970 and 2010 (Goldin and Katz, 2008), part of the measured upgrading in county employment composition could reflect changes in industry labels rather than changes in the underlying industrial mix. An industry may be reclassified as more skill-intensive simply because the education of its national workforce rises over time.

I address this concern using two alternative classifications. First, I fix each industry’s skill ranking at its 1990 mean level of educational attainment and apply that ranking to every census year. This prevents subsequent changes in national educational attainment from moving industries across skill categories. Second, I classify industries using the employment-weighted mean non-routine cognitive task content of their 1990-level occupational composition (Autor and Dorn, 2013). This alternative replaces an education-based measure of skill intensity with a task-based measure that does not mechanically depend on workers’ educational attainment.

Appendix Table C.23 reports employment share estimates under the two alternative industry-skill classifications. The fixed-1990 education classification produces similar patterns as the baseline specification, although the estimated responses of high-skill industries are somewhat more marked. Under the baseline time-varying classification, industries may move across skill categories as the educational attainment of their national workforce changes, even when the underlying industrial mix remains unchanged. Fixing industry assignments at their 1990 values removes this source of reclassification and isolates changes in employment across a stable set of industry-skill categories. The larger high-skill estimates under the fixed classification suggest that time-varying assignments attenuate part of the measured reallocation in the baseline specification. In this sense, the baseline estimates appear conservative, although this is an empirical feature of the results rather than a mechanical implication of the classification procedure.

The NRC task-based classification yields effects on low- and high-skill industry employment shares that closely resemble the baseline estimates. The coefficients for the medium-skill employment share are less precisely estimated and statistically insignificant, but this difference is not necessarily inconsistent with the main findings, since the education- and task-based classifications need not assign the same industries to the middle category. The education-based measure groups industries according to workers’ schooling, whereas the NRC measure ranks industries by cognitive task content, so the task-based middle group may contain workers drawn from a relatively

wide range of educational backgrounds. Overall, the central pattern is robust across both alternative classifications: exogenous increases in medium- and high-skill population shares shift county employment away from low-skill industries and towards high-skill industries.

Appendix Table C.24 examines whether the baseline results are sensitive to heterogeneity within the college-educated population. Rather than combining all college-educated workers into a single high-skill category, I consider two relative-supply margins separately. The first is the difference between the population shares of workers with a bachelor’s degree (BA) and those with some college (SC), $n_{BA} - n_{SC}$, which I instrument with the corresponding difference in their exogenous form, $\hat{n}_{BA} - \hat{n}_{SC}$. The second is defined analogously as the difference between the shares of workers with an advanced degree (ADV) and a bachelor’s degree, $n_{ADV} - n_{BA}$. The outcome industries are classified using the non-routine cognitive task index. For both margins, the estimates are directionally and quantitatively similar to the baseline effects of high-skill population share on employment shares of low- and high-skill industries. The findings, therefore, suggest that the baseline findings are not driven solely by treating all college-educated workers as a homogeneous skill group.

Baseline Model with Controls. A natural concern raised in Section 3 is that predicted ancestry composition may capture persistence in cultural factors rather than human capital. To address this, I augment the baseline model with a full set of pre-period controls, including (i) newspaper issues in 1800-1950 per 1950 county population (Library of Congress, 2025), (ii) average shares of Catholics and Protestants in 1936–1952 (Department of Commerce and Labor, Bureau of the Census, 1940; National Council of Churches, 1956), (iii) average shares of votes for Democratic candidates in 1920–1960 presidential elections (CQ Press, 2025), (iv) employment share in mining and manufacturing in 1950, and (v) urban population share in 1950 (both from the Census). If the cultural channels captured by these proxies were driving the baseline results, the coefficients would attenuate towards zero and lose significance. However, Appendix Table C.25 shows that the signs and magnitudes of the estimated coefficients remain close to the baseline estimates and are statistically significant, which mitigates the concern that the cultural channel dominates the human capital mechanism.

Additional Exclusion Restriction Tests. The pre-period cultural controls discussed above address the concern that the instruments capture persistent cultural influences rather than human capital. To probe the exclusion restriction further, I implement three additional exercises that isolate the identifying variation more narrowly.

The first exercise removes the variation in predicted ancestry composition at the macro-

region level. I residualize each baseline instrument on county-year predicted ancestry shares aggregated to seven world regions, together with state and year fixed effects.³⁸ The residualized instruments exclude the component of identifying variation associated with differences in broad regional ancestry composition. This exercise addresses the concern that the baseline estimates reflect cultural persistence, migration networks, or other ancestry-related influences common to macro-origin regions rather than differences in embedded human capital. Appendix Table C.26 reports the corresponding estimates for employment and establishment shares across all industry-skill types. The results are very similar to the baseline estimates in both sign and magnitude, providing little evidence that broad regional cultural or network variation accounts for the main findings and lending further support to the human-capital interpretation.

The second exercise assesses whether the identifying variation is systematically related to cultural outcomes. Appendix Table C.27 reports the results for four placebo outcomes. The most informative contemporaneous outcome is the county share of non-English speakers. The estimated coefficients are statistically insignificant, and the AR test fails to reject their joint null at conventional significance levels. Since a cultural-enclave or co-ethnic network mechanism could generate greater linguistic concentration in locations with a larger ancestry-driven population presence, the absence of a detectable response weakens this alternative interpretation. It does not rule out all cultural or network channels, but it provides no evidence that linguistic concentration is an important mechanism behind the baseline results. The table also reports estimates for pre-period measures of cultural and political composition, including the average county Catholic share between 1936 and 1952 and the Democratic presidential vote share between 1920 and 1960. The coefficients are statistically insignificant throughout, providing no evidence that the identifying variation is systematically linked to these dimensions of long-run local culture.

The self-employment share in ethnic-network-intensive industries, requires a more nuanced interpretation.³⁹ The coefficient on high-skill population share is individually significant, although the AR test fails to reject the joint null that medium- and high-skill population shares have no effect. The evidence is not a clean placebo null, but neither does it reveal a robust joint relationship between ancestry-driven human capital and self-employment in ethnic-network-intensive

³⁸The regions are Northwestern Europe, Southern Europe, Eastern Europe, East Asia, South Asia, Latin America, and Africa.

³⁹Ethnic-network-intensive industries are sectors, in which co-ethnic business and social networks may reduce information frictions and facilitate entry into self-employment, including food retail, eating and drinking establishments, personal services, and repair trades (Rauch, 2001). The outcome variable is the county-level self-employment share within these industries.

sectors. Appendix Table C.28 further separates the estimates by instrument group. The high-skill coefficient is statistically significant under the LED instruments but insignificant under the HED instruments, even though the point estimates are similar in magnitude. This difference appears to reflect precision rather than a systematic contrast across instrument groups. Moreover, the joint AR test fails to reject under either specification. Therefore, the results provide limited evidence that ethnic-network self-employment constitutes an important independent channel behind the baseline findings, although the individually significant high-skill coefficient warrants a more cautious interpretation than the other placebo outcomes.

The third exercise exploits differences in educational dispersion among culturally proximate ancestry groups. I partition origin countries into 15 clusters using k -means based on weighted genetic distance, which serves as a proxy for general cultural proximity (Spolaore and Wacziarg, 2009). I then classify clusters according to the dispersion of schooling across their constituent origin countries, measured using the within-cluster standard deviation of educational attainment (Barro and Lee, 2013). Clusters with above-median schooling dispersion form the high-education-dispersion (HED) group, while the remaining clusters form the low-education-dispersion (LED) group. This classification allows me to compare estimates generated by ancestry groups that are overall similar in cultural proximity but differ in the extent of educational heterogeneity across origins. HED clusters contain culturally proximate origin countries with relatively dispersed schooling levels, whereas LED clusters contain origin countries that are comparatively similar along both dimensions. For instance, the HED group includes pairings such as Germans and Irish during 1880–1920 and Eastern European Jews and Southern Italians during 1900–1920, whereas the LED group includes Scandinavian origin groups over the same period. Appendix Tables C.29 and C.30 report the employment and establishment share estimates obtained from the HED and LED instrument groups, respectively. The estimated responses of high-skill employment and establishment shares are similar across the two groups, and the differences are not statistically distinguishable. This similarity provides little evidence that the baseline results are driven by systematic differences in coarse cultural proximity across origin groups. Instead, the findings are consistent with the identifying variation operating through differences in human capital embedded within ancestry groups.

Collectively, these three exercises address the cultural channel concern from complementary directions and yield a consistent pattern. The orthogonalized instruments remove the component of identifying variation associated with macro-region ancestry composition. The placebo regressions provide little evidence that the instruments predict linguistic concentration, ethnic-network self-

employment, or pre-period measures of cultural and political composition. Finally, the HED and LED instrument groups, which compare ancestry groups with similar overall cultural proximity but different degrees of educational dispersion, produce generally similar estimates for the main outcomes. Thus, these results further reduce the concern that cultural proximity or ethnic-network formation drives the baseline findings and strengthen the interpretation that the relevant ancestry-based variation operates primarily through human capital.

Conley Spatial-HAC Standard Errors. The baseline standard errors are clustered at the county level, allowing for arbitrary serial correlation within counties but treating errors as independent across counties. This approach may understate uncertainty if residual correlation extends across county boundaries and declines gradually with geographic distance. Therefore, I recompute inference using the Conley spatial-HAC estimator (Conley, 1999).⁴⁰ I consider distance cutoffs of 50, 100, 200, 500, and 1,000 kilometers and allow for serial correlation in the time dimension.

Appendix Tables C.31 and C.32 report estimates for employment and establishment shares, respectively, split by industry-skill type in three panels. The county-level results are qualitatively unchanged under the spatial-HAC correction. The estimated effects retain their signs and remain statistically significant across all distance cutoffs, although the standard errors generally increase as the cutoff is widened. Thus, the results remain robust even when residual dependence is allowed to extend well beyond county boundaries. The Kleibergen-Paap rk Wald first-stage F-statistics decline at larger distance cutoffs but continue to indicate substantial instrument strength throughout.

Commuting Zone Replication and Conley Spatial-HAC Standard Errors. Workers and firms frequently cross-county boundaries, so county-level estimates may be affected by spillovers across neighboring counties within the same local labor market. Such interference may violate the stable unit treatment value assumption (SUTVA) and create a mismatch between the geographic unit of analysis and the relevant labor market. To address this concern, I re-estimate the baseline analysis at the commuting zone level. Commuting zones are designed to approximate self-contained local labor markets (Autor and Dorn, 2013), making them a natural alternative geographic unit for evaluating whether the county-level results are driven by geographic aggregation.

I reconstruct the instrument at the commuting zone level using predicted ancestry stocks. County-level distance measures are aggregated using ancestry population weights, so that the commuting-zone measure reflects the locations in which the relevant ancestral communities are concentrated. The method follows the same two-step procedure used in the county analysis. Ancestry

⁴⁰I implement it via Stata's `acreg` command (Colella et al., 2023).

try stocks are first regressed on the orthogonalized push-pull predictors and distance controls, with division-by-origin-country and commuting-zone-by-continent fixed effects. Skill-specific population is then regressed on the resulting predicted ancestry component, with state and year fixed effects.

Appendix Tables C.33 and C.34 demonstrate that the commuting zone estimates preserve the central qualitative pattern found at the county level. Increases in medium- and high-skill population shares reduce the employment and establishment shares of low-skill industries, while increasing the shares of high-skill industries. Several of the estimated effects are larger in magnitude when the analysis is conducted at the commuting zone level, particularly for the low- and high-skill outcomes. Some secondary coefficients become less precisely estimated as the geographic unit is widened and the first stage weakens. Overall, the main findings are not an artifact of using counties as the unit of analysis and remain evident when local labor markets are defined more broadly as commuting zones.

Permutation Tests. To rule out spurious correlation between the instruments and unobserved local trends, I implement permutation tests. I randomly re-assign the instruments within various sets of observations including (i) within the entire sample, (ii) within census year t , and (iii) within census year t and division $r(d)$. For each design, I perform 1,000 placebo runs and plot the distribution of the estimated coefficients. Appendix Figure D.4 displays density plots of the first-stage permutation estimates, and Appendix Figures D.5 and D.6 illustrate density plots of the reduced-form permutation estimates for employment and establishment shares. Across all versions, I find no statistically significant effects for either first-stage or reduced-form estimates. The placebo distributions are tightly centered at zero, and the baseline estimates lie far in their tails. The implied one-sided permutation p -values are below 0.01 across all specifications, supporting the credibility of the design.

8 Conclusion

There exist substantial differences in the skill composition of industries across the United States, yet the role of local population human capital in shaping these patterns remains less understood. This paper examines how exogenous shifts in local human capital reshape county-level employment and establishment shares across industries with different skill intensities.

Since local skill supply and industrial composition are jointly determined, I use a two-step ancestry-based strategy that predicts county ancestry stocks from historical origin-by-destination

immigration patterns over 1860-2010 and maps them into education-specific working-age populations over 1970-2010. A constant elasticity of substitution (CES) model guides the analysis and predicts reallocation towards sectors that use the increasingly abundant skill group more intensively.

The estimates reveal systematic reallocation. Relative to low-skill population share, a one-percentage-point increase in medium- and high-skill population shares raises employment share of high-skill industries by 0.18 and 0.54 percentage points, respectively, while reducing low-skill industry share by 0.39 and 0.32 percentage points. Establishment shares follow the same broad pattern. Positive responses arise mainly in nontradable industries, whereas contractions in less skill-intensive activity are borne primarily by tradable industries. Small establishments account for much of the decline in low-skill activity, while medium-sized establishments exhibit the strongest expansion in more skill-intensive industries. Between-industry reallocation accounts for approximately 26 percent of the combined within- and between-industry change in mean schooling, with within-industry upgrading accounting for the remaining 74 percent. The between-industry component accords with the CES benchmark, although the estimates do not identify the elasticity of substitution.

The external validity of the estimates is shaped by the slow-moving, ancestry-predicted shifts in county human capital observed from 1970 to 2010. The results are most informative about persistent demographic changes that allow workers and firms time to adjust, rather than short-run training programs or abrupt migration inflows. Similar findings at the commuting zone level indicate that the central pattern is not driven by county boundaries, although remote work and the growing tradability of business services may have weakened the link between worker location and skill-intensive production since 2010.

The policy implications are more specific than the claim that greater human capital automatically creates jobs or firms. Changes in workforce composition can reshape industries in which local employment and establishments are concentrated without generating aggregate growth. Education, training, and worker-attraction policies may support more skill-intensive activity when local employers can adjust, but may also impose costs on workers and businesses tied to contracting activities. Such policies should, therefore, be evaluated in terms of compositional change, adjustment capacity, and transition costs. Future research could combine these quantity responses with wage data and matched employer-employee records to study incidence, firm dynamics, worker transitions, productivity, inequality, and innovation.

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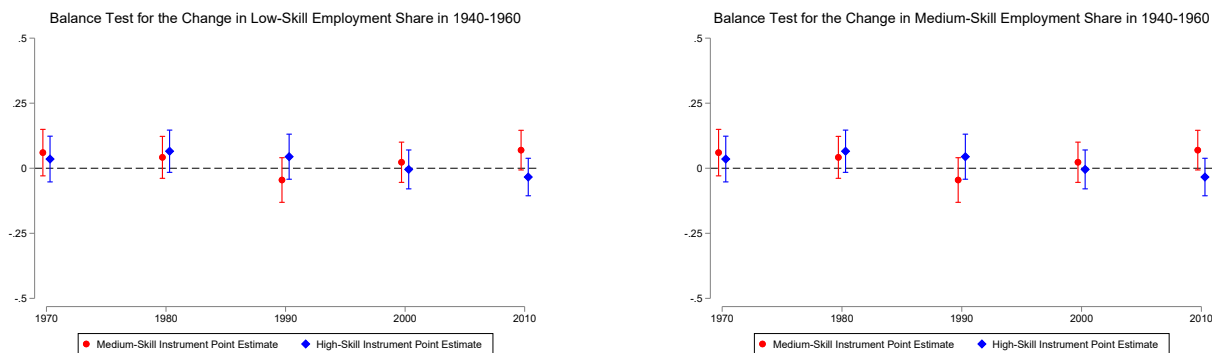


Figure 1: BALANCE TESTS: THE CHANGES IN EMPLOYMENT SHARES IN 1940-1960

Notes: These figures display the results of the balance tests specified in equation (16). Coefficients report the effect of a one-standard-deviation increase in each instrument, human-capital-specific exogenous population, for each year separately over the 1970-2010 period on a standardized measure of the changes in low- and medium-skill employment shares between 1940 and 1960. All regressions include state fixed effects, and standard errors are clustered by county for all specifications. Whiskers indicate 95% confidence intervals computed using standard errors clustered by county.

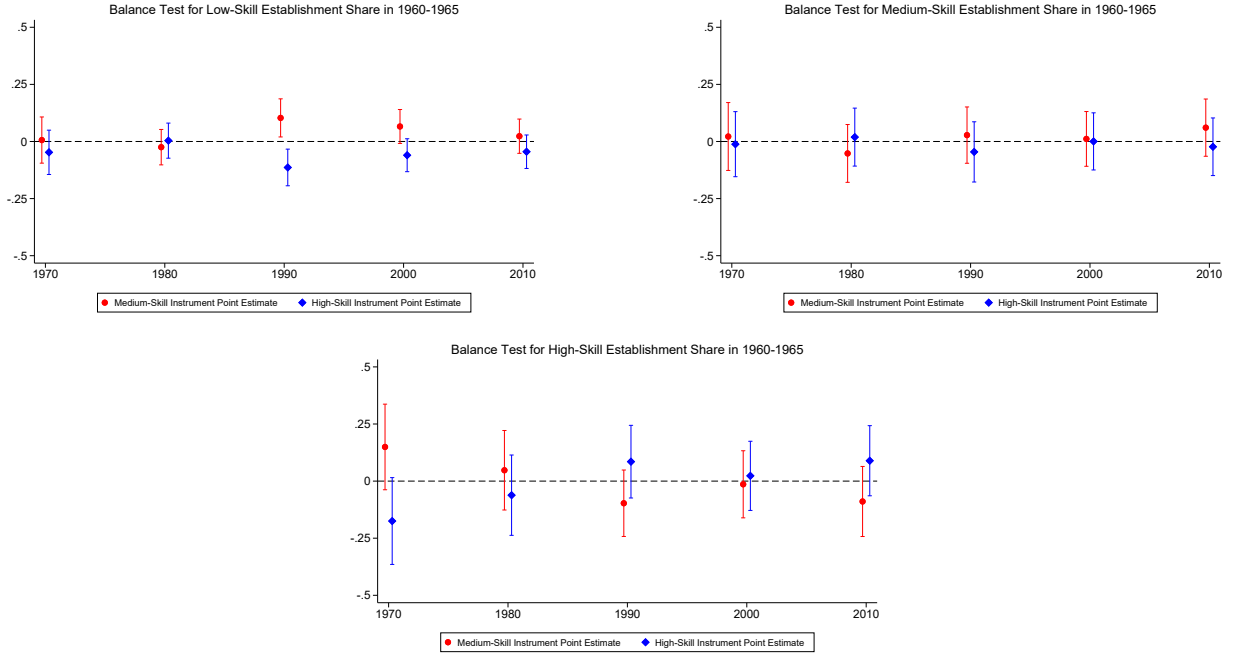


Figure 2: BALANCE TESTS: THE CHANGES IN ESTABLISHMENT SHARES IN 1960-1965

Notes: These figures display the results of the balance tests specified in equation (16). Coefficients report the effect of a one-standard-deviation increase in each instrument, human-capital-specific exogenous population, for each year separately over the 1970-2010 period on a standardized measure of the changes in low-, medium-, and high-skill establishment shares between 1960 and 1965. All regressions include state fixed effects, and standard errors are clustered by county for all specifications. Whiskers indicate 95% confidence intervals computed using standard errors clustered by county.

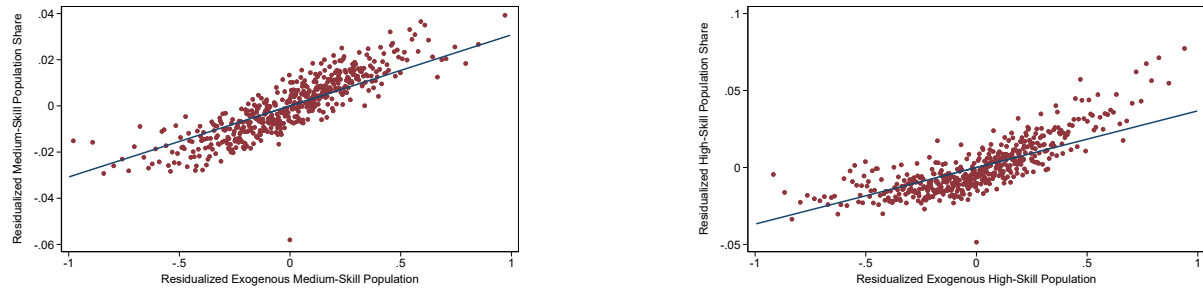


Figure 3: FIRST-STAGE RELATIONSHIP

Notes: These figures display binned scatter plots of the first-stage estimates, whereby residualized exogenous working-age population of each human capital level constructed in equation (12) is plotted against residualized endogenous working-age population of each human capital level. Residuals of each variable have been obtained conditional on state and time fixed effects. The number of bins has been fixed at 500. The figures have been plotted using the “binsreg” package in Stata (Cattaneo et al., 2024).

Table 1: SUMMARY STATISTICS

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Population			
<i>Working – Age Population Share_{e,d,t} (%)</i>	59.57 (13.29)	25.61 (8.00)	14.73 (6.68)
<i>Mean Education_{e,d,t}</i>	10.80 (0.63)	13.55 (0.11)	16.77 (0.19)
Panel B: Industry			
Baseline			
<i>Employment Share_{e,d,t} (%)</i>	31.79 (25.53)	45.99 (17.13)	22.22 (11.75)
<i>Establishment Share_{e,d,t} (%)</i>	23.98 (20.16)	44.09 (15.15)	31.82 (16.18)
<i>Mean Education_{e,d,t}</i>	11.41 (0.46)	12.76 (0.14)	14.82 (0.14)
Tradable Sector			
<i>Employment Share_{e,d,t} (%)</i>	17.69 (13.75)	12.36 (7.95)	0.70 (1.34)
<i>Establishment Share_{e,d,t} (%)</i>	11.10 (19.97)	9.56 (13.54)	2.44 (5.36)
<i>Mean Education_{e,d,t}</i>	11.27 (0.54)	12.62 (0.22)	14.44 (0.21)
Nontradable Sector			
<i>Employment Share_{e,d,t} (%)</i>	15.15 (13.20)	31.61 (10.67)	22.49 (10.59)
<i>Establishment Share_{e,d,t} (%)</i>	14.68 (11.55)	31.59 (14.06)	30.62 (16.84)
<i>Mean Education_{e,d,t}</i>	11.56 (0.39)	12.82 (0.12)	14.86 (0.16)
<i>N</i>	15,705	15,705	15,705

Notes: This table presents the mean, standard deviation (in parentheses), and the number of observations for both independent and dependent variables of all skill types (e) at the county (d) level for the decennial 1970-2010 period (t). Panel A displays the human-capital-specific variables. The working-age population share variable of each skill type at the county level in a certain decennial census year equals the number of working-age individuals with a given mean education level by county relative to total county population in that year. The employment share variable of each skill type at the county level in a given census year represents the number of workers of each skill type by county relative to the entire workforce in that county and census year. The establishment share variable of each skill type at the county level in a certain year is normalized by the total number of establishments by county in that year. Panel B reports the industry-skill-specific variables utilized in the baseline and tradability analyses. The determinant of the continuous skill measure corresponds to mean education of workers employed by each industry in each year. The measure of tradability is based on [Burstein et al. \(2020\)](#). The tradable sector includes the agriculture, mining, and manufacturing industries, whereas the nontradable sector covers all services industries. The employment share variable of each skill type in each sector and census year is normalized by the entire workforce in each sector and year. The establishment share of each skill type in each sector and year is normalized by the total number of establishments in each sector and year.

Table 2: OLS ESTIMATES OF EMPLOYMENT AND ESTABLISHMENT SHARES

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> _{MS,d,t}	-0.610*** (0.016)	0.548*** (0.022)	0.062*** (0.015)
<i>Population Share</i> _{HS,d,t}	-0.362*** (0.009)	-0.227*** (0.014)	0.588*** (0.011)
R^2	0.973	0.913	0.943
Panel B: Establishment Shares			
<i>Population Share</i> _{MS,d,t}	-0.145*** (0.039)	0.086** (0.036)	0.059** (0.030)
<i>Population Share</i> _{HS,d,t}	-0.270*** (0.019)	-0.300*** (0.019)	0.565*** (0.021)
R^2	0.310	0.064	0.454
N	15,705	15,705	15,705
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table reports the OLS results of estimating equation (8). The dependent variables are employment and establishment shares of each industrial skill type at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are medium- and high-skill working-age population shares at the county level. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 3: IV ESTIMATES OF EMPLOYMENT AND ESTABLISHMENT SHARES

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.389*** (0.047)	0.212*** (0.059)	0.177*** (0.036)
<i>Population Share</i> $_{HS,d,t}$	-0.315*** (0.022)	-0.229*** (0.027)	0.544*** (0.020)
AR Wald F-Test P-value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.745*** (0.135)	-0.056 (0.133)	0.955*** (0.114)
<i>Population Share</i> $_{HS,d,t}$	-0.110* (0.065)	-0.175*** (0.061)	0.331*** (0.055)
AR Wald F-Test P-value	0.000	0.006	0.000
SW First-Stage F-Stats	133; 364	133; 364	133; 364
<i>N</i>	15,705	15,705	15,705
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table reports the IV results of estimating equation (8). The dependent variables are employment and establishment shares of each industrial skill type at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4: EMPLOYMENT ESTIMATES IN THE TRADABLE AND NONTRADABLE SECTORS

	Employment Shares			
	All	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)
Panel A: Tradable Sector				
<i>Population Share</i> _{MS,d,t}	-0.444*** (0.076)	-0.473*** (0.062)	0.020 (0.035)	0.011 (0.012)
<i>Population Share</i> _{HS,d,t}	-0.601*** (0.035)	-0.290*** (0.029)	-0.343*** (0.020)	0.035*** (0.006)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000
<i>N</i>	15,705	15,705	15,705	12,564
Panel B: Nontradable Sector				
<i>Population Share</i> _{MS,d,t}	0.444*** (0.076)	0.015 (0.030)	0.246*** (0.051)	0.183*** (0.038)
<i>Population Share</i> _{HS,d,t}	0.601*** (0.035)	-0.060*** (0.014)	0.100*** (0.025)	0.561*** (0.021)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000
<i>N</i>	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Notes: This table reports the IV results of estimating equation (8). The dependent variables are employment shares of each industrial skill type in the tradable and nontradable sectors at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5: ESTABLISHMENT ESTIMATES IN THE TRADABLE AND NONTRADABLE SECTORS

	Establishment Shares			
	All	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)
Panel A: Tradable Sector				
<i>Population Share_{MS,d,t}</i>	-1.030*** (0.172)	-0.927*** (0.140)	-0.061 (0.094)	-0.042 (0.045)
<i>Population Share_{HS,d,t}</i>	-0.181** (0.074)	-0.109* (0.064)	0.005 (0.043)	-0.077*** (0.021)
AR Wald F-Test P-value	0.000	0.000	0.716	0.000
Panel B: Nontradable Sector				
<i>Population Share_{MS,d,t}</i>	1.030*** (0.172)	0.204*** (0.066)	-0.133 (0.103)	0.960*** (0.115)
<i>Population Share_{HS,d,t}</i>	0.181** (0.074)	-0.098*** (0.030)	-0.205*** (0.047)	0.484*** (0.054)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000
<i>N</i>	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Notes: This table reports the IV results of estimating equation (8). The dependent variables are employment shares of each industrial skill type in the tradable and nontradable sectors at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: DECOMPOSITION OF THE COUNTY EMPLOYMENT SKILL INDEX

	$\Delta \bar{e}$	Within ($\mathcal{W}$)	Between ($\mathcal{B}$)	Cross ($\mathcal{C}$)	Between Share (λ)
	(1)	(2)	(3)	(4)	(5)
<i>Population Share</i> $_{MS,d,t}$	2.062*** (0.646)	2.293*** (0.235)	0.214 (0.451)	-0.444*** (0.068)	-13.097*** (2.612)
<i>Population Share</i> $_{HS,d,t}$	1.555*** (0.213)	1.175*** (0.092)	0.416*** (0.152)	-0.036 (0.027)	-2.691*** (0.965)
AR Wald F-Test P-value	0.000	0.000	0.002	0.000	0.000
SW First-Stage F-Stats	116; 353	116; 353	116; 353	116; 353	116; 353
N	12,558	12,558	12,558	12,558	12,558
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV results of estimating equation (22). The dependent variables are the change in the county-level employment skill index, $\Delta \bar{e}_{d,t}$, and its within-industry ($\mathcal{W}_{d,t}$), between-industry ($\mathcal{B}_{d,t}$), and cross-term ($\mathcal{C}_{d,t}$) components, along with the between-share $\lambda_{d,t} = \mathcal{B}_{d,t} / \bar{e}_{d,t}$ obtained from the FHK decomposition. Column (5) uses $\lambda_{d,t}$ winsorized at the 1st and 99th percentiles to limit the influence of extreme values arising in county-years for which $\Delta \bar{e}_{d,t}$ is close to zero. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the p -value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7: INITIAL INDUSTRY COMPOSITION AND THE MAGNITUDE OF REALLOCATION

	Employment Share of High-Skill Industries
<i>Population Share</i> $_{MS,d,t}$	0.061 (0.077)
<i>Population Share</i> $_{HS,d,t}$	0.844*** (0.089)
<i>Population Share</i> $_{MS,d,t} \times s_{w,d}^0$	1.049 (0.741)
<i>Population Share</i> $_{HS,d,t} \times s_{w,d}^0$	−2.687*** (0.818)
AR Wald F-Test P-value	0.000
<i>N</i>	15,705
State FE	Yes
Year FE	Yes

Notes: This table reports IV results of estimating equation (23). The dependent variable is employment share of high-skill industries, while the independent variables are medium- and high-skill population shares and their interactions with initial industry composition in 1970, $s_{w,d}^0$, which is treated as a predetermined initial condition. These four endogenous variables are instrumented with medium- and high-skill exogenous working-age population and their interactions with $s_{w,d}^0$, yielding an exactly identified system. The specification presents the p -value for the Anderson-Rubin Wald F-test. All regressions include state and year fixed effects. Standard errors are clustered by county, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendices

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A Model Derivations

This appendix derives the equilibrium labor allocations and uses them to obtain the comparative statics presented in Section 2.

A.1 Firm Optimization and Relative Skill Wages

Each representative firm type $i \in \{b, w\}$ produces a sector-specific output, Q_i , which it sells at price p_i in a perfectly competitive market. Production combines high-skill labor ($L_{i,h}$) and low-skill labor ($L_{i,l}$) according to a constant elasticity of substitution (CES) technology

$$Q_i = \left(\beta_{i,h} L_{i,h}^{\frac{\sigma-1}{\sigma}} + \beta_{i,l} L_{i,l}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (\text{A.1})$$

where $1 < \sigma < \infty$ denotes the elasticity of substitution between high- and low-skill labor inputs, and $\beta_{i,h}$ and $\beta_{i,l}$ represent relative productivity parameters for each labor type within firm i . The productivity parameters differ across firm types such that $\beta_{w,h} > \beta_{b,h}$ and $\beta_{b,l} > \beta_{w,l}$, with $\beta_{i,h} + \beta_{i,l} = 1$. Within production, white-collar firms employ high-skill labor more intensively ($\beta_{w,h} > \beta_{w,l}$), whereas blue-collar firms rely more heavily on low-skill labor ($\beta_{b,l} > \beta_{b,h}$).

Each representative firm type chooses its high- and low-skill labor inputs to maximize profit

$$\max_{L_{i,h}, L_{i,l}} \{p_i Q_i - \omega_h L_{i,h} - \omega_l L_{i,l}\}. \quad (\text{A.2})$$

Since each firm is a price taker in the labor market, it takes the common skill wages ω_h and ω_l as given. The equilibrium skill wages are determined jointly by the firms' first-order conditions and the economy-wide labor-market-clearing conditions

$$L_{b,h} + L_{w,h} = N_h, \quad (\text{A.3})$$

$$L_{b,l} + L_{w,l} = N_l, \quad (\text{A.4})$$

where N_h and N_l represent the total labor supplies of high- and low-skill workers, respectively. The aggregate labor supply conditions are imposed at the market level and, therefore, do not enter either firm's maximization problem. In addition, no firm-specific multiplier appears in either firm's profit maximization problem. If the aggregate labor resource constraints were instead embedded in a planner or equivalent joint allocation problem, their shadow values would correspond to the

equilibrium wages.

The first-order conditions (FOCs) equate the value of the marginal product of each skill type to the corresponding market wage:

$$p_i \frac{\partial Q_i}{\partial L_{i,h}} = \omega_h, \quad (\text{A.5})$$

$$p_i \frac{\partial Q_i}{\partial L_{i,l}} = \omega_l. \quad (\text{A.6})$$

Taking their ratio eliminates the firm-specific output price and gives

$$\frac{L_{i,l}}{L_{i,h}} = \left(\frac{\beta_{i,l}}{\beta_{i,h}} \right)^\sigma \left(\frac{\omega_h}{\omega_l} \right)^\sigma. \quad (\text{A.7})$$

Perfect competition and constant returns to scale imply zero profits, so the unit cost of each firm type (c_b, c_w) equals its output price:

$$c_b(\omega_h, \omega_l) = p_b, \quad (\text{A.8})$$

$$c_w(\omega_h, \omega_l) = p_w, \quad (\text{A.9})$$

Under the normalization $(p_b = p_w = 1)$, these conditions imply equality of unit costs across firm types:

$$c_b(\omega_h, \omega_l) = c_w(\omega_h, \omega_l). \quad (\text{A.10})$$

Define

$$B_b = \left(\frac{\beta_{b,l}}{\beta_{b,h}} \right)^\sigma, \quad B_w = \left(\frac{\beta_{w,l}}{\beta_{w,h}} \right)^\sigma, \quad Z = \left(\frac{\beta_{b,l}^\sigma - \beta_{w,l}^\sigma}{\beta_{w,h}^\sigma - \beta_{b,h}^\sigma} \right)^{\frac{\sigma}{\sigma-1}}.$$

The firms' input-ratio conditions in equation (A.7) may then be written as

$$L_{b,l} = \frac{B_b}{Z} L_{b,h}, \quad L_{w,l} = \frac{B_w}{Z} L_{w,h}. \quad (\text{A.11})$$

A.2 Equilibrium Labor Allocations

Combining the input ratios in equation (A.11) with the labor-market-clearing conditions in equations (A.3) and (A.4) yields

$$L_{w,h} = \frac{B_b N_h - Z N_l}{B_b - B_w}, \quad L_{w,l} = \frac{B_w (B_b N_h - Z N_l)}{Z (B_b - B_w)}, \quad (\text{A.12})$$

$$L_{b,h} = \frac{ZN_l - B_w N_h}{B_b - B_w}, \quad L_{b,l} = \frac{B_b (ZN_l - B_w N_h)}{Z(B_b - B_w)}. \quad (\text{A.13})$$

The constant Z pins down the equilibrium relative wage (ω_h/ω_l) and collects the productivity terms arising from the unit-cost-equality condition in equation (A.10).

Let total labor supply be $N = N_h + N_l$ and sectoral employment shares be $n_h = N_h/(N_h + N_l) = N_h/N$ and $n_l = N_l/(N_h + N_l) = N_l/N$ with $n_h + n_l = 1$. Since the allocations are homogeneous of degree one in (N_h, N_l) , the corresponding shares are homogeneous of degree zero. This indicates that sectoral employment shares depend only on the composition of labor supply. Total labor supply can, therefore, be normalized to one ($N = 1$). Under this normalization, $N_h = n_h$ and $N_l = n_l = 1 - n_h$. The equilibrium labor allocations become

$$L_{w,h} = \frac{B_b n_h - Z n_l}{B_b - B_w}, \quad L_{w,l} = \frac{B_w (B_b n_h - Z n_l)}{Z(B_b - B_w)}, \quad (\text{A.14})$$

$$L_{b,h} = \frac{Z n_l - B_w n_h}{B_b - B_w}, \quad L_{b,l} = \frac{B_b (Z n_l - B_w n_h)}{Z(B_b - B_w)}. \quad (\text{A.15})$$

A.3 Sectoral Employment Shares

The sectoral labor-demand shares are defined as

$$s_w = \frac{L_{w,h} + L_{w,l}}{\sum_{i \in \{b,w\}} (L_{i,h} + L_{i,l})}, \quad s_b = \frac{L_{b,h} + L_{b,l}}{\sum_{i \in \{b,w\}} (L_{i,h} + L_{i,l})},$$

where s_w is the share of total employment allocated to the white-collar sector, while s_b is the corresponding share allocated to the blue-collar sector, and $s_w + s_b = 1$.

Labor-market clearing condition implies that total labor demand equals total labor supply:

$$\sum_{i \in \{b,w\}} (L_{i,h} + L_{i,l}) = N_h + N_l. \quad (\text{A.16})$$

Since total employment is normalized to one, the sectoral employment shares simplify to

$$s_w = L_{w,h} + L_{w,l}, \quad s_b = L_{b,h} + L_{b,l}.$$

Substituting the equilibrium labor allocations in equations (A.14) and (A.15) yields

$$s_w = \frac{(Z + B_w)(B_b n_h - Z n_l)}{Z(B_b - B_w)}, \quad s_b = \frac{(Z + B_b)(Z n_l - B_w n_h)}{Z(B_b - B_w)}. \quad (\text{A.17})$$

Since $n_h + n_l = 1$, there exists only one independent labor supply share. Once $n_l = 1 - n_h$ is imposed, the relevant comparative statics are, therefore, total derivatives rather than partial derivatives.

Differentiating the white-collar employment share with respect to n_h gives

$$\frac{ds_w}{dn_h} = \frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} > 0. \quad (\text{A.18})$$

The corresponding effect on the blue-collar employment share is

$$\frac{ds_b}{dn_h} = -\frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} < 0. \quad (\text{A.19})$$

The differences in the relative productivity parameters and the imperfect substitutability between labor types ensure that $B_b > 1 > B_w$, $B_b - B_w > 0$, and $Z > 0$. Thus, an increase in the high-skill labor supply share reallocates total employment towards the white-collar sector and away from the blue-collar sector.

Equivalently, because $dn_l = -dn_h$, an increase in n_l represents movement in the opposite direction along the same labor supply share margin:

$$\frac{ds_b}{dn_l} = \frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} > 0, \quad (\text{A.20})$$

and

$$\frac{ds_w}{dn_l} = -\frac{(B_b + Z)(B_w + Z)}{Z(B_b - B_w)} < 0. \quad (\text{A.21})$$

Therefore, a larger low-skill labor supply share reallocates employment towards the blue-collar sector and away from the white-collar sector.

A.4 Skill-Specific Sectoral Allocation Shares

The sectoral employment shares s_w and s_b describe how the entire workforce is divided between the two firm types. A related set of comparative statics shows how each skill group is allocated across the two sectors.

For high-skill workers, define

$$\delta_{w,h} = \frac{L_{w,h}}{N_h}, \quad \delta_{b,h} = \frac{L_{b,h}}{N_h}.$$

Here, $\delta_{w,h}$ is the fraction of all high-skill workers employed in the white-collar sector, whereas $\delta_{b,h}$ is the fraction employed in the blue-collar sector. By construction, $\delta_{w,h} + \delta_{b,h} = 1$.

Similarly, define the sectoral allocation shares of low-skill workers as

$$\delta_{w,l} = \frac{L_{w,l}}{N_l}, \quad \delta_{b,l} = \frac{L_{b,l}}{N_l}.$$

These shares also satisfy $\delta_{w,l} + \delta_{b,l} = 1$.

Holding total labor supply fixed, the high-skill allocation shares become

$$\delta_{w,h} = \frac{B_b n_h - Z n_l}{n_h (B_b - B_w)} \quad (\text{A.22})$$

and

$$\delta_{b,h} = \frac{Z n_l - B_w n_h}{n_h (B_b - B_w)}, \quad (\text{A.23})$$

The low-skill allocation shares become

$$\delta_{w,l} = \frac{B_w (B_b n_h - Z n_l)}{Z n_l (B_b - B_w)} \quad (\text{A.24})$$

and

$$\delta_{b,l} = \frac{B_b (Z n_l - B_w n_h)}{Z n_l (B_b - B_w)}. \quad (\text{A.25})$$

For high-skill workers, imposing $n_l = 1 - n_h$ and differentiating with respect to n_h gives

$$\frac{d\delta_{w,h}}{dn_h} = \frac{Z}{n_h^2 (B_b - B_w)} > 0 \quad (\text{A.26})$$

and

$$\frac{d\delta_{b,h}}{dn_h} = -\frac{Z}{n_h^2 (B_b - B_w)} < 0. \quad (\text{A.27})$$

Thus, as the high-skill supply share rises, a larger fraction of high-skill workers is employed in the white-collar sector, and a smaller fraction is employed in the blue-collar sector.

For low-skill workers,

$$\frac{d\delta_{w,l}}{dn_h} = \frac{B_b B_w}{Z n_l^2 (B_b - B_w)} > 0 \quad (\text{A.28})$$

and

$$\frac{d\delta_{b,l}}{dn_h} = -\frac{B_b B_w}{Z n_l^2 (B_b - B_w)} < 0. \quad (\text{A.29})$$

Therefore, a rise in the high-skill supply share increases the fraction of low-skill workers employed in the white-collar sector and reduces the fraction employed in the blue-collar sector. The intuition is that a greater high-skill supply share expands the white-collar sector as a whole, causing it to absorb a larger proportion of both skill groups. These results pertain to the allocation of the low-skill workforce across sectors and do not, by themselves, determine changes in sector-specific low-skill employment levels. Under the normalization of fixed total labor supply, however, the increase in n_h is accompanied by a decline in the aggregate number of low-skill workers.

The effects of an increase in the low-skill supply share yield the opposite results:

$$\frac{d\delta_{w,h}}{dn_l} = -\frac{Z}{n_h^2 (B_b - B_w)} < 0, \quad (\text{A.30})$$

$$\frac{d\delta_{b,h}}{dn_l} = \frac{Z}{n_h^2 (B_b - B_w)} > 0, \quad (\text{A.31})$$

$$\frac{d\delta_{w,l}}{dn_l} = -\frac{B_b B_w}{Z n_l^2 (B_b - B_w)} < 0, \quad (\text{A.32})$$

and

$$\frac{d\delta_{b,l}}{dn_l} = \frac{B_b B_w}{Z n_l^2 (B_b - B_w)} > 0. \quad (\text{A.33})$$

Accordingly, an increase in the low-skill supply share raises the fraction of both high- and low-skill workers employed in the blue-collar sector and lowers the corresponding fractions employed in the white-collar sector.

These allocation shares describe how each skill group is distributed across sectors. They are distinct from within-sector skill-composition shares and, therefore, do not directly measure within-industry skill upgrading.

B Data Appendix

B.1 Construction of Immigration Stock Data

I construct county-level data on immigration utilizing the individual census files of the IPUMS USA and NHGIS samples of the 1850 (full sample), 1860 (full sample), 1870 (10% sample), 1880 (10% sample), 1890 (10% sample), 1900 (5% sample), 1910 (1% sample), 1920 (10% sample), 1930 (5% sample), 1940 (1% sample), 1950 (1% sample), 1960 (5% sample), 1970 (1% Form 1 Metro sample), 1980 (5% State sample), 1990 (5% State sample), and 2000 (5% sample) waves, as well as the 2006-2010 five-year sample of the ACS. I define immigrants as “foreign-born” individuals born outside of the US and its territories in each decennial wave beginning from 1850 to 2000 in the census and the 2006-2010 five-year sample of the ACS. I define the measure of immigration stock, $I_{o,d,t}$, as the number of immigrants (foreign-born people) from origin (country) o , who resides in US destination (county) d in census year t . Countries of origin are based on the detailed birthplace variable in the census.

Both origins and destinations are either historical (prior to 1990) or modern (post-1990) geographical units. Therefore, to have consistent geographic units of countries and counties, I follow the method highlighted in [Burchardi et al. \(2019\)](#), in that I transform non-1990-level foreign countries of origin or detailed birthplaces to 1990-level foreign countries as well as non-1990-level counties, county groups, and PUMAs to 1990-level counties. Initially, I transition non-1990-level counties, county groups, and PUMAs to 1990-level counties in each census year utilizing the transition matrices created in [Burchardi et al. \(2019\)](#) and complementing it with county crosswalk files generated in [Eckert et al. \(2020\)](#).⁴¹ Afterwards, I generate a country crosswalk file and transform non-1990-level countries to 1990-level countries in each census wave. All transition matrices and crosswalk files rely on population-based weights.

B.2 Construction of Working-Age Population Data

I construct variables for human-capital-specific working-age population shares using each decennial census wave in 1970, 1980, 1990, 2000, and the 2006-2010 five-year sample of the ACS in the following steps. First, I restrict the sample to individuals aged 16-64. Second, I allocate them into skill cells based on their educational attainment (*i.e.*, years of schooling). Finally, I take the

⁴¹The additional details on the construction of county crosswalk files and the weighting scheme can be found in those papers alongside the following websites: [\(1\)](#) and [\(2\)](#).

number of individuals with a given human capital level by county relative to the entire working-age population by county. To have consistent geographic units of counties, I follow the procedure described above.

B.3 Construction of Employment Data

I construct variables for all, native, and immigrant employment shares of each industry-skill type utilizing each decennial census wave in 1970, 1980, 1990, 2000, and the 2006-2010 five-year sample of the ACS. The measure of employment is the number of individuals employed, and the continuous skill measure corresponds to average years of education of workers employed by each industry. I further distinguish employment shares by industry tradability and discuss how I generate these shares below.

Initially, I restrict the sample to only households and drop individuals working in the military occupations along with those classified as “unemployed” and “not in the labor force.” Afterwards, I aggregate the 1990-level census industry codes to a balanced panel of industries utilizing an industry concordance outlined in [Autor et al. \(2019\)](#).⁴² Then I allocate workers to three industry-skill cells based on mean education of workers in industries. Later, I take the number of workers by county and industry-skill types relative to the entire workforce by county. I proceed with the construction of immigrant and native employment shares similarly, in that I take the number of workers by county, industry-skill types, and nativity relative to the entire workforce by county. Lastly, I generate the employment shares for workers of all industry-skill types in the tradable and nontradable sectors through distinguishing them by industry tradability.

B.4 Construction of Establishments Data

I construct establishment share of each industry skill type by county along with industry tradability using the 1970, 1980, 1990, 2000, and 2010 extracts of the County Business Patterns (CBP) data published by the Census Bureau and supplement them with those assembled in [Eckert et al. \(2020, 2022\)](#).⁴³ The data provide further important details on establishment size, defined as the number of workers employed in a given establishment, and industry, in which an establishment operates. Apart from establishment and industry characteristics, they also provide employment figures dur-

⁴²For additional details, see their paper and David Dorn’s [website](#). The crosswalk file I utilize is “[C4] Census *ind1990 to ind1990dd*.”

⁴³For further details on digitization, cleaning, imputation on employment counts, and consolidation of these panels under consistent industry codes, see their papers, and Fabian Eckert’s [website](#).

ing the pay period including March 12, first-quarter payroll, and annual payroll. Consistent with the literature, I define small establishments as those employing fewer than 20 workers, medium establishments as those employing between 20 and 500 workers, and large establishments as those employing more than 500 workers. I carry out the following steps to create establishment shares.

First, using the industry concordance files generated in [Eckert et al. \(2020, 2022\)](#), I crosswalk all industry codes to a unique industry classification, *SIC87*. Afterwards, I map the *SIC87* industry codes to a balanced panel of industries compiled in [Autor et al. \(2019\)](#).⁴⁴ This step is necessary for not only making them compatible with the census industry codes but also identifying the industry-skill rankings discussed in the text. Next I take the number of establishments by county and industry-skill types relative to the total number of establishments by county. I further differentiate establishment share of each industry-skill type by the industry tradability.

⁴⁴See David Dorn’s [website](#) for additional details. The crosswalk file I utilize is “[C8] *SIC87* to *ind1990ddx*.”

C Appendix Tables

Table C.1: FIRST-STAGE ESTIMATES

	Medium-Skill Population Share	High-Skill Population Share
	(1)	(2)
<i>Exogenous Population</i> _{MS,d,t}	0.072*** (0.003)	-0.134*** (0.006)
<i>Exogenous Population</i> _{HS,d,t}	-0.045*** (0.002)	0.151*** (0.005)
R^2	0.836	0.641
SW First-Stage F-Stats	133; 364	133; 364
N	15,705	15,705
State FE	Yes	Yes
Time FE	Yes	Yes

Notes: This table presents the first-stage estimates. The dependent variables are medium- and high-skill population shares at the county level in separate specifications as displayed in columns (1) and (2), respectively. The independent variables are exogenous working-age population of medium- and high-skill types at the county level. Each specification reports the Sanderson-Windmeijer F-statistic for each endogenous variable. Both regressions include state and time fixed effects. Standard errors are clustered by county for both specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.2: LOW-SKILL INDUSTRIES

Census Industry Code	Census Industry Title
10	Agricultural Production, Crops
20	Landscape and Horticultural Services
100	Meat Products
141	Carpets and Rugs
151	Apparel and Accessories, except Knit
230	Logging
531	Scrap and Waste Materials
722	Services to Dwellings and Other Buildings
750	Automobile Parking and Carwashes
761	Private Households

Notes: This time-varying industry-skill classification is based on the mean education of workers employed by a given industry as described in the text. The table presents skill types evaluated for 2010. These industry codes and titles correspond to the 1990-level three-digit census industry codes harmonized to a balanced panel of industries using the concordance in [Autor et al. \(2019\)](#). “Low-skill” industries are those whose average education corresponds to a high-school degree or less.

Table C.3: MEDIUM-SKILL INDUSTRIES

Census Industry Code	Census Industry Title	Census Industry Code	Census Industry Title
11	Agricultural Production, Livestock	32	Fishing, Hunting, and Trapping
40	Metal Mining	41	Coal Mining
42	Oil and Gas Extraction	50	Nonmetallic Mining and Quarrying, except Fuels
60	All Construction	100	Meat Products
101	Dairy Products	102	Canned, Frozen, and Preserved Fruits and Vegetables
110	Grain Mill Products	111	Bakery Products
112	Sugar and Confectionery Products	120	Beverage Industries
121	Misc. Food Preparations and Kindred Products	122	Food Industries, n.s
130	Tobacco Manufactures	132	Knitting Mills
140	Dyeing and Finishing Textiles, except Wool and Knit Goods	141	Carpets and Rugs
142	Yarn, Thread, and Fabric Mills	151	Apparel and Accessories, except Knit
152	Miscellaneous Fabricated Textile Products	160	Pulp, Paper, and Paperboard Mills
161	Miscellaneous Paper and Pulp Products	162	Paperboard Containers and Boxes
171	Newspaper Publishing and Printing	172	Printing, Publishing, and Allied Industries, except Newspapers
180	Plastics, Synthetics, and Resins	181	Drugs
182	Soaps and Cosmetics	190	Paints, Varnishes, and Related Products
191	Agricultural Chemicals	201	Miscellaneous Petroleum and Coal Products
210	Tires and Inner Tubes	211	Other Rubber Products, and Plastics Footwear and Belting
212	Miscellaneous Plastics Products	221	Footwear, except Rubber and Plastic
222	Leather Products, except Footwear	230	Logging
231	Sawmills, Planing Mills, and Millwork	232	Wood Buildings and Mobile Homes
242	Furniture and Fixtures	250	Glass and Glass Products
251	Cement, Concrete, Gypsum, and Plaster Products	252	Structural Clay Products
261	Pottery and Related Products	262	Misc. Nonmetallic Mineral and Stone Products
270	Blast Furnaces, Steelworks, Rolling and Finishing Mills	271	Iron and Steel Foundries
272	Primary Aluminum Industries	280	Other Primary Metal Industries
281	Cutlery, Handtools, and General Hardware	282	Fabricated Structural Metal Products
291	Metal Forgings and Stampings	300	Miscellaneous Fabricated Metal Products
301	Metal Industries, n.s	310	Engines and Turbines
311	Farm Machinery and Equipment	312	Construction and Material Handling Machines
320	Metalworking Machinery	331	Machinery, except Electrical, n.e.c
332	Machinery, n.s	340	Household Appliances
351	Motor Vehicles and Motor Vehicle Equipment	360	Ship and Boat Building and Repairing
361	Railroad Locomotives and Equipment	370	Cycles and Miscellaneous Transportation Equipment
390	Toys, Amusement, and Sporting Goods	391	Miscellaneous Manufacturing Industries
392	Manufacturing Industries, n.s	400	Railroads
401	Bus Service and Urban Transit	402	Taxicab Service
410	Trucking Service	411	Warehousing and Storage
412	U.S. Postal Service	420	Water Transportation
432	Services Incidental to Transportation	450	Electric Light and Power
451	Gas and Steam Supply Systems	470	Water Supply and Irrigation
471	Sanitary Services	472	Utilities, n.s
500	Motor Vehicles and Equipment	501	Furniture and Home Furnishings
502	Lumber and Construction Materials	511	Metals and Minerals, except Petroleum
512	Electrical Goods	521	Hardware, Plumbing and Heating Supplies
530	Machinery, Equipment, and Supplies	531	Scrap and Waste Materials
532	Miscellaneous Wholesale, Durable Goods	540	Paper and Paper Products
541	Drugs, Chemicals, and Allied Products	542	Apparel, Fabrics, and Notions
550	Groceries and Related Products	551	Farm-product Raw Materials
552	Petroleum Products	560	Alcoholic Beverages
561	Farm Supplies	562	Miscellaneous Wholesale, Nondurable Goods
571	Wholesale Trade, n.s	580	Lumber and Building Material Retailing
581	Hardware Stores	582	Retail Nurseries and Garden Stores
601	Grocery Stores	610	Retail Bakeries
611	Food Stores, n.e.c	612	Motor Vehicle Dealers
620	Auto and Home Supply Stores	621	Gasoline Service Stations
622	Miscellaneous Vehicle Dealers	623	Apparel and Accessory Stores, except Shoe
630	Shoe Stores	631	Furniture and Home Furnishings Stores
632	Household Appliance Stores	640	Music Stores
641	Eating and Drinking Places	650	Liquor Stores
651	Sporting Goods, Bicycles, and Hobby Stores	652	Book and Stationery Stores
660	Jewelry Stores	661	Gift, Novelty, and Souvenir Shops
662	Sewing, Needlework, and Piece Goods Stores	670	Vending Machine Operators
671	Direct Selling Establishments	672	Fuel Dealers
681	Retail Florists	682	Miscellaneous Retail Stores
691	Retail Trade, n.s	701	Savings Institutions, Including Credit Unions
731	Personnel Supply Services	740	Detective and Protective Services
742	Automotive Rental and Leasing, without Drivers	751	Automotive Repair and Related Services
752	Electrical Repair Shops	760	Miscellaneous Repair Services
771	Laundry, Cleaning, and Garment Services	772	Beauty Shops
780	Barber Shops	791	Miscellaneous Personal Services
801	Video Tape Rental	802	Bowling Centers
810	Miscellaneous Entertainment and Recreation Services	832	Nursing and Personal Care Facilities
860	Child Day Care Services	861	Job Training and Vocational Rehabilitation Services
870	Residential Care Facilities, without Nursing	882	Engineering, Architectural, and Surveying Services

Notes: This time-varying industry-skill classification is based on the mean education of workers employed by a given industry as described in the text. The table presents skill types evaluated for 2010. These industry codes and titles correspond to the 1990-level three-digit census industry codes harmonized to a balanced panel of industries using the concordance in [Autor et al. \(2019\)](#). “Medium-skill” industries are those whose average education corresponds to some college.

Table C.4: HIGH-SKILL INDUSTRIES

<i>Census Industry Code</i>	<i>Census Industry Title</i>	<i>Census Industry Code</i>	<i>Census Industry Title</i>
12	Veterinary Services	31	Forestry
181	Drugs	192	Industrial and Miscellaneous Chemicals
200	Petroleum Refining	322	Computers and Related Equipment
341	Radio, TV, and Communication Equipment	342	Electrical Machinery, Equipment, and Supplies, n.e.c
352	Aircraft and Parts	362	Guided Missiles, Space Vehicles, and Parts
371	Scientific and Controlling Instruments	372	Medical, Dental, and Optical Instruments and Supplies
421	Air Transportation	440	Radio and Television Broadcasting and Cable
441	Telephone Communications	452	Electric and Gas, and Other Combinations
510	Professional and Commercial Equipment and Supplies	541	Drugs, Chemicals, and Allied Products
633	Radio, TV, and Computer Stores	642	Drug Stores
663	Catalog and Mail Order Houses	700	Banking
702	Credit Agencies, n.e.c	710	Security, Commodity Brokerage, and Investment Companies
711	Insurance	712	Real Estate, Including Real Estate-insurance Offices
721	Advertising	741	Business Services, n.e.c
800	Theaters and Motion Pictures	812	Offices and Clinics of Physicians
820	Offices and Clinics of Dentists	821	Offices and Clinics of Chiropractors
822	Offices and Clinics of Optometrists	831	Hospitals
840	Health Services, n.e.c	841	Legal Services
842	Elementary and Secondary Schools	850	Colleges and Universities
852	Libraries	860	Educational Services, n.e.c
871	Social Services, n.e.c	872	Museums, Art Galleries, and Zoos
880	Religious Organizations	881	Membership Organizations, n.e.c
882	Engineering, Architectural, and Surveying Services	890	Accounting, Auditing, and Bookkeeping Services
891	Research, Development, and Testing Services	892	Management and Public Relations Services
893	Miscellaneous Professional and Related Services	901	General Government, n.e.c
910	Justice, Public Order, and Safety	921	Public Finance, Taxation, and Monetary Policy
930	Administration of Environmental Quality and Housing Programs	931	Administration of Economic Programs
932	National Security and International Affairs		

Notes: This time-varying industry-skill classification is based on the mean education of workers employed by a given industry as described in the text. The table presents skill types evaluated for 2010. These industry codes and titles correspond to the 1990-level three-digit census industry codes harmonized to a balanced panel of industries using the concordance in [Autor et al. \(2019\)](#). “High-skill” industries are those whose average education corresponds to a college degree and above.

Table C.5: TRADABLE INDUSTRIES

Census Industry Code	Census Industry Title	Census Industry Code	Census Industry Title
10	Agricultural Production, Crops	221	Footwear, except Rubber and Plastic
11	Agricultural Production, Livestock	222	Leather Products, except Footwear
20	Veterinary Services	230	Logging
31	Forestry	231	Sawmills, Planing Mills, and Millwork
32	Fishing, Hunting, and Trapping	232	Wood Buildings and Mobile Homes
40	Metal Mining	242	Furniture and Fixtures
41	Coal Mining	250	Glass and Glass Products
42	Oil and Gas Extraction	251	Cement, Concrete, Gypsum, and Plaster Products
50	Nonmetallic Mining and Quarrying, except Fuels	252	Structural Clay Products
100	Meat Products	261	Pottery and Related Products
101	Dairy Products	262	Misc. Nonmetallic Mineral and Stone Products
102	Canned, Frozen, and Preserved Fruits and Vegetables	270	Blast Furnaces, Steelworks, Rolling and Finishing Mills
110	Grain Mill Products	271	Iron and Steel Foundries
111	Bakery Products	272	Primary Aluminum Industries
112	Sugar and Confectionery Products	281	Cutlery, Handtools, and General Hardware
120	Beverage Industries	282	Fabricated Structural Metal Products
121	Misc. Food Preparations and Kindred Products	291	Metal Forgings and Stampings
122	Food Industries, n.s	300	Miscellaneous Fabricated Metal Products
130	Tobacco Manufactures	301	Metal Industries, n.s
132	Knitting Mills	310	Engines and Turbines
140	Dyeing and Finishing Textiles, except Wool and Knit Goods	311	Farm Machinery and Equipment
141	Carpets and Rugs	312	Construction and Material Handling Machines
142	Yarn, Thread, and Fabric Mills	320	Metalworking Machinery
151	Apparel and Accessories, except Knit	331	Machinery, except Electrical, n.e.c
152	Miscellaneous Fabricated Textile Products	332	Machinery, n.s
160	Pulp, Paper, and Paperboard Mills	340	Household Appliances
161	Miscellaneous Paper and Pulp Products	351	Motor Vehicles and Motor Vehicle Equipment
162	Paperboard Containers and Boxes	360	Ship and Boat Building and Repairing
171	Newspaper Publishing and Printing	361	Railroad Locomotives and Equipment
172	Printing, Publishing, and Allied Industries, except Newspapers	370	Cycles and Miscellaneous Transportation Equipment
180	Plastics, Synthetics, and Resins	390	Toys, Amusement, and Sporting Goods
181	Drugs	391	Miscellaneous Manufacturing Industries
182	Soaps and Cosmetics	392	Manufacturing Industries, n.s
190	Paints, Varnishes, and Related Products	510	Professional and Commercial Equipment and Supplies
191	Agricultural Chemicals	540	Paper and Paper Products
200	Petroleum Refining	571	Wholesale Trade, n.s
201	Miscellaneous Petroleum and Coal Products	632	Household Appliance Stores
210	Tires and Inner Tubes	640	Music Stores
211	Other Rubber Products, and Plastics Footwear and Belting	660	Jewelry Stores
212	Miscellaneous Plastics Products	671	Direct Selling Establishments
220	Footwear, except Rubber and Plastic		

Notes: This table presents the tradable industries. Following [Burstein et al. \(2020\)](#), all goods industries (agriculture, mining, and manufacturing) are classified as tradable, while the construction industry is dropped from the analysis. These industry codes and titles correspond to the 1990-level three-digit census industry codes harmonized to a balanced panel of industries using the concordance in [Autor et al. \(2019\)](#).

Table C.6: NONTRADABLE INDUSTRIES

Census Industry Code	Census Industry Title	Census Industry Code	Census Industry Title
232	Wood Buildings and Mobile Homes	410	Trucking Service
400	Railroads	411	Warehousing and Storage
401	Bus Service and Urban Transit	412	U.S. Postal Service
402	Taxicab Service	420	Water Transportation
421	Air Transportation	432	Services Incidental to Transportation
440	Radio and Television Broadcasting and Cable	450	Electric Light and Power
441	Telephone Communications	451	Gas and Steam Supply Systems
470	Water Supply and Irrigation	471	Sanitary Services
472	Utilities, n.s	500	Motor Vehicles and Equipment
501	Furniture and Home Furnishings	502	Lumber and Construction Materials
511	Metals and Minerals, except Petroleum	512	Electrical Goods
521	Hardware, Plumbing and Heating Supplies	530	Machinery, Equipment, and Supplies
531	Scrap and Waste Materials	532	Miscellaneous Wholesale, Durable Goods
540	Paper and Paper Products	541	Drugs, Chemicals, and Allied Products
542	Apparel, Fabrics, and Notions	550	Groceries and Related Products
551	Farm-product Raw Materials	552	Petroleum Products
560	Alcoholic Beverages	561	Farm Supplies
562	Miscellaneous Wholesale, Nondurable Goods	580	Lumber and Building Material Retailing
581	Hardware Stores	582	Retail Nurseries and Garden Stores
601	Grocery Stores	610	Retail Bakeries
611	Food Stores, n.e.c	612	Motor Vehicle Dealers
620	Auto and Home Supply Stores	621	Gasoline Service Stations
622	Miscellaneous Vehicle Dealers	623	Apparel and Accessory Stores, except Shoe
630	Shoe Stores	631	Furniture and Home Furnishings Stores
641	Eating and Drinking Places	650	Liquor Stores
651	Sporting Goods, Bicycles, and Hobby Stores	652	Book and Stationery Stores
661	Gift, Novelty, and Souvenir Shops	662	Sewing, Needlework, and Piece Goods Stores
670	Vending Machine Operators	681	Retail Florists
682	Miscellaneous Retail Stores	691	Retail Trade, n.s
700	Banking	701	Savings Institutions, Including Credit Unions
702	Credit Agencies, n.e.c	710	Security, Commodity Brokerage, and Investment Companies
711	Insurance	721	Advertising
731	Personnel Supply Services	740	Detective and Protective Services
741	Business Services, n.e.c	742	Automotive Rental and Leasing, without Drivers
751	Automotive Repair and Related Services	752	Electrical Repair Shops
760	Miscellaneous Repair Services	771	Laundry, Cleaning, and Garment Services
772	Beauty Shops	780	Barber Shops
791	Miscellaneous Personal Services	800	Theaters and Motion Pictures
801	Video Tape Rental	802	Bowling Centers
810	Miscellaneous Entertainment and Recreation Services	812	Offices and Clinics of Physicians
820	Offices and Clinics of Dentists	821	Offices and Clinics of Chiropractors
822	Offices and Clinics of Optometrists	831	Hospitals
840	Health Services, n.e.c	841	Legal Services
842	Elementary and Secondary Schools	850	Colleges and Universities
852	Libraries	860	Educational Services, n.e.c
871	Social Services, n.e.c	872	Museums, Art Galleries, and Zoos
880	Religious Organizations	881	Membership Organizations, n.e.c
882	Engineering, Architectural, and Surveying Services	891	Research, Development, and Testing Services
892	Management and Public Relations Services	893	Miscellaneous Professional and Related Services
901	General Government, n.e.c	910	Justice, Public Order, and Safety
921	Public Finance, Taxation, and Monetary Policy	930	Administration of Environmental Quality and Housing Programs
931	Administration of Economic Programs	932	National Security and International Affairs

Notes: This table presents the nontradable industries. Following [Burstein et al. \(2020\)](#), all services industries are classified as nontradable, while the construction industry is dropped from the analysis. These industry codes and titles correspond to the 1990-level three-digit census industry codes harmonized to a balanced panel of industries using the concordance in [Autor et al. \(2019\)](#).

Table C.7: LOW-SKILL TRADABLE AND NONTRADABLE INDUSTRIES

Tradable Industries		Nontradable Industries	
Census Industry Code	Census Industry Title	Census Industry Code	Census Industry Title
10	Agricultural Production, Crops	531	Scrap and Waste Materials
100	Meat Products	722	Services to Dwellings and Other Buildings
141	Carpets and Rugs	761	Private Households
151	Apparel and Accessories, Except Knit		
230	Logging		

Notes: This time-varying industry-skill classification across both tradable and nontradable sectors is based on the mean education of workers employed by a given industry as described in the text. The table presents skill types evaluated for 2010. “Low-skill” industries are those whose average education corresponds to a high-school degree or less in both sectors.

Table C.8: MEDIUM-SKILL TRADABLE AND NONTRADABLE INDUSTRIES

Tradable Industries		Nontradable Industries	
Census Industry Code	Census Industry Title	Census Industry Code	Census Industry Title
11	Agricultural Production, Livestock	400	Railroads
32	Fishing, Hunting, and Trapping	401	Bus Service and Urban Transit
40	Metal Mining	402	Taxicab Service
41	Coal Mining	410	Trucking Service
42	Oil and Gas Extraction	411	Warehousing and Storage
50	Nonmetallic Mining and Quarrying, Except Fuels	412	U.S. Postal Service
101	Dairy Products	420	Water Transportation
102	Canned, Frozen, and Preserved Fruits and Vegetables	432	Services Incidental to Transportation
110	Grain Mill Products	450	Electric Light and Power
111	Bakery Products	451	Gas and Steam Supply Systems
112	Sugar and Confectionery Products	470	Water Supply and Irrigation
120	Beverage Industries	471	Sanitary Services
121	Misc. Food Preparations and Kindred Products	472	Utilities, n.s
122	Food Industries, n.s	500	Motor Vehicles and Equipment
130	Tobacco Manufactures	501	Furniture and Home Furnishings
132	Knitting Mills	502	Lumber and Construction Materials
140	Dyeing and Finishing Textiles, Except Wool and Knit Goods	511	Metals and Minerals, Except Petroleum
142	Yarn, Thread, and Fabric Mills	512	Electrical Goods
152	Miscellaneous Fabricated Textile Products	530	Machinery, Equipment, and Supplies
160	Pulp, Paper, and Paperboard Mills	532	Miscellaneous Wholesale, Durable Goods
161	Miscellaneous Paper and Pulp Products	540	Paper and Paper Products
162	Paperboard Containers and Boxes	542	Apparel, Fabrics, and Notions
171	Newspaper Publishing and Printing	550	Groceries and Related Products
172	Printing, Publishing, and Allied Industries, Except Newspapers	551	Farm-product Raw Materials
180	Plastics, Synthetics, and Resins	552	Petroleum Products
182	Soaps and Cosmetics	560	Alcoholic Beverages
190	Paints, Varnishes, and Related Products	561	Farm Supplies
191	Agricultural Chemicals	562	Miscellaneous Wholesale, Nondurable Goods
201	Miscellaneous Petroleum and Coal Products	571	Wholesale Trade, n.s
210	Tires and Inner Tubes	580	Lumber and Building Material Retailing
211	Other Rubber Products, and Plastics Footwear and Belting	581	Hardware Stores
212	Miscellaneous Plastics Products	582	Retail Nurseries and Garden Stores
221	Footwear, Except Rubber and Plastic	591	Department Stores
222	Leather Products, Except Footwear	600	Miscellaneous General Merchandise Stores
231	Sawmills, Planing Mills, and Millwork	601	Grocery Stores
232	Wood Buildings and Mobile Homes	610	Retail Bakeries
242	Furniture and Fixtures	611	Food Stores, n.e.c
250	Glass and Glass Products	612	Motor Vehicle Dealers
251	Cement, Concrete, Gypsum, and Plaster Products	620	Auto and Home Supply Stores
252	Structural Clay Products	621	Gasoline Service Stations
261	Pottery and Related Products	622	Miscellaneous Vehicle Dealers
262	Misc. Nonmetallic Mineral and Stone Products	623	Apparel and Accessory Stores, Except Shoe
270	Blast Furnaces, Steelworks, Rolling and Finishing Mills	630	Shoe Stores
271	Iron and Steel Foundries	631	Furniture and Home Furnishings Stores
272	Primary Aluminum Industries	640	Household Appliance Stores
281	Cutlery, Handtools, and General Hardware	641	Eating and Drinking Places
282	Fabricated Structural Metal Products	650	Liquor Stores
291	Metal Forgings and Stampings	651	Sporting Goods, Bicycles, and Hobby Stores
292	Ordnance	652	Book and Stationery Stores
300	Miscellaneous Fabricated Metal Products	660	Jewelry Stores
301	Metal Industries, n.s	662	Sewing, Needlework, and Piece Goods Stores
310	Engines and Turbines	670	Vending Machine Operators
311	Farm Machinery and Equipment	671	Direct Selling Establishments
312	Construction and Material Handling Machines	672	Fuel Dealers
320	Metalworking Machinery	681	Retail Florists
331	Machinery, Except Electrical, n.e.c	682	Miscellaneous Retail Stores
332	Machinery, n.s	691	Retail Trade, n.s
340	Household Appliances	731	Personnel Supply Services
351	Motor Vehicles and Motor Vehicle Equipment	740	Detective and Protective Services
360	Ship and Boat Building and Repairing	750	Automotive Rental and Leasing, Without Drivers
361	Railroad Locomotives and Equipment	752	Electrical Repair Shops
370	Cycles and Miscellaneous Transportation Equipment	760	Miscellaneous Repair Services
390	Toys, Amusement, and Sporting Goods	762	Hotels and Motels
392	Manufacturing Industries, n.s	770	Lodging Places, Except Hotels and Motels
		771	Laundry, Cleaning, and Garment Services
		772	Beauty Shops
		780	Barber Shops
		781	Funeral Service and Crematories
		791	Miscellaneous Personal Services
		802	Bowling Centers
		810	Miscellaneous Entertainment and Recreation Services
		832	Nursing and Personal Care Facilities
		861	Job Training and Vocational Rehabilitation Services
		862	Child Day Care Services
		870	Residential Care Facilities, Without Nursing
		881	Labor Unions

Notes: This time-varying industry-skill classification across both tradable and nontradable sectors is based on the mean education of workers employed by a given industry as described in the text. The table presents skill types evaluated for 2010. "Medium-skill" industries are those whose average education corresponds to some college in both sectors.

Table C.9: HIGH-SKILL TRADABLE AND NONTRADABLE INDUSTRIES

Tradable Industries		Nontradable Industries	
Census Industry Code	Census Industry Title	Census Industry Code	Census Industry Title
20	Veterinary Services	421	Air Transportation
31	Forestry	440	Radio and Television Broadcasting and Cable
181	Drugs	441	Telephone Communications
192	Industrial and Miscellaneous Chemicals	452	Electric and Gas, and Other Combinations
200	Petroleum Refining	510	Professional and Commercial Equipment and Supplies
322	Computers and Related Equipment	541	Drugs, Chemicals, and Allied Products
341	Radio, TV, and Communication Equipment	642	Drug Stores
342	Electrical Machinery, Equipment, and Supplies, n.e.c	663	Catalog and Mail Order Houses
352	Aircraft and Parts	700	Banking
362	Guided Missiles, Space Vehicles, and Parts	702	Credit Agencies, n.e.c
371	Scientific and Controlling Instruments	710	Security, Commodity Brokerage, and Investment Companies
372	Medical, Dental, and Optical Instruments and Supplies	711	Insurance
		712	Real Estate, Including Real Estate-insurance Offices
		721	Advertising
		732	Computer and Data Processing Services
		741	Business Services, n.e.c
		800	Theaters and Motion Pictures
		812	Offices and Clinics of Physicians
		820	Offices and Clinics of Dentists
		821	Offices and Clinics of Chiropractors
		822	Offices and Clinics of Optometrists
		831	Hospitals
		840	Health Services, n.e.c
		841	Legal Services
		842	Elementary and Secondary Schools
		850	Colleges and Universities
		852	Libraries
		860	Educational Services, n.e.c
		872	Museums, Art Galleries, and Zoos
		880	Religious Organizations
		882	Engineering, Architectural, and Surveying Services
		890	Accounting, Auditing, and Bookkeeping Services
		891	Research, Development, and Testing Services
		892	Management and Public Relations Services
		893	Miscellaneous Professional and Related Services

Notes: This time-varying industry-skill classification across both tradable and nontradable sectors is based on the mean education of workers employed by a given industry as described in the text. The table presents skill types evaluated for 2010. “High-skill” industries are those whose average education corresponds to a college degree and above in both sectors.

Table C.10: ALTERNATIVE MEASURES OF ECONOMIC MAGNITUDE

		Baseline IV Coefficient	Instrument-Scaled Standardized Effect	Implied Elasticity
		(1)	(2)	(3)
Panel A: Employment Shares				
Low Skill	<i>Population Share_{MS,d,t}</i>	−0.389	−0.098	−0.313
	<i>Population Share_{HS,d,t}</i>	−0.315	−0.155	−0.146
Medium Skill	<i>Population Share_{MS,d,t}</i>	0.212	0.052	0.118
	<i>Population Share_{HS,d,t}</i>	−0.229	−0.109	−0.073
High Skill	<i>Population Share_{MS,d,t}</i>	0.177	0.058	0.204
	<i>Population Share_{HS,d,t}</i>	0.544	0.349	0.361
Panel B: Establishment Shares				
Low Skill	<i>Population Share_{MS,d,t}</i>	−0.745	−0.059	−0.796
	<i>Population Share_{HS,d,t}</i>	−0.110	−0.017	−0.067
Medium Skill	<i>Population Share_{MS,d,t}</i>	−0.056	−0.005	−0.033
	<i>Population Share_{HS,d,t}</i>	−0.175	−0.031	−0.059
High Skill	<i>Population Share_{MS,d,t}</i>	0.955	0.103	0.769
	<i>Population Share_{HS,d,t}</i>	0.331	0.070	0.153

Notes: This table presents alternative measures of the economic magnitude of the baseline estimates reported in Table 3. Column (1) reproduces the baseline IV coefficients. Since both the dependent variables and the population-share regressors are measured as shares, each coefficient gives the percentage-point change in the outcome share associated with a one-percentage-point increase in the corresponding population share. Column (2) reports the instrument-scaled standardized effect, $\hat{S}_{y,x} = \hat{\beta}_{IV} \frac{SD(\hat{x}^\perp)}{SD(y^\perp)}$, where $\hat{x}^\perp$ denotes the first-stage fitted component of the endogenous population share after removing state and year fixed effects, and $y^\perp$ denotes the corresponding residualized outcome. This measure gives the change in the outcome, in residual standard-deviation units, associated with a one-standard-deviation increase in the instrument-predicted component of the population share. Column (3) reports the elasticity implied by the linear specification, $\hat{\mathcal{E}}_{y,x} = \hat{\beta}_{IV} \frac{\bar{x}}{\bar{y}}$, evaluated at the estimation-sample means. It gives the percentage change in the outcome share associated with a one-percent change in the population share at the sample means. All calculations use the common estimation sample of 15,705 county-year observations. The underlying specifications include state and year fixed effects, with standard errors clustered by county.

Table C.11: EMPLOYMENT ESTIMATES BY NATIVITY

	All	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)
Panel A: Native Employment Shares				
<i>Population Share_{MS,d,t}</i>	0.045 (0.076)	-0.332*** (0.046)	0.091 (0.070)	0.044 (0.067)
<i>Population Share_{HS,d,t}</i>	-0.088** (0.037)	-0.192*** (0.021)	-0.384*** (0.032)	0.496*** (0.030)
AR Wald F-Test P-value	0.006	0.000	0.000	0.000
Panel B: Immigrant Employment Shares				
<i>Population Share_{MS,d,t}</i>	-0.045 (0.076)	-0.709*** (0.088)	-0.091 (0.075)	0.800*** (0.088)
<i>Population Share_{HS,d,t}</i>	0.088** (0.037)	-1.012*** (0.051)	0.167*** (0.035)	0.845*** (0.045)
AR Wald F-Test P-value	0.006	0.000	0.000	0.000
<i>N</i>	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are native and immigrant employment shares of each industrial skill type at the county level in separate regressions as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.12: HETEROGENEITY - ESTABLISHMENT ESTIMATES BY SIZE

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Small Establishments			
<i>Population Share</i> _{MS,d,t}	-0.820*** (0.179)	-0.380 (0.230)	1.607*** (0.244)
<i>Population Share</i> _{HS,d,t}	-0.228*** (0.036)	-0.217*** (0.047)	0.335*** (0.043)
Panel B: Medium Establishments			
<i>Population Share</i> _{MS,d,t}	-0.285*** (0.109)	0.272*** (0.084)	0.187*** (0.039)
<i>Population Share</i> _{HS,d,t}	-0.015 (0.050)	0.061* (0.037)	0.048** (0.019)
Panel C: Large Establishments			
<i>Population Share</i> _{MS,d,t}	-0.055* (0.033)	-0.016 (0.032)	-0.010 (0.015)
<i>Population Share</i> _{HS,d,t}	-0.002 (0.013)	0.004 (0.014)	0.014* (0.008)
<i>N</i>	14,896	14,896	14,896
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are small, medium, and large establishment shares of each industrial skill type at the county level in separate regressions as displayed in Panels A, B, and C, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population at the county level. Consistent with the literature, small establishments are defined as those employing fewer than 20 workers, medium establishments are defined as those employing between 20 and 500 workers, and large establishments are defined as those employing more than 500 workers. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.13: HETEROGENEITY - ESTABLISHMENT ESTIMATES BY SIZE IN THE TRADABLE SECTOR

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Small Establishments			
<i>Population Share</i> _{MS,d,t}	-0.416*** (0.065)	-0.067** (0.034)	-0.024** (0.010)
<i>Population Share</i> _{HS,d,t}	-0.120*** (0.030)	-0.030* (0.017)	-0.009* (0.005)
Panel B: Medium Establishments			
<i>Population Share</i> _{MS,d,t}	-0.544*** (0.112)	0.077 (0.079)	0.020 (0.025)
<i>Population Share</i> _{HS,d,t}	-0.055 (0.050)	0.027 (0.033)	-0.002 (0.012)
Panel C: Large Establishments			
<i>Population Share</i> _{MS,d,t}	-0.046 (0.033)	-0.009 (0.032)	-0.011 (0.014)
<i>Population Share</i> _{HS,d,t}	-0.001 (0.013)	0.003 (0.015)	0.010 (0.008)
<i>N</i>	14,896	14,896	14,896
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are small, medium, and large establishment shares of each industrial skill type in the tradable sector in separate regressions as displayed in Panels A, B, and C, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population at the county level. Consistent with the literature, small establishments are defined as those employing fewer than 20 workers, medium establishments are defined as those employing between 20 and 500 workers, and large establishments are defined as those employing more than 500 workers. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.14: HETEROGENEITY - ESTABLISHMENT ESTIMATES BY SIZE IN THE NONTRADABLE SECTOR

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Small Establishments			
<i>Population Share</i> _{MS,d,t}	-0.098* (0.060)	-0.258*** (0.094)	0.833*** (0.098)
<i>Population Share</i> _{HS,d,t}	-0.099*** (0.026)	-0.292*** (0.042)	0.417*** (0.044)
Panel B: Medium Establishments			
<i>Population Share</i> _{MS,d,t}	0.248*** (0.024)	0.162*** (0.040)	0.136*** (0.046)
<i>Population Share</i> _{HS,d,t}	0.060*** (0.012)	0.035* (0.020)	0.048*** (0.018)
Panel C: Large Establishments			
<i>Population Share</i> _{MS,d,t}	0.000 (0.000)	-0.004 (0.003)	0.002 (0.006)
<i>Population Share</i> _{HS,d,t}	-0.000 (0.000)	0.002 (0.001)	0.005** (0.002)
<i>N</i>	14,896	14,896	14,896
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are small, medium, and large establishment shares of each industrial skill type in the nontradable sector in separate regressions as displayed in Panels A, B, and C, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population at the county level. Consistent with the literature, small establishments are defined as those employing fewer than 20 workers, medium establishments are defined as those employing between 20 and 500 workers, and large establishments are defined as those employing more than 500 workers. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.15: INITIAL INDUSTRY COMPOSITION AND THE MAGNITUDE OF REALLOCATION: TERCILE ESTIMATES

	Employment Share of High-Skill Industries			
	Full Sample	T1: Low $s_{w,d}^0$	T2: Medium $s_{w,d}^0$	T3: High $s_{w,d}^0$
<i>Population Share</i> $_{MS,d,t}$	0.177*** (0.036)	0.113*** (0.040)	0.185*** (0.062)	0.067 (0.077)
<i>Population Share</i> $_{HS,d,t}$	0.544*** (0.020)	0.617*** (0.026)	0.580*** (0.031)	0.415*** (0.037)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000
SW First-Stage F-Stats	133; 364	188; 168	37; 87	144; 234
<i>N</i>	15,705	5,260	5,210	5,235
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of the baseline specification in equation (8) separately by terciles of the county's initial (1970) employment share of high-skill-intensive industries, $s_{w,d}^0$. Column (1) reports estimates for the full sample. Columns (2)–(4) report estimates for counties in the bottom, middle, and top terciles of $s_{w,d}^0$, respectively. The dependent variable is the employment share of high-skill-intensive industries. The endogenous medium- and high-skill population shares are instrumented with their respective exogenous working-age population counterparts. Each specification is exactly identified and includes state and year fixed effects. The table reports the p -value from the Anderson–Rubin Wald F-test and the skill-specific Sanderson–Windmeijer first-stage F-statistics. Standard errors are clustered by county, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

**Table C.16: INITIAL INDUSTRY COMPOSITION AND AGGREGATE EMPLOYMENT:
TERCILE ESTIMATES**

	Aggregate County Employment			
	Full Sample	T1: Low $s_{w,d}^0$	T2: Medium $s_{w,d}^0$	T3: High $s_{w,d}^0$
<i>Population Share</i> $_{MS,d,t}$	0.258*** (0.016)	0.201*** (0.022)	0.342*** (0.034)	0.294*** (0.030)
<i>Population Share</i> $_{HS,d,t}$	0.096*** (0.008)	0.074*** (0.011)	0.151*** (0.018)	0.095*** (0.012)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000
SW First-Stage F-Stats	133; 364	188; 168	37; 87	144; 234
<i>N</i>	15,705	5,260	5,210	5,235
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of the baseline in equation (8) specification separately by terciles of the county's initial (1970) employment share of high-skill-intensive industries, $s_{w,d}^0$. Column (1) reports estimates for the full sample. Columns (2)–(4) report estimates for counties in the bottom, middle, and top terciles of $s_{w,d}^0$, respectively. The dependent variable is the natural log of aggregate county employment. The endogenous medium- and high-skill population shares are instrumented with their respective exogenous working-age population shares. Each specification is exactly identified and includes state and year fixed effects. The table reports the p -value from the Anderson–Rubin Wald F-test and the Sanderson–Windmeijer first-stage F-statistics for the medium- and high-skill population shares, respectively. Standard errors are clustered by county, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.17: ROBUSTNESS - EMPLOYMENT AND ESTABLISHMENT ESTIMATES USING INSTRUMENTS FOR THE 1860-1900 PERIOD

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.381*** (0.048)	0.187*** (0.061)	0.194*** (0.037)
<i>Population Share</i> $_{HS,d,t}$	-0.312*** (0.022)	-0.238*** (0.027)	0.550*** (0.020)
AR Wald F-Test P-Value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.731*** (0.138)	-0.080 (0.137)	0.958*** (0.119)
<i>Population Share</i> $_{HS,d,t}$	-0.086 (0.065)	-0.201*** (0.062)	0.333*** (0.055)
AR Wald F-Test P-Value	0.00	0.000	0.000
SW First-Stage F-Stats	147; 374	147; 374	147; 374
<i>N</i>	15,705	15,705	15,705
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are employment and establishment shares of each industrial skill type at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population. Exogenous working-age population of each skill type has been generated using the predicted ancestry stocks based on the immigrant leave-out push-pull variables for only the 1860-1900 period. Each specification reports the p-value for the Anderson-Rubin Wald F-Test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.18: ROBUSTNESS - EMPLOYMENT AND ESTABLISHMENT ESTIMATES USING INSTRUMENTS FOR THE 1860-1950 PERIOD

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.391*** (0.048)	0.206*** (0.060)	0.185*** (0.037)
<i>Population Share</i> $_{HS,d,t}$	-0.317*** (0.022)	-0.229*** (0.027)	0.546*** (0.020)
AR Wald F-Test P-Value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.744*** (0.136)	-0.070 (0.135)	0.958*** (0.117)
<i>Population Share</i> $_{HS,d,t}$	-0.097 (0.064)	-0.193*** (0.061)	0.334*** (0.055)
AR Wald F-Test P-Value	0.000	0.000	0.000
SW First-Stage F-Stats	138; 365	138; 365	138; 365
<i>N</i>	15,705	15,705	15,705
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are employment and establishment shares of each industrial skill type at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population. Exogenous working-age population of each skill type has been generated using the predicted immigration stocks based on the immigrant leave-out push-pull variables for only the 1860-1950 period. Each specification reports the *p*-value for the Anderson-Rubin Wald F-Test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.19: ROBUSTNESS - EXCLUDING THE FIVE LARGEST ANCESTRY GROUPS

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> _{MS,d,t}	-0.350*** (0.049)	0.152** (0.061)	0.198*** (0.037)
<i>Population Share</i> _{HS,d,t}	-0.330*** (0.022)	-0.211*** (0.027)	0.540*** (0.020)
AR Wald F-Test P-Value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> _{MS,d,t}	-0.723*** (0.138)	-0.109 (0.136)	0.988*** (0.117)
<i>Population Share</i> _{HS,d,t}	-0.111* (0.063)	-0.162*** (0.060)	0.318*** (0.054)
AR Wald F-Test P-Value	0.000	0.019	0.000
SW First-Stage F-Stats	149; 355	149; 355	149; 355
<i>N</i>	15,705	15,705	15,705
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are employment and establishment shares of each industrial skill type in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population. While isolating exogenous working-age population of each skill type, I exclude the five largest ancestry groups from the list of predicted ancestry stocks. These groups are English, German, Irish, Italian, and Mexican. Each specification reports the *p*-value for the Anderson-Rubin Wald F-Test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.20: ROBUSTNESS - EXCLUDING THE FIVE LARGEST COUNTIES

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> _{MS,d,t}	-0.395*** (0.047)	0.219*** (0.058)	0.176*** (0.036)
<i>Population Share</i> _{HS,d,t}	-0.318*** (0.022)	-0.226*** (0.026)	0.544*** (0.020)
AR Wald F-Test P-Value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> _{MS,d,t}	-0.762*** (0.134)	-0.036 (0.131)	0.989*** (0.111)
<i>Population Share</i> _{HS,d,t}	-0.116* (0.064)	-0.164*** (0.060)	0.347*** (0.053)
AR Wald F-Test P-Value	0.000	0.008	0.000
SW First-Stage F-Stats	132; 372	132; 372	132; 372
<i>N</i>	15,680	15,680	15,680
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variables are employment and establishment shares of each industrial skill type at the county level excluding the five most “immigrant-receiving” counties in separate regressions as reported in Panels A and B, respectively. The excluded counties are Los Angeles, Cook, Harris, Miami-Dade, and Kings counties. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population. Each specification reports the *p*-value for the Anderson-Rubin Wald F-Test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.21: ROBUSTNESS - ALTERNATIVE TRADABILITY CLASSIFICATION

	Employment Shares			
	All	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)
Panel A: Tradable Sector				
<i>Population Share_{MS,d,t}</i>	-0.491*** (0.098)	-0.559*** (0.080)	0.171*** (0.049)	-0.102*** (0.032)
<i>Population Share_{HS,d,t}</i>	-0.483*** (0.049)	-0.277*** (0.037)	-0.428*** (0.028)	0.222*** (0.020)
Panel B: Nontradable Sector				
<i>Population Share_{MS,d,t}</i>	0.491*** (0.098)	0.354*** (0.045)	0.012 (0.043)	0.125*** (0.040)
<i>Population Share_{HS,d,t}</i>	0.483*** (0.049)	-0.020 (0.022)	0.030 (0.022)	0.474*** (0.022)
<i>N</i>	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variable is employment share of each industrial skill type in the tradable and nontradable sectors at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population. In classifying workers by the tradability domain, I use an alternative tradability measure highlighted in [Mian and Sufi \(2014\)](#) based on geographical Herfindahl-Hirschman Indices (HHIs). All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.22: ROBUSTNESS - ALTERNATIVE TRADABILITY CLASSIFICATION

	Establishment Shares			
	All	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)
Panel A: Tradable Sector				
<i>Population Share_{MS,d,t}</i>	-0.971*** (0.148)	-1.052*** (0.155)	0.076 (0.126)	0.006 (0.062)
<i>Population Share_{HS,d,t}</i>	-0.099 (0.068)	-0.168** (0.070)	0.076 (0.055)	-0.006 (0.026)
Panel B: Nontradable Sector				
<i>Population Share_{MS,d,t}</i>	0.971*** (0.148)	-0.099 (0.081)	-0.104 (0.098)	1.174*** (0.089)
<i>Population Share_{HS,d,t}</i>	0.099 (0.068)	-0.155*** (0.037)	-0.213*** (0.046)	0.467*** (0.047)
<i>N</i>	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Notes: This table presents the IV results of estimating equation (8). The dependent variable is establishment share of each industrial skill type in the tradable and nontradable sectors at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population. In classifying establishments by the tradability domain, I use an alternative tradability measure highlighted in [Mian and Sufi \(2014\)](#) based on geographical Herfindahl-Hirschman Indices (HHIs). All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.23: ROBUSTNESS - ALTERNATIVE SKILL CLASSIFICATIONS: EMPLOYMENT SHARES

	Panel A: Frozen-1990 Education			Panel B: NRC Task Score		
	Low Skill	Medium Skill	High Skill	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Population Share_{MS,d,t}</i>	-0.398*** (0.073)	0.145*** (0.032)	0.254*** (0.064)	-0.183*** (0.060)	-0.016 (0.049)	0.199*** (0.037)
<i>Population Share_{HS,d,t}</i>	-0.575*** (0.035)	-0.147*** (0.016)	0.722*** (0.034)	-0.545*** (0.028)	0.007 (0.022)	0.538*** (0.023)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000	0.722	0.000
SW First-Stage F-Stats	133; 364	133; 364	133; 364	133; 364	133; 364	133; 364
<i>N</i>	15,705	15,705	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) under two alternative industry-skill classifications. Panel A freezes the industry-skill ranking at the 1990-level cross-section, assigning each industry its average years of schooling as a fixed skill rank evaluated in 1990 across all census years, restricting subsequent attainment growth from relabeling any industry. Panel B classifies industries by the employment-weighted mean non-routine cognitive (NRC) task score of their 1990-level occupational composition (Autor and Dorn, 2013), replacing credential-based ranking with a task-content measure independent of worker education levels. The dependent variables are employment shares of each industrial skill type at the county level. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.24: ROBUSTNESS - SUBSTITUTABILITY AT FINER EDUCATION MARGINS: EMPLOYMENT SHARES

	Panel A: BA – Some College Margin			Panel B: Advanced – BA Margin		
	Low NRC	Medium NRC	High NRC	Low NRC	Medium NRC	High NRC
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Pop Share_{BA,d,t} – Pop Share_{SC,d,t}</i>	-0.518*** (0.046)	-0.094*** (0.025)	0.596*** (0.038)			
<i>Pop Share_{ADV,d,t} – Pop Share_{BA,d,t}</i>				-0.657*** (0.058)	-0.033 (0.050)	0.562*** (0.067)
KP rk Wald F-Stat	524	524	524	2,574	2,574	2,574
<i>N</i>	15,705	15,705	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates using finer education margins to assess substitutability across adjacent skill groups. The dependent variables are county-level employment shares classified by non-routine cognitive (NRC) task intensity (Autor and Dorn, 2013). Panel A instruments the difference between the BA and some-college population shares with the corresponding difference in ancestry-predicted exogenous population stocks. Panel B instruments the difference between the advanced-degree and BA population shares in the same way. Each specification presents the Kleibergen-Paap rk Wald F-statistic. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.25: ROBUSTNESS - BASELINE REGRESSIONS WITH CONTROLS

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> _{MS,d,t}	-0.359*** (0.048)	0.178*** (0.062)	0.180*** (0.040)
<i>Population Share</i> _{HS,d,t}	-0.140*** (0.044)	-0.429*** (0.051)	0.569*** (0.036)
AR Wald F-Test P-value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> _{MS,d,t}	-0.672*** (0.141)	-0.048 (0.134)	0.838*** (0.120)
<i>Population Share</i> _{HS,d,t}	0.150 (0.123)	-0.319*** (0.110)	0.221** (0.100)
AR Wald F-Test P-value	0.000	0.001	0.000
SW First-Stage F-Stats	75; 117	75; 117	75; 117
<i>N</i>	15,705	15,705	15,705
Controls	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) with a full set of controls, which includes newspaper issues (1800–1950) per 1950 county population, average Catholic and Protestant shares (1936–1952), average Democratic presidential vote share (1920–1960), the mining and manufacturing employment share (1950), and the urban population share (1950). The dependent variables are employment and establishment shares of each industrial skill type at the county level in separate specifications as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and time fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.26: ROBUSTNESS - ORTHOGONALIZED INSTRUMENTS: EMPLOYMENT AND ESTABLISHMENT SHARES

	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)
Panel A: Employment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.388*** (0.047)	0.212*** (0.059)	0.176*** (0.036)
<i>Population Share</i> $_{HS,d,t}$	-0.315*** (0.022)	-0.228*** (0.027)	0.543*** (0.020)
AR Wald F-Test P-value	0.000	0.000	0.000
Panel B: Establishment Shares			
<i>Population Share</i> $_{MS,d,t}$	-0.745*** (0.135)	-0.056 (0.133)	0.955*** (0.114)
<i>Population Share</i> $_{HS,d,t}$	-0.109* (0.065)	-0.175*** (0.061)	0.330*** (0.055)
AR Wald F-Test P-value	0.000	0.006	0.000
SW First-Stage F-Stats	133; 363	133; 363	133; 363
<i>N</i>	15,705	15,705	15,705
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) using orthogonalized shift-share instruments. Each baseline instrument is residualized on county-year shares of predicted ancestry aggregated to seven world regions, plus state and year fixed effects; the residuals are orthogonal by construction to any cultural or network effect operating at the macro-region level. The dependent variables are employment and establishment shares of each industrial skill type at the county level, as displayed in Panels A and B, respectively. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with the orthogonalized medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.27: ROBUSTNESS - EXCLUSION RESTRICTION TESTS: PLACEBO OUTCOMES

	Self-Empl. Share	Non-Eng. Speaker Sh.	Catholic Share	Dem. Vote Share
	(1)	(2)	(3)	(4)
<i>Population Share_{MS,d,t}</i>	0.159 (0.125)	2.278 (3.972)	0.125 (0.209)	25.666 (62.541)
<i>Population Share_{HS,d,t}</i>	0.050** (0.025)	0.383 (0.813)	-0.047 (0.054)	11.477 (13.610)
AR Wald F-Test P-value	0.114	0.846	0.188	0.568
SW First-Stage F-Stats	127; 340	127; 340	108; 263	108; 263
<i>N</i>	9,417	9,417	3,140	3,140
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No

Notes: This table reports IV estimates of equation (8) using placebo outcomes to test the exclusion restriction. The dependent variables in columns (1) and (2) are the self-employment share in ethnic-network-intensive industries and the non-English speaker share, respectively, measured over the 1990–2010 period. The dependent variables in columns (3) and (4) are county-level pre-period averages of the Catholic population share (1936–1952) and the Democratic presidential vote share (1920–1960), respectively; these specifications are cross-sectional and do not include year fixed effects. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with medium- and high-skill exogenous working-age population at the county level. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.28: IDENTIFICATION - SELF-EMPLOYMENT IN ETHNIC-NETWORK
INDUSTRIES: HED VS. LED INSTRUMENTS

	HED Instruments	LED Instruments
	(1)	(2)
<i>Population Share</i> $_{MS,d,t}$	0.283 (0.198)	0.181 (0.113)
<i>Population Share</i> $_{HS,d,t}$	0.052 (0.047)	0.050* (0.026)
AR Wald F-Test P-value	0.288	0.136
SW First-Stage F-Stats	110; 187	124; 357
N	9,417	9,417
State FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table reports IV estimates using the self-employment share in ethnic-network-intensive industries as the placebo outcome over the 1990–2010 period. Column (1) instruments with the HED shift-share and column (2) with the LED shift-share. Origin countries are partitioned into 15 clusters by k-means on Spolaore-Wacziarg weighted F_{ST} genetic distance; countries in clusters whose within-cluster standard deviation of Barro-Lee schooling exceeds the median form the HED group, and the remainder form the LED group. Each specification presents the p -value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.29: ROBUSTNESS - HIGH VS. LOW EDUCATION DISPERSION INSTRUMENTS: EMPLOYMENT SHARES

	Panel A: HED Instruments			Panel B: LED Instruments		
	Low Skill	Medium Skill	High Skill	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Population Share_{MS,d,t}</i>	-0.410*** (0.052)	0.302*** (0.061)	0.108*** (0.038)	-0.345*** (0.053)	0.158** (0.064)	0.186*** (0.039)
<i>Population Share_{HS,d,t}</i>	-0.311*** (0.023)	-0.200*** (0.027)	0.511*** (0.020)	-0.339*** (0.022)	-0.203*** (0.025)	0.542*** (0.019)
AR Wald F-Test P-value	0.000	0.000	0.000	0.000	0.000	0.000
SW First-Stage F-Stats	132; 323	132; 323	132; 323	129; 294	129; 294	129; 294
<i>p</i> : HED=LED (<i>MS</i>)				0.381	0.106	0.148
<i>p</i> : HED=LED (<i>HS</i>)				0.363	0.951	0.268
<i>N</i>	15,705	15,705	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) using separate shift-share instruments constructed from high-education-dispersion (HED) and low-education-dispersion (LED) origin countries. Origin countries are partitioned into 15 clusters by k-means on Spolaore-Wacziarg weighted F_{ST} genetic distance; countries in clusters whose within-cluster standard deviation of Barro-Lee schooling exceeds the median from the HED group, and the remainder form the LED group. Countries in the same cluster share similar cultural proximity but differ in education variation. The dependent variables are employment shares of each industrial skill type at the county level. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with HED or LED shift-share instruments, respectively. The equality test rows report the *p*-value from a conservative *t*-test of $H_0: \beta_{HED} = \beta_{LED}$, with $t = (\hat{\beta}_{HED} - \hat{\beta}_{LED}) / \sqrt{SE_{HED}^2 + SE_{LED}^2}$. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.30: ROBUSTNESS - HIGH VS. LOW EDUCATION DISPERSION INSTRUMENTS: ESTABLISHMENT SHARES

	Panel A: HED Instruments			Panel B: LED Instruments		
	Low Skill	Medium Skill	High Skill	Low Skill	Medium Skill	High Skill
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Population Share_{MS,d,t}</i>	-0.945*** (0.144)	0.007 (0.142)	1.083*** (0.124)	-0.807*** (0.142)	-0.155 (0.144)	1.124*** (0.121)
<i>Population Share_{HS,d,t}</i>	-0.174*** (0.067)	-0.165*** (0.063)	0.383*** (0.057)	-0.230*** (0.060)	-0.104* (0.057)	0.369*** (0.053)
AR Wald F-Test P-value	0.000	0.003	0.000	0.000	0.200	0.000
SW First-Stage F-Stats	132; 323	132; 323	132; 323	129; 294	129; 294	129; 294
<i>p</i> : HED=LED (<i>MS</i>)				0.495	0.423	0.810
<i>p</i> : HED=LED (<i>HS</i>)				0.532	0.475	0.865
<i>N</i>	15,705	15,705	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) using separate shift-share instruments constructed from high-education-dispersion (HED) and low-education-dispersion (LED) origin countries. Origin countries are partitioned into 15 clusters by k-means on Spolaore-Wacziarg weighted F_{ST} genetic distance; countries in clusters whose within-cluster standard deviation of Barro-Lee schooling exceeds the median from the HED group, and the remainder form the LED group. Countries in the same cluster share similar cultural proximity but differ in education variation. The dependent variables are establishment shares of each industrial skill type at the county level. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with HED or LED shift-share instruments, respectively. The equality test rows report the *p*-value from a conservative *t*-test of $H_0: \beta_{HED} = \beta_{LED}$, with $t = (\hat{\beta}_{HED} - \hat{\beta}_{LED}) / \sqrt{SE_{HED}^2 + SE_{LED}^2}$. Each specification presents the *p*-value for the Anderson-Rubin Wald F-test and the Sanderson-Windmeijer F-statistic for each endogenous variable. All regressions include state and year fixed effects. Standard errors are clustered by county for all specifications, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.31: IV ESTIMATES OF EMPLOYMENT SHARES WITH CONLEY SPATIAL-HAC INFERENCE

	50 km	100 km	200 km	500 km	1,000 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Low Skill					
<i>Population Share</i> $_{MS,d,t}$	-0.389*** (0.060)	-0.389*** (0.070)	-0.389*** (0.079)	-0.389*** (0.095)	-0.389*** (0.080)
<i>Population Share</i> $_{HS,d,t}$	-0.315*** (0.026)	-0.315*** (0.033)	-0.315*** (0.040)	-0.315*** (0.052)	-0.315*** (0.078)
Panel B: Medium Skill					
<i>Population Share</i> $_{MS,d,t}$	0.212*** (0.075)	0.212** (0.085)	0.212** (0.097)	0.212* (0.111)	0.212** (0.093)
<i>Population Share</i> $_{HS,d,t}$	-0.229*** (0.033)	-0.229*** (0.042)	-0.229*** (0.050)	-0.229*** (0.071)	-0.229** (0.104)
Panel C: High Skill					
<i>Population Share</i> $_{MS,d,t}$	0.177*** (0.038)	0.177*** (0.042)	0.177*** (0.050)	0.177*** (0.058)	0.177*** (0.058)
<i>Population Share</i> $_{HS,d,t}$	0.544*** (0.021)	0.544*** (0.026)	0.544*** (0.031)	0.544*** (0.041)	0.544*** (0.052)
KP rk Wald F-Stat	69	59	44	23	20
<i>N</i>	15,705	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) using Conley spatial-HAC inference. Panels A, B, and C report results for the employment shares of low-, medium-, and high-skill industries, respectively. Columns (1)-(5) report results using spatial distance cutoffs of 50, 100, 200, 500, and 1,000 kilometers, respectively, with a serial-correlation correction in the time dimension. The independent variables are the endogenous medium- and high-skill working-age population shares, instrumented with their respective exogenous counterparts. Since the specification and estimation sample remain unchanged across columns, the coefficient estimates are identical, while the spatial-HAC standard errors and corresponding statistical significance vary across distance cutoffs. Each panel reports the Kleibergen-Paap rk Wald F-statistic for the corresponding cutoff. All regressions include state and year fixed effects. Standard errors are reported in parentheses, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.32: IV ESTIMATES OF ESTABLISHMENT SHARES WITH CONLEY SPATIAL-HAC INFERENCE

	50 km	100 km	200 km	500 km	1,000 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Low Skill					
<i>Population Share</i> $_{MS,d,t}$	-0.745*** (0.165)	-0.745*** (0.191)	-0.745*** (0.233)	-0.745** (0.300)	-0.745** (0.304)
<i>Population Share</i> $_{HS,d,t}$	-0.110 (0.078)	-0.110 (0.097)	-0.110 (0.113)	-0.110 (0.094)	-0.110 (0.099)
Panel B: Medium Skill					
<i>Population Share</i> $_{MS,d,t}$	-0.056 (0.142)	-0.056 (0.158)	-0.056 (0.193)	-0.056 (0.247)	-0.056 (0.227)
<i>Population Share</i> $_{HS,d,t}$	-0.175*** (0.067)	-0.175** (0.078)	-0.175** (0.087)	-0.175** (0.081)	-0.175** (0.072)
Panel C: High Skill					
<i>Population Share</i> $_{MS,d,t}$	0.955*** (0.126)	0.955*** (0.139)	0.955*** (0.152)	0.955*** (0.178)	0.955*** (0.224)
<i>Population Share</i> $_{HS,d,t}$	0.331*** (0.059)	0.331*** (0.067)	0.331*** (0.070)	0.331*** (0.059)	0.331*** (0.068)
KP rk Wald F-Stat	69	59	44	23	20
<i>N</i>	15,705	15,705	15,705	15,705	15,705
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates of equation (8) using Conley spatial-HAC inference. Panels A, B, and C report results for the establishment shares of low-, medium-, and high-skill industries, respectively. Columns (1)-(5) report results using spatial distance cutoffs of 50, 100, 200, 500, and 1,000 kilometers, respectively, with a serial-correlation correction in the time dimension. The independent variables are the endogenous medium- and high-skill working-age population shares, instrumented with their respective exogenous counterparts. Since the specification and estimation sample remain unchanged across columns, the coefficient estimates are identical, while the spatial-HAC standard errors and corresponding statistical significance vary across distance cutoffs. The table reports the Kleibergen-Paap rk Wald F-statistic for each cutoff. All regressions include state and year fixed effects. Standard errors are reported in parentheses, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.33: IV ESTIMATES OF EMPLOYMENT SHARES AT THE COMMUTING ZONE LEVEL WITH CONLEY SPATIAL-HAC INFERENCE

	50 km	100 km	200 km	500 km	1,000 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Low Skill					
<i>Population Share</i> _{MS,d,t}	−0.571*** (0.161)	−0.571*** (0.163)	−0.571*** (0.167)	−0.571*** (0.189)	−0.571*** (0.209)
<i>Population Share</i> _{HS,d,t}	−0.407*** (0.072)	−0.407*** (0.076)	−0.407*** (0.084)	−0.407*** (0.101)	−0.407*** (0.130)
Panel B: Medium Skill					
<i>Population Share</i> _{MS,d,t}	0.416** (0.193)	0.416** (0.195)	0.416** (0.201)	0.416* (0.216)	0.416* (0.225)
<i>Population Share</i> _{HS,d,t}	−0.000 (0.083)	−0.000 (0.086)	−0.000 (0.097)	−0.000 (0.111)	−0.000 (0.138)
Panel C: High Skill					
<i>Population Share</i> _{MS,d,t}	0.155 (0.141)	0.155 (0.144)	0.155 (0.151)	0.155 (0.182)	0.155 (0.196)
<i>Population Share</i> _{HS,d,t}	0.407*** (0.066)	0.407*** (0.068)	0.407*** (0.076)	0.407*** (0.087)	0.407*** (0.103)
KP rk Wald F-Stat	17	16	15	14	9
<i>N</i>	3,705	3,705	3,705	3,705	3,705
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table reports commuting-zone-level IV estimates of equation (8) using Conley spatial-HAC inference. Panels A, B, and C report results for the employment shares of low-, medium-, and high-skill industries, respectively. Columns (1)-(5) report results using spatial distance cutoffs of 50, 100, 200, 500, and 1,000 kilometers, respectively, with a serial-correlation correction in the time dimension. The independent variables are endogenous medium- and high-skill working-age population shares at the commuting-zone level, instrumented with their respective exogenous counterparts. Since the specification and estimation sample remain unchanged across columns, the coefficient estimates are identical, while the spatial-HAC standard errors and corresponding statistical significance vary across distance cutoffs. The table reports the Kleibergen-Paap rk Wald F-statistic for each cutoff, rounded to the nearest whole number. All regressions include state and year fixed effects. Standard errors are reported in parentheses, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table C.34: IV ESTIMATES OF ESTABLISHMENT SHARES AT THE COMMUTING ZONE LEVEL WITH CONLEY SPATIAL-HAC INFERENCE

	50 km	100 km	200 km	500 km	1,000 km
	(1)	(2)	(3)	(4)	(5)
Panel A: Low Skill					
<i>Population Share</i> $_{MS,d,t}$	−1.008** (0.428)	−1.008** (0.448)	−1.008** (0.491)	−1.008 (0.627)	−1.008 (0.685)
<i>Population Share</i> $_{HS,d,t}$	−0.164 (0.192)	−0.164 (0.219)	−0.164 (0.233)	−0.164 (0.288)	−0.164 (0.299)
Panel B: Medium Skill					
<i>Population Share</i> $_{MS,d,t}$	0.211 (0.392)	0.211 (0.397)	0.211 (0.414)	0.211 (0.481)	0.211 (0.508)
<i>Population Share</i> $_{HS,d,t}$	0.105 (0.172)	0.105 (0.187)	0.105 (0.197)	0.105 (0.231)	0.105 (0.275)
Panel C: High Skill					
<i>Population Share</i> $_{MS,d,t}$	1.678*** (0.342)	1.678*** (0.353)	1.678*** (0.376)	1.678*** (0.409)	1.678*** (0.538)
<i>Population Share</i> $_{HS,d,t}$	0.614*** (0.157)	0.614*** (0.164)	0.614*** (0.178)	0.614*** (0.204)	0.614*** (0.210)
KP rk Wald F-Stat	17	16	15	14	9
<i>N</i>	3,705	3,705	3,705	3,705	3,705
State FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table reports commuting-zone-level IV estimates of equation (8) using Conley spatial-HAC inference. Panels A, B, and C report results for the establishment shares of low-, medium-, and high-skill industries, respectively. Columns (1)-(5) report results using spatial distance cutoffs of 50, 100, 200, 500, and 1,000 kilometers, respectively, with a serial-correlation correction in the time dimension. The independent variables are endogenous medium- and high-skill working-age population shares at the commuting-zone level, instrumented with their respective exogenous counterparts. Since the specification and estimation sample remain unchanged across columns, the coefficient estimates are identical, while the spatial-HAC standard errors and corresponding statistical significance vary across distance cutoffs. The table reports the Kleibergen-Paap rk Wald F-statistic for each cutoff, rounded to the nearest whole number. All regressions include state and year fixed effects. Standard errors are reported in parentheses, and *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

D Appendix Figures

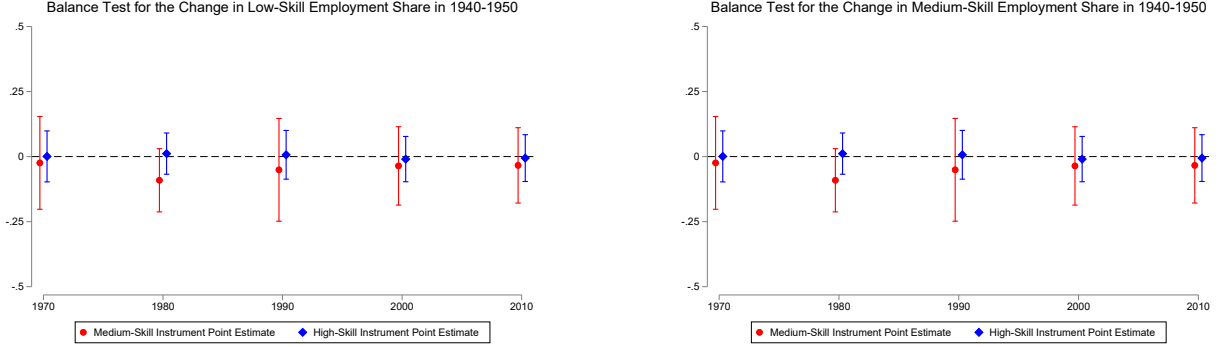


Figure D.1: ADDITIONAL BALANCE TESTS: THE CHANGES IN EMPLOYMENT SHARES IN 1940-1950

Notes: These figures display the results of the balance tests specified in equation (16). Coefficients report the effect of a one-standard-deviation increase in each instrument, human-capital-specific exogenous population, for each year separately over the 1970-2010 period on a standardized measure of the changes in low- and medium-skill employment shares between 1940 and 1950. All regressions include state fixed effects, and standard errors are clustered by county for all specifications. Whiskers indicate 95% confidence intervals computed using standard errors clustered by county.

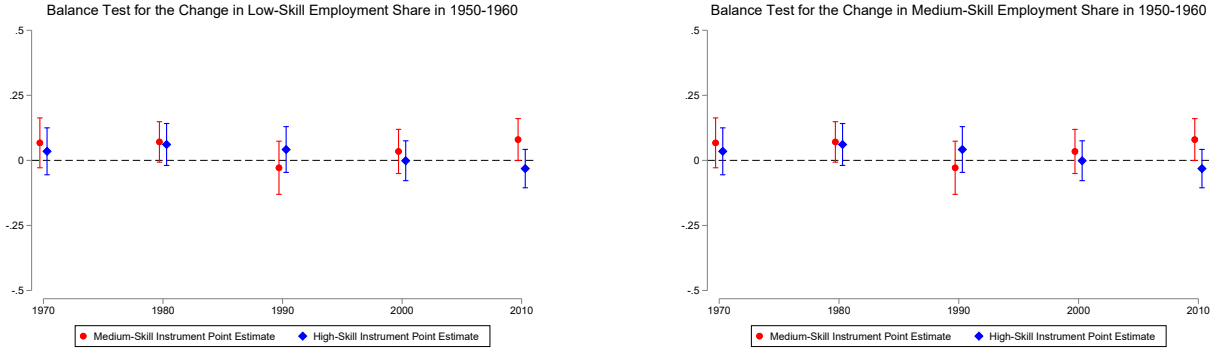


Figure D.2: ADDITIONAL BALANCE TESTS: THE CHANGES IN EMPLOYMENT SHARES IN 1950-1960

Notes: These figures display the results of the balance tests specified in equation (16). Coefficients report the effect of a one-standard-deviation increase in each instrument, human-capital-specific exogenous population, for each year separately over the 1970-2010 period on a standardized measure of the changes in low- and medium-skill employment shares between 1950 and 1960. All regressions include state fixed effects, and standard errors are clustered by county for all specifications. Whiskers indicate 95% confidence intervals computed using standard errors clustered by county.

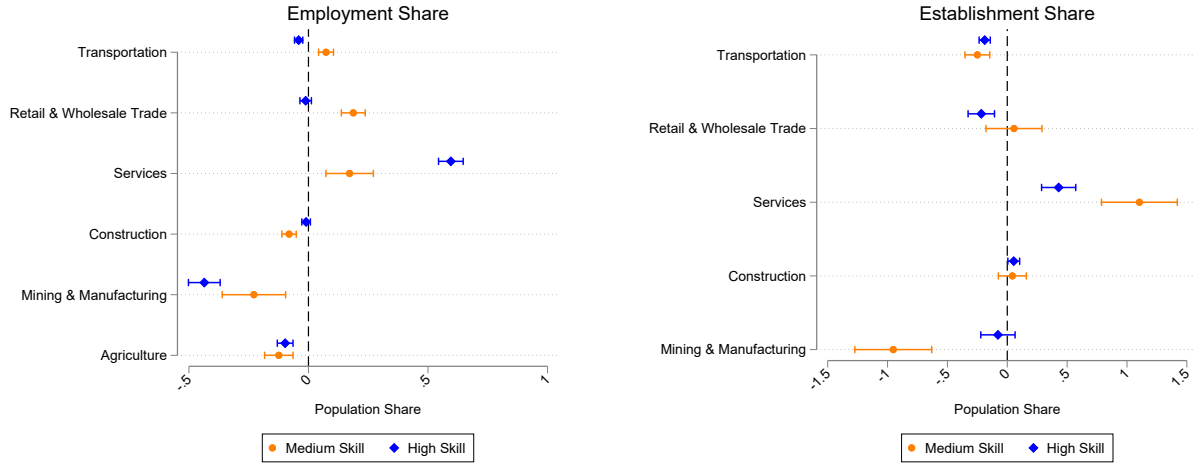


Figure D.3: EMPLOYMENT AND ESTABLISHMENT ESTIMATES BY INDUSTRY

Notes: These figures display the coefficient plots of estimating equation (8), but the outcome variables are not disaggregated by industrial skill types. The dependent variables are employment and establishment shares of major industries at the county level in separate specifications. The independent variables are endogenous medium- and high-skill working-age population shares instrumented with exogenous medium- and high-skill working-age population at the county level. The “Services” sector also includes “Finance, Insurance, and Real Estate” along with “Public Administration,” and the “Transportation” sector includes “Communications” and “Other Public Utilities.” Since the CBP data do not provide the full coverage of establishments operating in agriculture during the study period, I exclude them from the analysis. All regressions include state and time fixed effects, and standard errors are clustered by county for all specifications. Whiskers indicate 95% confidence intervals computed using standard errors clustered by county.

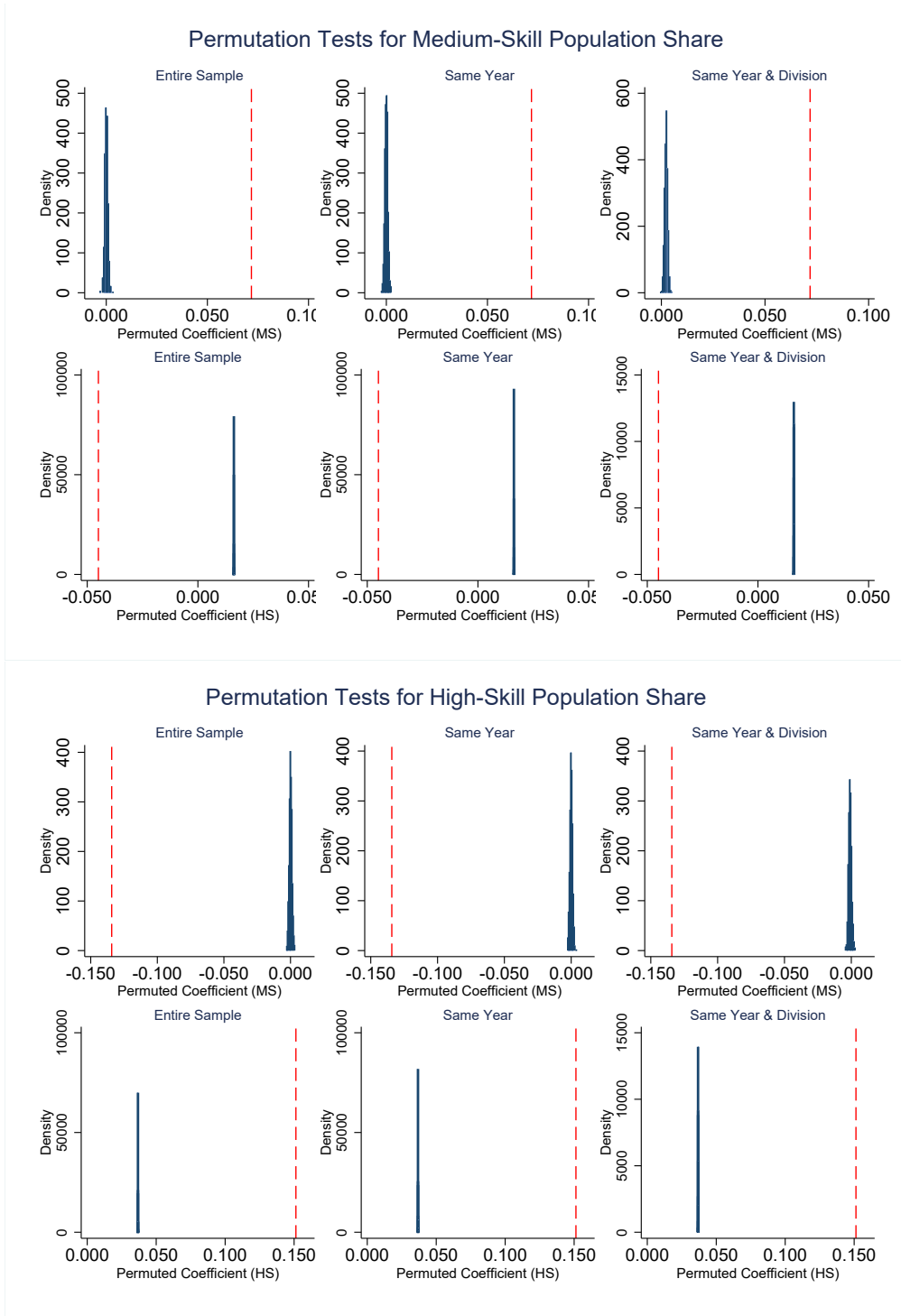


Figure D.4: PERMUTATION TESTS: FIRST-STAGE ESTIMATES

Notes: These figures display three permutation (placebo) tests for the first-stage estimates in Appendix Table C.1. In each test, I randomly re-assign the instruments across observations: (i) within the entire sample, (ii) within census year t , and (iii) within census year t and division $r(d)$. For each design, I run 1,000 placebo regressions and plot the distribution of the estimated coefficients. The dashed red lines mark the point estimates from the main first-stage results. Across all designs, the placebo distributions are tightly centered at zero and lie far from the main estimates; the implied one-sided permutation p -values are less than 0.01. All placebo regressions include state and year fixed effects, and standard errors are clustered by county for all specifications.

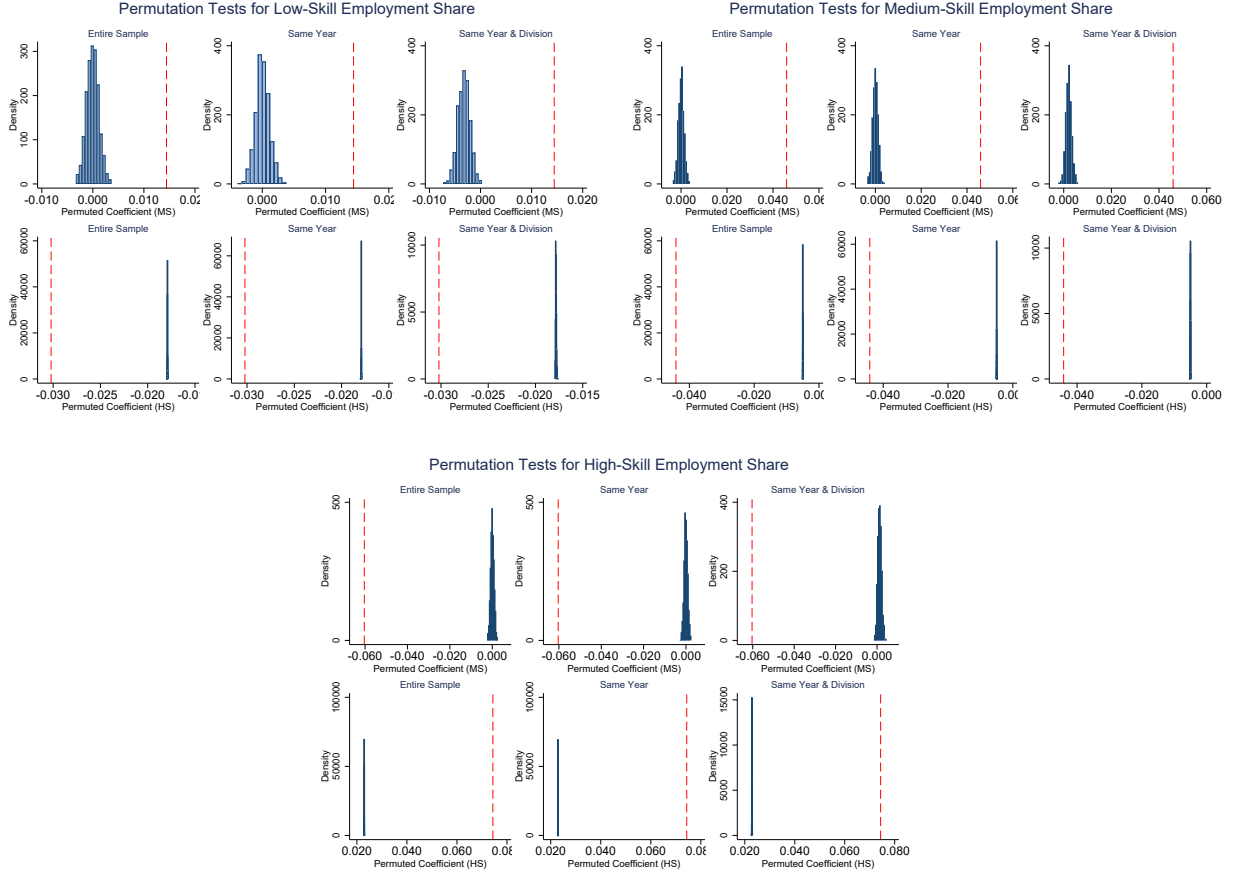


Figure D.5: PERMUTATION TESTS: REDUCED-FORM ESTIMATES FOR EMPLOYMENT SHARES

Notes: These figures display three permutation (placebo) tests for the reduced-form estimates for employment shares. In each test, I randomly re-assign the instruments across observations: (i) within the entire sample, (ii) within census year t , and (iii) within census year t and division $r(d)$. For each design, I run 1,000 placebo regressions and plot the distribution of the estimated coefficients. The dashed red lines mark the point estimates from the main reduced-form results. Across all designs, the placebo distributions are tightly centered at zero and lie far from the main estimates; the implied one-sided permutation p -values are less than 0.01. All placebo regressions include state and year fixed effects, and standard errors are clustered by county for all specifications.

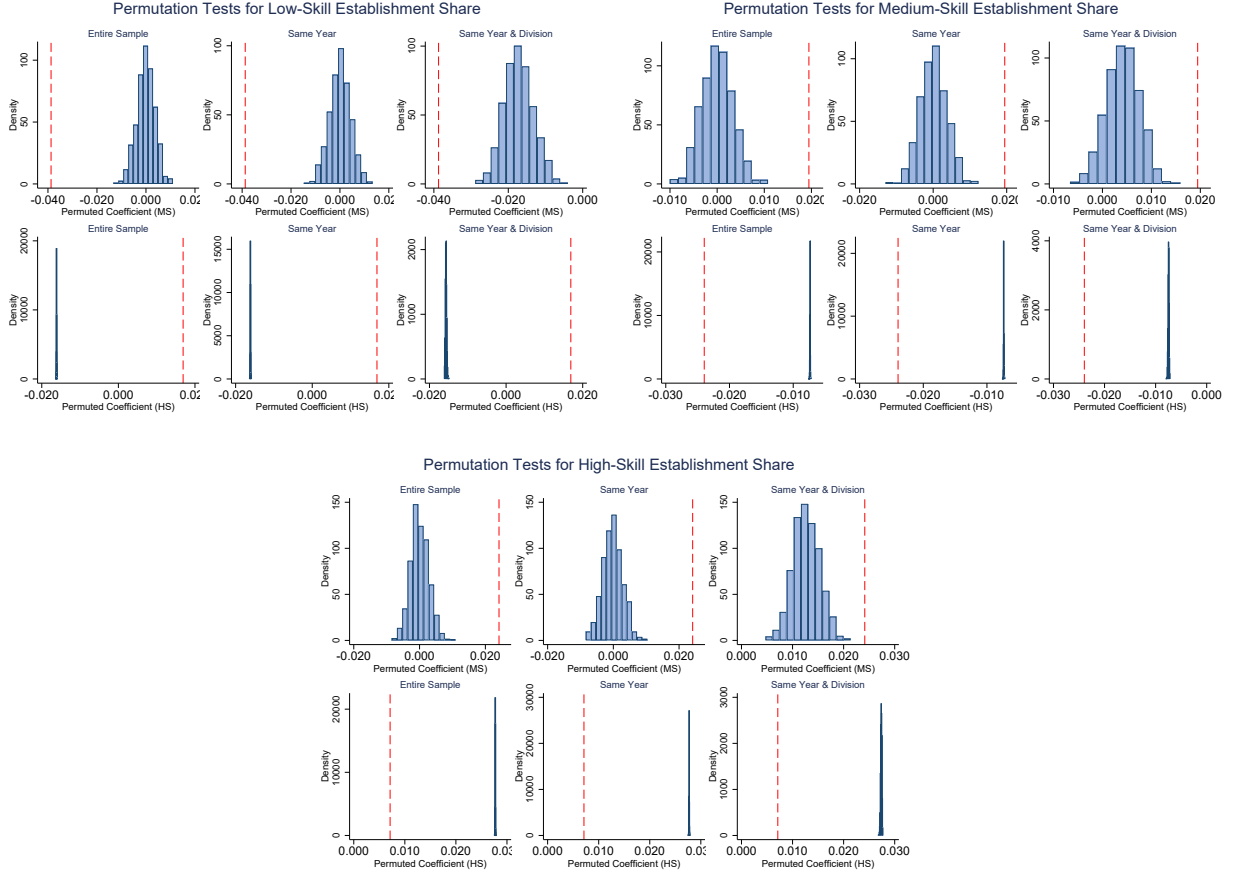


Figure D.6: PERMUTATION TESTS: REDUCED-FORM ESTIMATES FOR ESTABLISHMENT SHARES

Notes: These figures display three permutation (placebo) tests for the reduced-form estimates for establishment shares. In each test, I randomly re-assign the instruments across observations: (i) within the entire sample, (ii) within census year t , and (iii) within census year t and division $r(d)$. For each design, I run 1,000 placebo regressions and plot the distribution of the estimated coefficients. The dashed red lines mark the point estimates from the main reduced-form results. Across all designs, the placebo distributions are tightly centered at zero and lie far from the main estimates; the implied one-sided permutation p -values are less than 0.01. All placebo regressions include state and year fixed effects, and standard errors are clustered by county for all specifications.

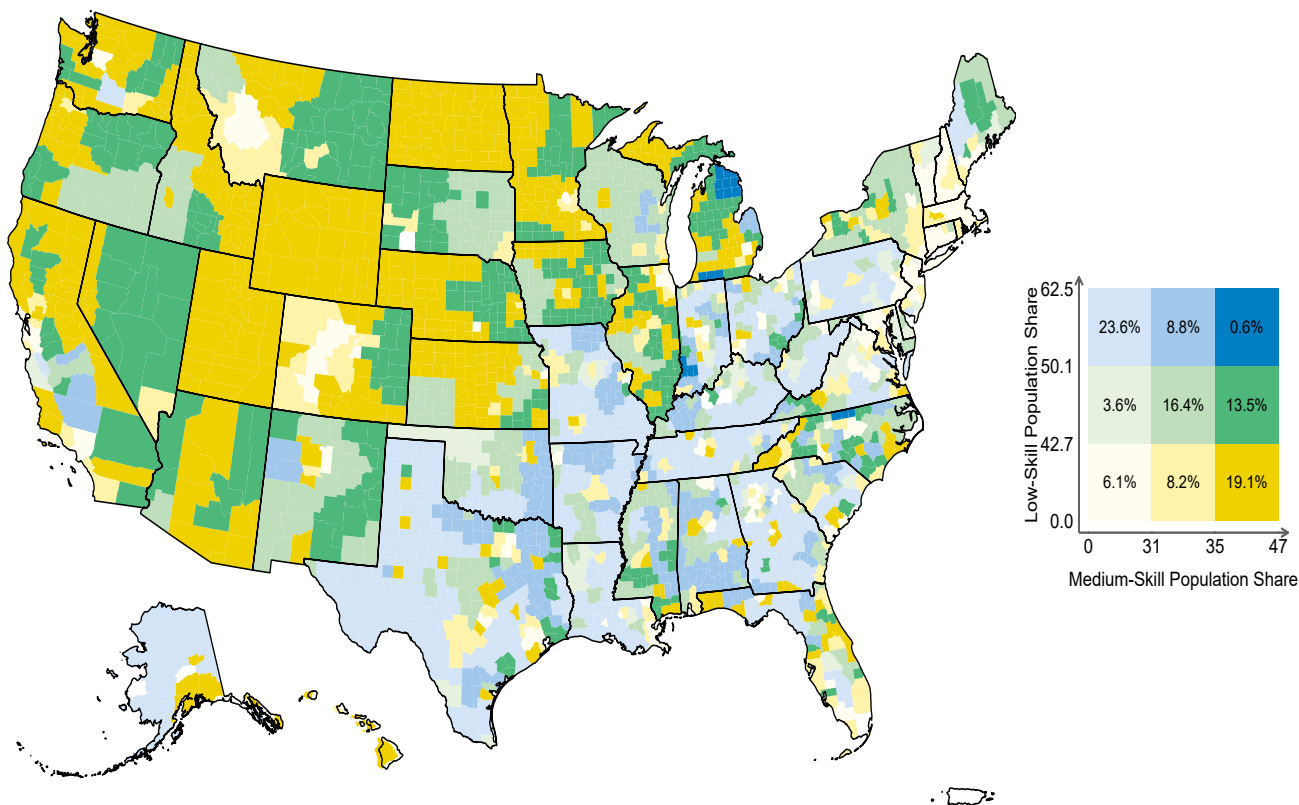


Figure D.7: THE BIVARIATE MAP OF LOW- AND MEDIUM-SKILL WORKING-AGE POPULATION SHARES IN 2010

Notes: This bivariate map illustrates the distribution of low- and medium-skill working-age population shares across US counties in 2010. Lighter colors indicate lower concentrations, while darker colors represent higher concentrations for each respective skill type. The map has been plotted using the “bimap” package in Stata ([Naqvi, 2024](#)).

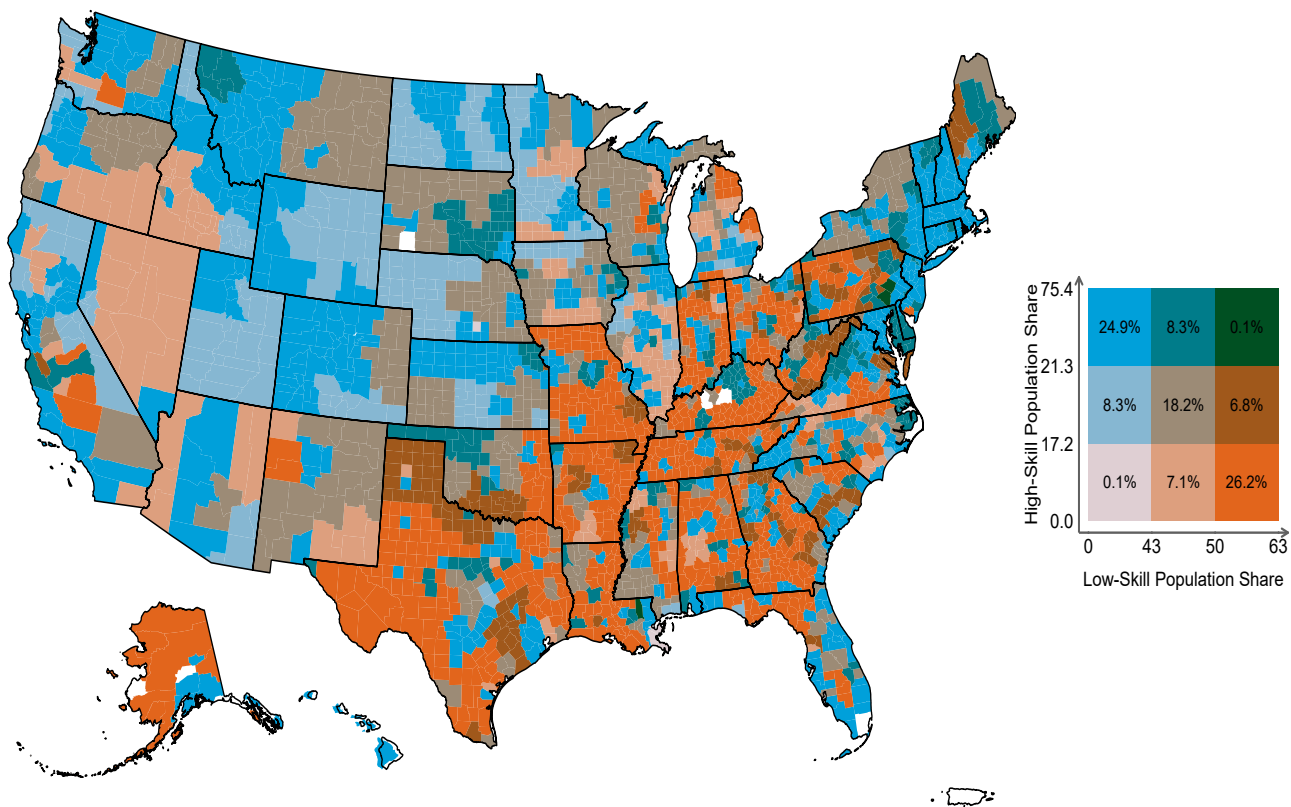


Figure D.8: THE BIVARIATE MAP OF LOW- AND HIGH-SKILL WORKING-AGE POPULATION SHARES IN 2010

Notes: This bivariate map illustrates the distribution of low- and high-skill working-age population shares across US counties in 2010. Lighter colors indicate lower concentrations, while darker colors represent higher concentrations for each respective skill type. The map has been plotted using the “bimap” package in Stata ([Naqvi, 2024](#)).

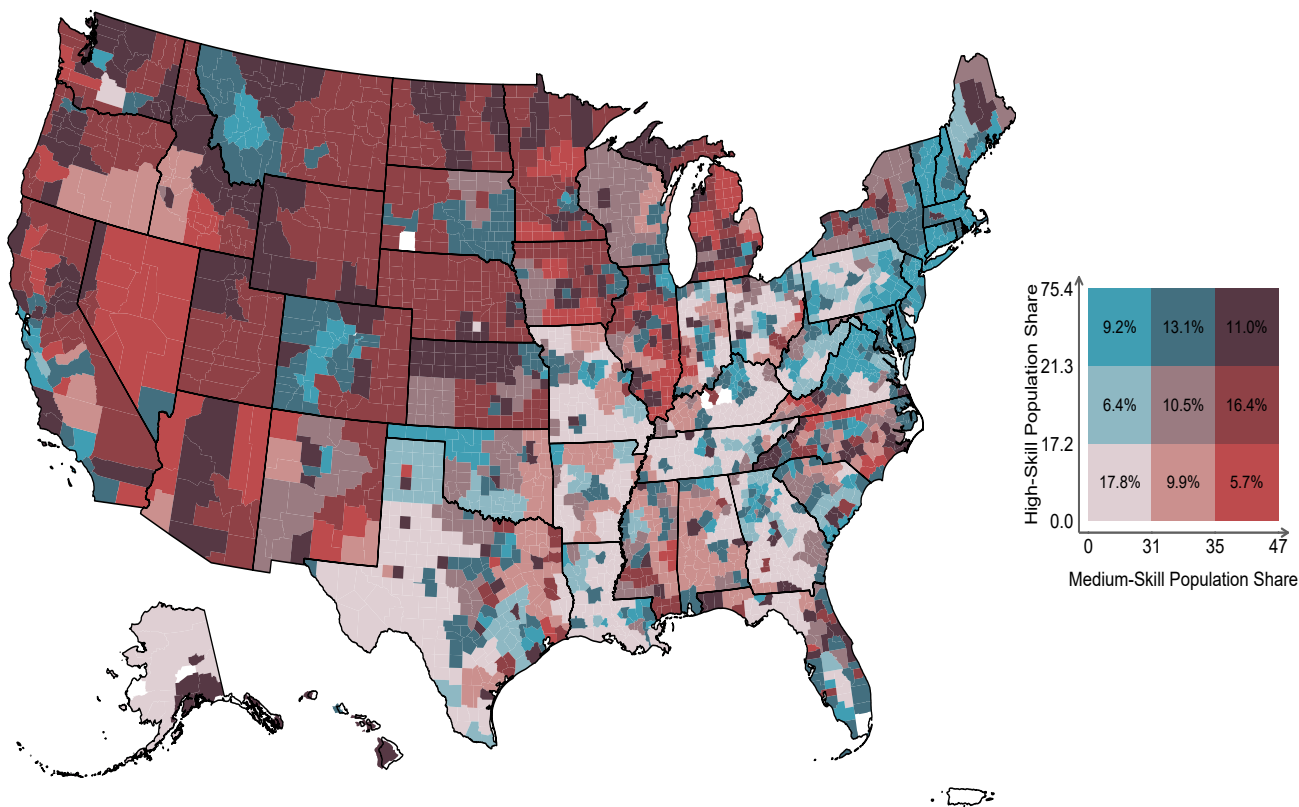
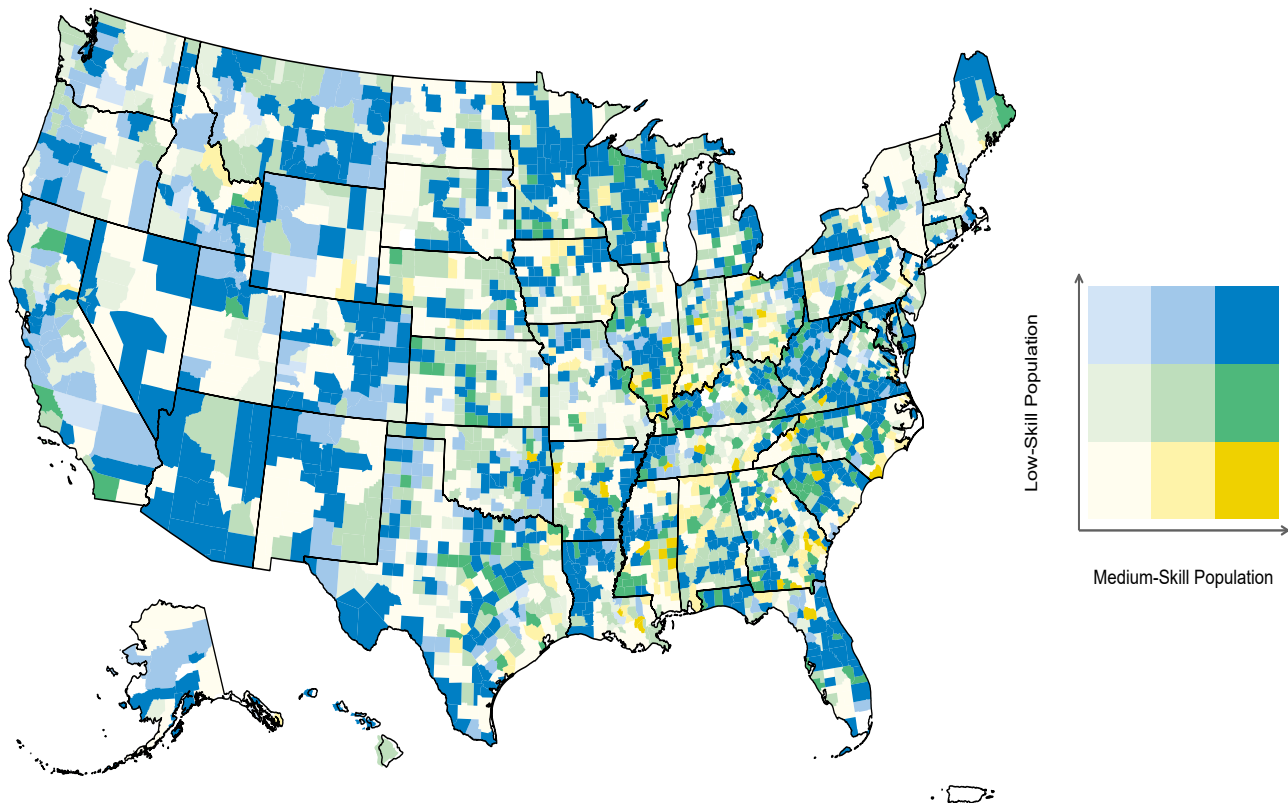


Figure D.9: THE BIVARIATE MAP OF LOW- AND HIGH-SKILL WORKING-AGE POPULATION SHARES IN 2010

Notes: This bivariate map illustrates the distribution of medium- and high-skill working-age population shares across US counties in 2010. Lighter colors indicate lower concentrations, while darker colors represent higher concentrations for each respective skill type. The map has been plotted using the “bimap” package in Stata ([Naqvi, 2024](#)).



**Figure D.10: THE BIVARIATE MAP OF LOW- AND MEDIUM-SKILL EXOGENOUS POPULATION
IN 2010**

Notes: This bivariate map illustrates the distribution of low- and medium-skill exogenous population across US counties in 2010. I regress each instrument on county and time fixed effects and obtain the residuals. The color coding visualizes the 200 quantiles of the residuals across counties. Lighter colors indicate lower quantiles, while darker colors represent higher quantiles for each respective skill type. The map has been plotted using the “`bimap`” package in Stata (Naqvi, 2024).

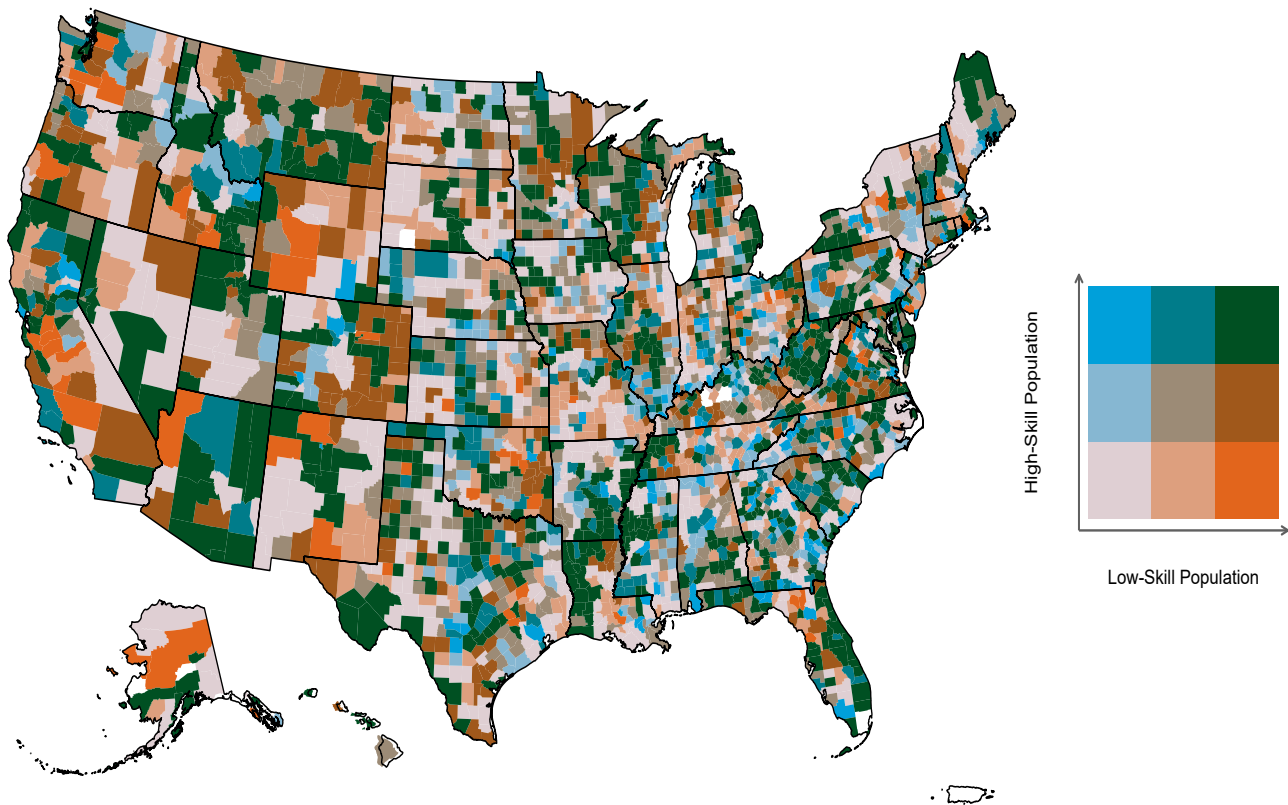
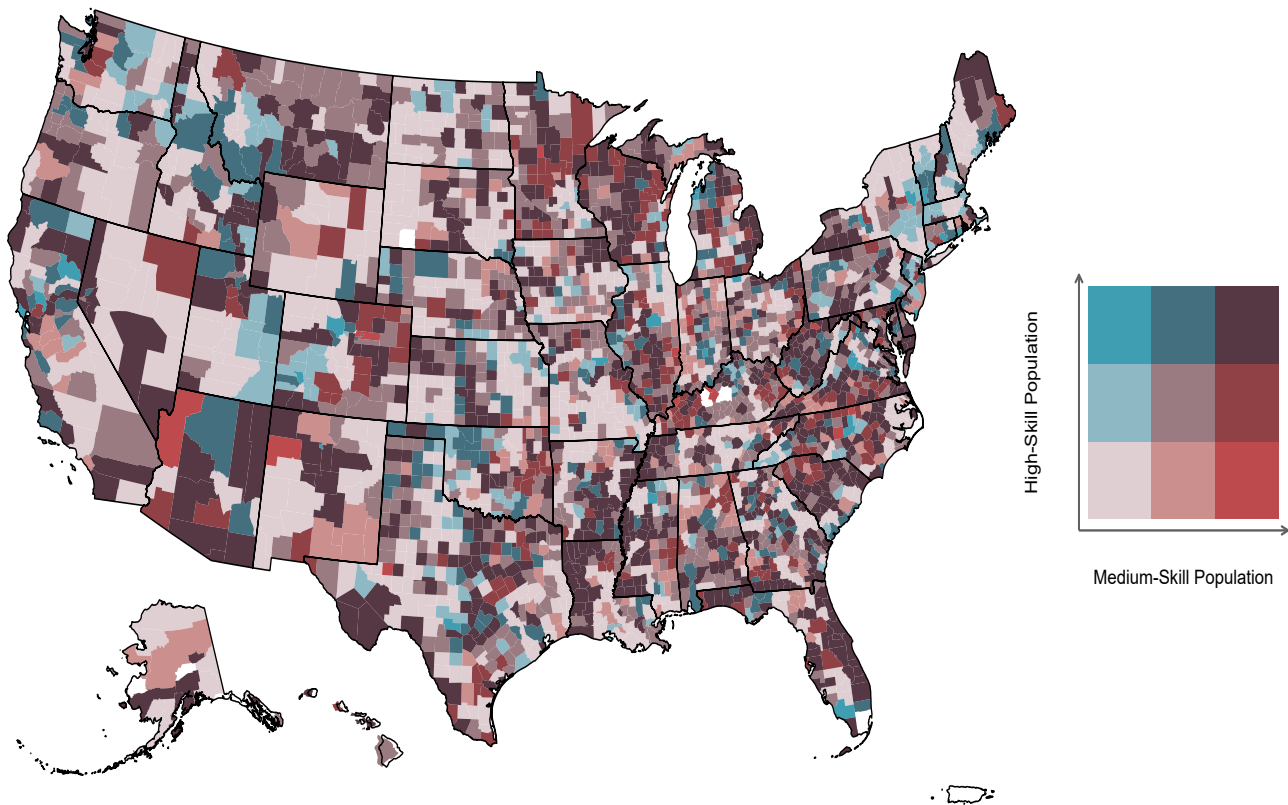


Figure D.11: THE BIVARIATE MAP OF LOW- AND HIGH-SKILL EXOGENOUS POPULATION IN 2010

Notes: This bivariate map illustrates the distribution of low- and high-skill exogenous population across US counties in 2010. I regress each instrument on county and time fixed effects and obtain the residuals. The color coding visualizes the 200 quantiles of the residuals across counties. Lighter colors indicate lower quantiles, while darker colors represent higher quantiles for each respective skill type. The map has been plotted using the “**bimap**” package in Stata ([Naqvi, 2024](#)).



**Figure D.12: THE BIVARIATE MAP OF MEDIUM- AND HIGH-SKILL EXOGENOUS POPULATION
IN 2010**

Notes: This bivariate map illustrates the distribution of medium- and high-skill exogenous population across US counties in 2010. I regress each instrument on state and time fixed effects and obtain the residuals. The color coding visualizes the 200 quantiles of the residuals across counties. Lighter colors indicate lower quantiles, while darker colors represent higher quantiles for each respective skill type. The map has been plotted using the “`bimap`” package in Stata (Naqvi, 2024).